IBA Cyclone® KIUBE
Site Requirement Guide
Vaulted & self-shielded

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Abstract:

The Site Requirement Guide for the Cyclone® KIUBE series vaulted or self-shielded version (Cyclone® KIUBE 100, Cyclone® KIUBE 150, Cyclone® KIUBE 180, Cyclone® KIUBE 300) provides information for site layout, utilities installation and IBA requirements for installation. It also provides information regarding the typical planning of a cyclotron installation and start-up. **This document is complete with the drawings annexed (listed in chap. 13)**

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1. LIST OF EQUIPMENT AND ROOMS

1.1 STANDARD EQUIPMENT CONFIGURATION

- Cyclotron and all related components for magnetic field, RF electric field, ion source power and vacuum generation
- Control system based on a PLC, Windows Server and Thin Client
- Helium cooling system, compressed air manifold and target accessories cabinet depending on target configuration
- Water cooling circulation system and manifold



Figure 1: Cyclone® KIUBE

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1.2 STANDARD ROOM CONFIGURATION

Refer to drawing SRG-0001 for vaulted cyclotron and drawing SRG-0300 and Chapter 9 for self-shielded version.

- **Cyclotron vault** for Cyclone® KIUBE and auxiliary equipment
- **Power supply room** for cyclotron power supply cabinets (RF, magnetic field, ion source, PLC, …)
- **Control room** for cyclotron control station. We recommend a separated room because of heat and noise generated by power supplies cabinets
- **Cyclotron lobby** with access to cyclotron vault
- **Heat exchanger room for water conditioner**, recommended as separate room.

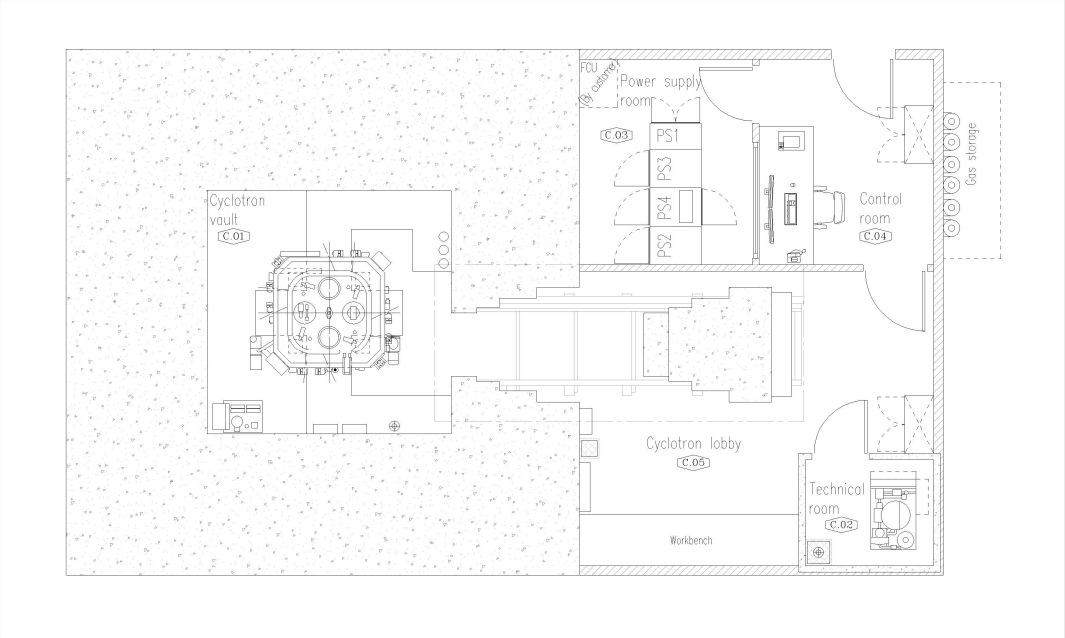


Figure 2: Standard layout of vaulted machine

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1.3 MAIN EQUIPMENT DIMENSIONS AND WEIGHT

Refer to drawing SRG-0001.

Table 1 : main equipment size and net weight

Description	Size WxDxH (mm)	*Weight* (kg)
Cyclotron		
A1 Cyclone® KIUBE	2000 x 2000 x 1850	18 000
Power supply room equipment:		
PS1 Power distribution cabinet	800 x 500 x 1880	215
PS2 RF Amplifier cabinet	600 x 800 x 1880	600
PS3 Ion sources & main coil cabinet	600 x 800 x 1880	500
PS4 Control cabinet (PLC)	600 x 800 x 1880	250
Heat exchanger room equipment		
A3 Water Conditioner	1300 x 700 x 1610	220
Auxiliary equipment in the vault		
A4 Hydraulic yoke lifting system	650 x 450 x 800	145
A7-A8 Target switching input-output modules (optional)	200 x 155 x 300 (each)	10 (each)
A9 Primary pump	690 x 400 x 500	36
A11 Helium compressor	690 x 400 x 500	90
A14 Gas target switching (optional)	200 x 155 x 300	10
Control room equipment:		
A2 Control PC + printer	400 x 510 x 160	10
A2 Screens (2)	410 x 200 x 510 (each)	5 (each)

1.4 ADDITIONAL EQUIPMENT

Some components are optional and will be installed depending on the Scope of Supply of Contract:

- Target systems: The cyclotron can be fitted with up to 8 target stations to produce the most common PET radioisotopes.
- Chemistry modules and hotcells: refer to dedicated datasheets
- Target shielded doors: local shielding to cover the exit ports and reduce the neutron flux near the cyclotron. The use of such doors will reduce the vault internal layer activation. Up to 4 ports can be equipped.
- Beam lines can be proposed as optional equipment; see chapter 7
- Cyclone® KIUBE could be self-shielded; see chapter 9

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- Syringe pump & (optional) Target Rinsing

The syringe pumps handle the liquid target material for $^{18}F^-$ (loading and rinsing), ^{68}Ga or $^{13}NH_3$ production targets.

All syringe pumps will be installed inside the PLC cabinet (PS4) in the Power Supply room or in a dedicated cabinet (optional, recommended for Gallium and Copper liquid targets). The electronic syringe pump is actuated through IBA controller to push the exact amount of liquid into each target system. Moreover, IBA's optimized design for target filling with target distribution valve on the cyclotron and filling bottle outside the vault reduces to almost zero (few µl) the enriched water losses in the line while giving the advantage of not having to enter the vault to replace enriched water bottle and check level.

For $^{18}F^-$ production, as an option, a second syringe pump can be added in order to rinse the target with $[^{16}O]$-H_2O after and/or before the run.

Liquid Target Switching Cabinet

(Optional, A7-A8 on drawing SRG-0001)

When multiple liquid targets for the same isotope are installed, liquid target switching boxes can be provided to be able to send activity produced with any of the targets to one out of multiple outputs. Swtching cabinet IN is usually installed on the vault wall, while switching cabinet OUT is either installed in the vault or in shielded trench closer to hotcells.

Inputs/outputs	8 in / 8 out (liquid target switching art. 220.021) 8 in / 16 out (liquid target switching plus art. 220.024)
Control	from cyclotron control system

Gas Switching System (optional)

(Optional, A14 on drawing SRG-0001)

The gas switching system has the same purpose as the liquid switching system. It allows sending activity from one of maximum two gas targets of the same type (CO_2 or CH_4) to one out of multiple destinations.

Inputs/outputs	2 in / 4 out
Control	from cyclotron control system

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2. SITE CONFIGURATION

The Buyer is responsible for obtaining all authorizations needed for construction of the facility. The Buyer should construct the building according to the requirements described in this guide in timely manner following the approval of work program by IBA. It is the responsibility of the Buyer to determine the appropriate level of shielding design and to execute the construction to fulfill the applicable local legislation.

The minimum dimensions are detailed in drawings:

- SRG-0001 for a vaulted machine with plug door
- SRG-0200 for a vaulted machine with a maze
- SRG-0300 for self-shielded configuration.
- SRG-0400, SRG-0410 and SRG-0420 for machine with beam line

2.1 CYCLOTRON VAULT

The Cyclone® KIUBE is also available with a self-shielding configuration. For all details about the cyclotron room with a self-shielded cyclotron, please refer to Chapter 8 and all related drawings (SRG-03XX). All requirements detailed in this Site Requirements Guide are valid also for the Self-Shielded configuration, except if specifically specified in Chapter 9.

2.1.1 TYPICAL ROOM REQUIREMENT FOR CYCLONE®KIUBE

Table 2: vault internal dimensions

		Cyclone® KIUBE 100/150/180/300
Vault internal dimensions	L x W	3800 x 3800 mm (minimum) 4000 x 4000 mm (recommended)
	Height	2500 mm

2.1.2 VAULT WALLS TYPICAL THICKNESS

Due to intense neutron and gamma flux generation during beam production on target, the Cyclone® KIUBE must be properly shielded. The concrete used for the vault must have a minimum density of 2350 kg/m^3. IBA provides a standard shielding report (MonteCarlo computation) and can provide as an option a customized report adapted to specific site conditions.

The final wall thickness must be validated by the Buyer according to applicable local regulations.

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Table 3: wall thickness

	Cyclone® KIUBE 100/150/180/300
Total wall thickness[a]	2000-2400 mm
Roof[a]	1700-2300 mm

a. to comply with EU public radiation protection; concrete density 2350 kg/m³

The inside layer of concrete in the vault will be activated by neutrons from the cyclotron operation, we call it decommissioning layer. Building such a layer, easily removable, will partly help the future disassembly of the cyclotron vault at the end of life of the equipment (decommissioning). Refer to section 2.1.8 for further information.

2.1.3 VAULT ACCESS

Access to the cyclotron vault can be either a plug door of sufficient concrete thickness or a maze with adequate attenuation (refer to drawings SRG-0001, SRG-0200 or SRG-0300).

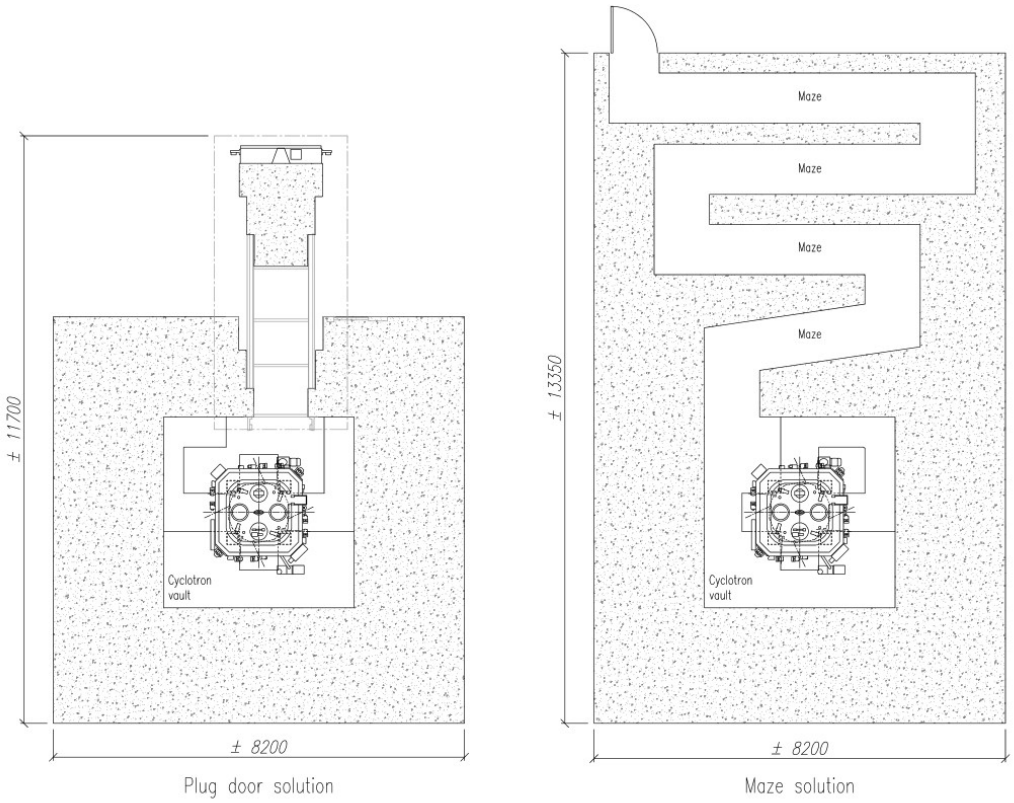


Figure 3: vault and maze access solution

- **Door minimum opening** for access to the vault is **1000mm wide and 2000mm high**.

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- The door will be **interlocked** in such a way that beam production is impossible if door is not properly closed after the search sequence (interlocks to be wired on IBA power cabinet, refer to section 5.1.1.1 for details).
- Door could be **opened from the inside** of the vault and must have a **manual operated override option** in case of power failure.
- In case of movable plug door, it must be equipped with **emergency systems** such as anti-crushing bars, emergency stop buttons, and warning lights.
- Moreover, no door movement should be possible without the presence of an operator; i.e. **the door movement should stop once the open/close button is released by the operator**.
- Finally, **an irradiation probe** must forbid the access to the vault when the radiation level is too high or if the probe is not working properly (refer to section 2.1.7).

2.1.3.1 SEARCH SEQUENCE

Before closing the vault door, an operator should enter the vault and check that nobody is still working inside: 'last man out' or 'search' procedure. At least one search button (SB on drawing SRG-0050) should be installed inside the vault. The vault door closing will be done during a short time span after the search sequence is accomplished: i.e. Search button is pushed and vault door is closed before the timer elapses.

The hardware and installation are Buyer responsibility and could be done by the vault door manufacturer.

2.1.4 RECOMMENDED LOCATION

Because of its weight and the weight of its shielding, the cyclotron is recommended to be located on the lowest level of the building. Installing it underground allows in some cases to reduce the thickness of the shielding using surrounding soil as shield.

2.1.5 VAULT FINISHING

The vault floor has to be leveled and smoothed and should be preferably coated with epoxy paint. The vault walls and ceiling shall be finished with washable polyurethane or epoxy paint on smooth concrete, light color to enhance safety.

The level difference between the finished floor under the four feet shall not exceed +/- 1mm with respect to horizontal plane.



Figure 4: vault finishing

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Ten 1mm stainless steel shims are available for levering the cyclotron during the rigging. The shims should be installed under cyclotron's feet in the vault: refer to drawing **SRG-0010**. These shims are required for optimal cyclotron weight repartition on the slab.

Table 4 : Cyclone® KIUBE weight

Cyclone®KIUBE weight	18,000 kg resting on four pillars
Pillar footprint	220 x 220 mm
Weight on floor	10 kg / cm^2
Finished floor level difference for cyclotron feet	+/- 1mm maximum

2.1.6 VAULT TRENCHES

A pit with trenches should be located under the cyclotron. The trenches will be connected by cable ducts to the power supply room and heat exchanger room. The cyclotron pit should be **400 mm deep** under the cyclotron. Dimensions are described on drawing **SRG-0010**.

Trenches should be preferably **coated with epoxy paint** and connected to a sump with suitable slope in all trenches (in case of water leakage). In accordance with local regulations, the sump should be connected to a liquid waste storage.

During normal operation, all trenches should be **covered with diamond plates or steel grid floor**, except under the cyclotron.



Figure 5: Vault floor & wall finishing

The required interconnection for IBA equipment is explained in chapter 3 and on drawing **SRG-0010**.

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2.1.7 RADIOPROTECTION BASIC REQUIREMENTS

The cyclotron vault is a High Radiation Area: some features should be included in the project to ensure safe operation, such as appropriate signs at cyclotron vault entrance and cyclotron status lights (as required by local regulations). The cyclotron status lights are provided by IBA (see chapter 5 Interlock).

Note: All equipements in the cyclotron vault are submitted to ageing by radiations and should be regularly checked and maintained. (Example: Inspection of insulation of lights and power outlet cables regularly)

At installation time, Buyer must have activity measurement devices and decontamination equipment available:

- **Fixed gamma probe** inside cyclotron vault with remote display and luminous alarm at vault entrance; minimum measuring range: 1 µSv/h - 10 mSv/h
- **Portable gamma and neutron detector**
- **Portable contamination (alpha/beta) monitor** and **a gamma counter** should be available
- **Hand and foot monitor** in the controlled zone access room
- **Activity measurement available** in the hotcells for target optimization
- **Decontamination equipment:** emergency shower and eye-wash

2.1.8 NOTES ON RADIATION PROTECTION AND DECOMMISSIONING

Radiation protection, calculation of the shielding and handling of induced and/or produced activity are the Buyer's responsibility in compliance with local and national requirements. IBA can provide a Montecarlo radiation simulation for the standard vault installation or for a particular application upon request.

2.1.8.1 RADIATION PROTECTION PLANS AND PERMITS

The Buyer will be responsible for compliance with relevant safety standards concerning radiation protection. Before installation, the buyer should provide:

- the shielding structure (cyclotron vault, laboratories transfer lines shielding …) required for operation of the supplies, even if based on IBA recommendation
- the installation drawing of environmental radiation monitoring system if not part of the IBA scope of supply
- all operating permits (installation and operating licenses).

IBA can produce beam at full power and full energy one week after installation start of the cyclotron.

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2.1.8.2 GENERAL INFORMATION FOR RADIATION PROTECTION

The Cyclone®KIUBE produces and accelerates negative ions: H^- and extracts protons H^+ to the target.

Table 5: Cyclone® KIUBE beam energy and current

	Proton energy	***Max Beam Current***
Cyclone® KIUBE 100	18 MeV	100 µA
Cyclone® KIUBE 150		150 µA
Cyclone® KIUBE 180		180 µA
Cyclone® KIUBE 300		300 µA

The negative ions are accelerated inside the vacuum chamber up to the extraction radius and are then stripped of the two electrons by passing through a very thin carbon foil (strippers). The resulting positive particles are directed outwards to one of the 8 exits. The beam is stopped in the target material and produces secondary particles (neutron or alpha) and radiations (gamma rays).

Two beams can be extracted simultaneously on two opposite ports (dual beam); the current of the two beams corresponds to the max beam current of the machine.

2.1.8.3 RADIONUCLIDES PRODUCED

- Standard PET isotopes: $^{18}F^-$, ^{68}Ga, ^{11}C, $^{15}O_2$, ^{13}N, $^{18}F_2$
- With solid target (option): ^{124}I, ^{123}I, ^{64}Cu, ^{89}Zr, ^{99m}Tc…

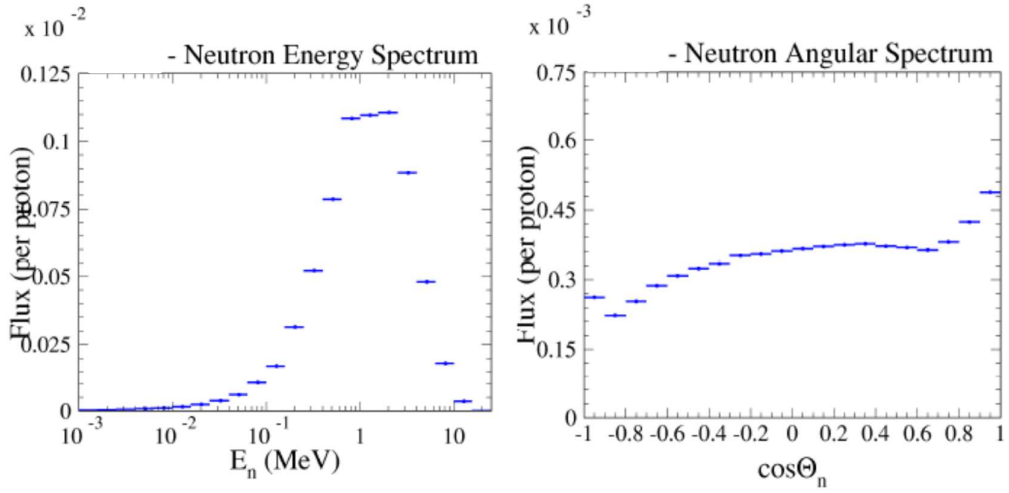
2.1.8.4 NEUTRON RADIATION

The source of neutron is mainly from (p,n) nuclear reaction on reaction on target material.

- Highest neutron flux: produced by ^{18}O (p,n) ^{18}F in target;
- Highest neutron Energy: 15 MeV (with a peak at ~2 MeV);
- Fluence n/p: 6.9 10^{-3}
- Fluence (40 µA on ^{18}O (p,n) ^{18}F): 3,55x10e^{11} neutrons/second.

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Table 6: neutron emission from $^{18}O(p,n)^{18}F$ with 18 MeV proton

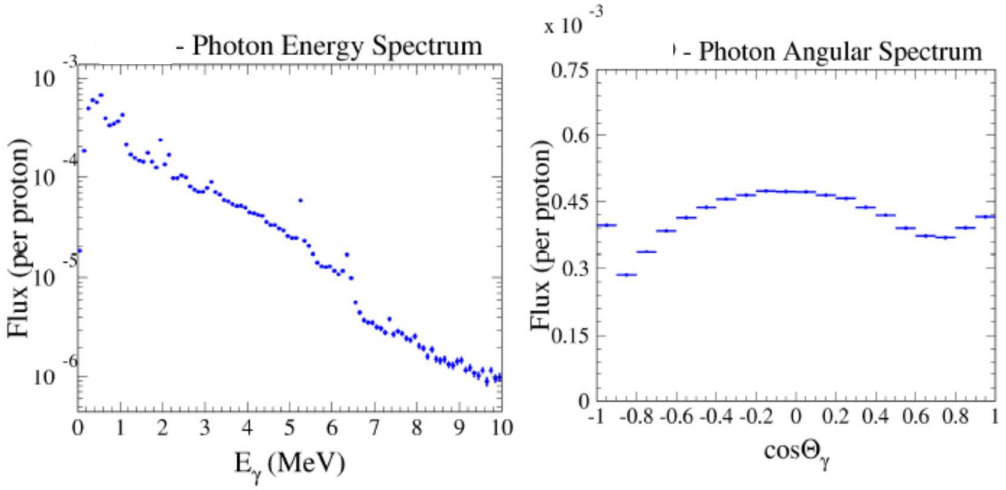


2.1.8.5 GAMMA RADIATION

The source of gamma radiation is secondary neutron capture and other effect in surrounding target materials.

- The gamma emission is isotropic, with a large energy range
- Fluence g/p: 8.3 10^{-3}

Table 7: gamma emission from$^{18}O(p,n)^{18}F$ with 18 MeV proton



2.1.8.6 CONCRETE SHIELDING

Special attention will be paid to make sure that no cracks or damages occur while the concrete is being poured and casted in the cyclotron vault. Proper concrete shielding is necessary because of the radiation hazards from neutron and gamma rays. IBA recommendations are based on a density of 2 350 kg/m³.

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Table 8: Dose rate at 1m from the Cyclotron (inside the vault)

Maximum Dose rate in front of the ^{18}F target at 1m from the Cyclotron (inside the vault)				
	KIUBE 100 single beam (1x 100 µA)	**KIUBE 150 dual beam (2x 75 µA)**	**KIUBE 180 dual beam (2x 100 µA)**	**KIUBE 300 dual beam (2x 150 µA)**
In front of the target (0°)	1.18 x 10^8 µSv/h	0.9 x 10^8 µSv/h	1.2 x 10^8 µSv/h	1.8 x 10^8 µSv/h
At 90° (perpendicular wall)	2.4 x10^7 µSv/h	0.2 x 10^8 µSv/h	0.27 x 10^8 µSv/h	0.41 x 10^8 µSv/h
At 180° (opposite wall)	5.5 x 10^6 µSv/h	0.9 x 10^8 µSv/h	1.2 x 10^8 µSv/h	1.8 x 10^8 µSv/h

Table 9: Wall thickness and dose rate

Maximum Contact dose rate (µSv/h) in front of the ^{18}F target after X (cm) of concrete wall				
Concrete Thickness (cm)	**KIUBE 100 single beam (1x 100 µA)**	**KIUBE 150 dual beam (2x 75 µA)**	**KIUBE 180 dual beam (2x 100 µA)**	**KIUBE 300 dual beam (2x 150 µA)**
200 cm	3.20 µSv/h	2.71 µSv/h	3.62 µSv/h	5.43 µSv/h
220 cm	**0.77 µSv/h**	**0.59 µSv/h**	**0.79 µSv/h**	1.19 µSv/h
230 cm	**0.36 µSv/h**	**0.3 µSv/h**	**0.4 µSv/h**	**0.6 µSv/h**

Special attention must be taken when pouring concrete. Successive horizontal flat layers must be avoided because it could create neutron leaks. Wood pieces on the top of the current drying layer can be used to create casting steps between the pours.

Moreover, steel bars are neutron channels; therefore, straight steel bars from inside to outside the bunker walls must be avoided.

Shielding validation with appropriate measurements, before starting the production, should be performed by the local Radiation Safety Officer (RSO). Local shielding should be added if required.

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2.1.8.7 ACTIVATION OF PARTS IN THE CYCLOTRON & MAINTENANCE

The main activated parts in the cyclotron are consumables like target foils and vacuum foils. Servicing the cyclotron will require care and attention in order to minimize radiation exposure. Always follow IBA maintenance procedure and planning of the work with preparation of the tools and workbench.

Recommendation:

- Use of proper local shielding (lead bags) on targets
- Have a dedicated workbench for maintenance close to cyclotron vault
- Use a workbench L-shield with lead glass for target foil maintenance
- Have a lead shielded container ready for target foil storage

2.1.8.8 DECOMMISSIONING RECOMMENDATIONS

National practices vary on clearance levels and waste management policies.

Modular or multi-layered constructions concepts can be implemented for new accelerator facilities in order to limit the waste volumes and cost of disposal at the end of life.

First inner wall made of concrete modular wall or debonded layer. This inner wall can be built in an independent casting. Iron reinforcement should be restricted to a minimum or replaced by iron-free reinforcement (such as glass fibers) in that part of the shielding. It should have no important structural function. The dismantling of concrete structures by only removing the activated layer can be done by gentle detonation or sawing.

This first layer can also be made in easily handled blocks fitting in containers for nuclear waste.

The thickness of this layer will be function of:

- The concrete and aggregates impurity content;
- Cyclotron usage and lifespan of the facility (usage and future decommissioning schedule);
- Local regulations on waste and decommissioning.

The responsibility of determining the required thickness of this so-called decommissioning layer belongs to the Buyer.

The outer wall will be the structural wall and it can contain normal reinforcement bars. As the outer wall will not be activated, it could be left in place or could be treated as normal demolition rubble.

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IBA can propose local shielding around the targets to reduce the neutron source (as most of neutrons are produced in the target). This solution will limit the volumes of activated waste by reducing the volume in which the neutrons can interact with infrastructure and shielding. The local target shielding is designed in composite materials. The shielding will include materials having large cross sections for inelastic scattering for maximal moderation of fast neutrons and hence reducing the mean energy, and materials with maximal caption (e.g. polyethylene). This local shielding could result in reduction of the neutron flux by a factor of 10. However, even if only partial absorption is reached, lower specific activation of concrete walls will result, hence important reduction of waste volumes.

This local target shielding ("shielded door") can be installed on four beam exit ports as shown on Figure 6. High activity ^{18}F targets shall be allocated on those exit ports to reduce overall activation of the bunker concrete.

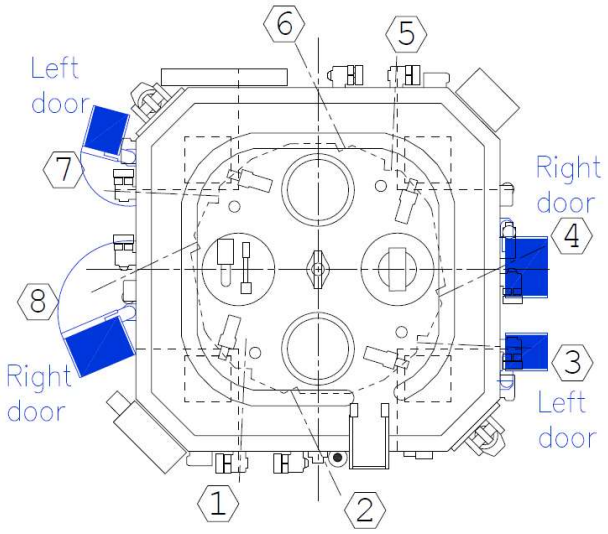


Figure 6 : local target shielding

2.1.8.9 GENERAL RADIOPROTECTION RULES

The general radio-protection rules shall be applicable simultaneously on site:

1. Ambient dose rate in working area < 1 mSv/h;
2. Derivative limit for airborne activity:
   - contamination level < 100Bq/m^3 if unknown radioisotope,
   - contamination level < 45000Bq/m^3 for ^{18}F,
   - contamination level < 3.78MBq/m^3 for ^{11}C, ^{13}N, ^{15}O;
3. individual dose irradiation limited to 1 mSv per person and per week or per intervention (if this one lasts for more than 1 week);
4. beta, gamma dose rate at 10 cm of the intervention area < 10 mSv/h;

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5. no production allowed within 24 hours before IBA intervention in the cyclotron vault begins

When one of these conditions is not fulfilled, IBA staff will not enter the vault.

2.2 CONTROL ROOM

The control room will be used by the operators of the cyclotron. We recommend locating it next to the power supply room, with a fixed window (recommended dimensions: 1800 x 800 mm) in order to see power supply cabinets, as shown on drawing SRG-0001 / SRG-0200 / SRG-300. The window is recommended but not mandatory.

IBA supplies the following equipment: accelerator control station, printer and equipment documentation (additional control stations can be installed upon order).

It is a good practice to have the environmental/radiation monitoring central monitoring system in the same control room.

Some office furniture must be provided by the Buyer:

- Table : recommended dimensions for cyclotron control equipment: 2000 x 800 mm
- Chairs
- Storage (cupboards) for cyclotron documentation and log files

The control room should have an international phone line and internet access (with minimum bandwith 10Mbps download, and minimum 5Mbps upload (This could be tested via a speed test such as https://www.speedtest.net), RJ-45 socket) for IBA remote support / IBA installation team. Alternatively, a wifi with internet access can be accepted.

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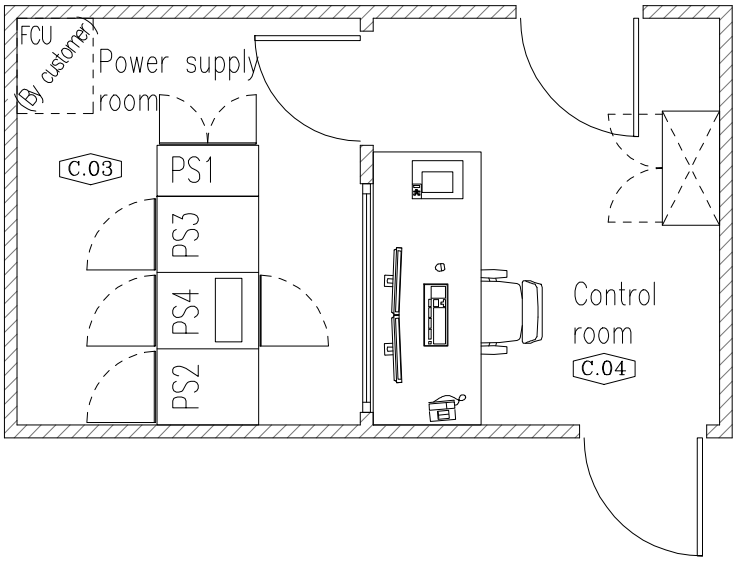


Figure 7 : control and power supply room typical layout (SRG 0001)



Figure 8: window between control and power supply room

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2.3 POWER SUPPLY ROOM

2.3.1 ROOM CONFIGURATION

The Power supply room should be **at least 3000 mm in height** (under slab height for RF cable) and **minimum 2400 mm under the finished ceiling panels** and has preferably **a raised floor system (400mm over the full room)**.

Minimal room surface is: 2700 x 3200 mm.

As an example, refer to drawing SRG-0001 as guideline for the power supply room layout.



Sufficient height in false ceiling will be foreseen to accommodate the overhead HVAC ductwork, electrical lighting fixtures and other building utilities. The room dimension will comply with local electrical codes clearance requirements between, around, above and in front of the equipment cabinet.

Figure 9: RF cable from amplifier to the cyclotron

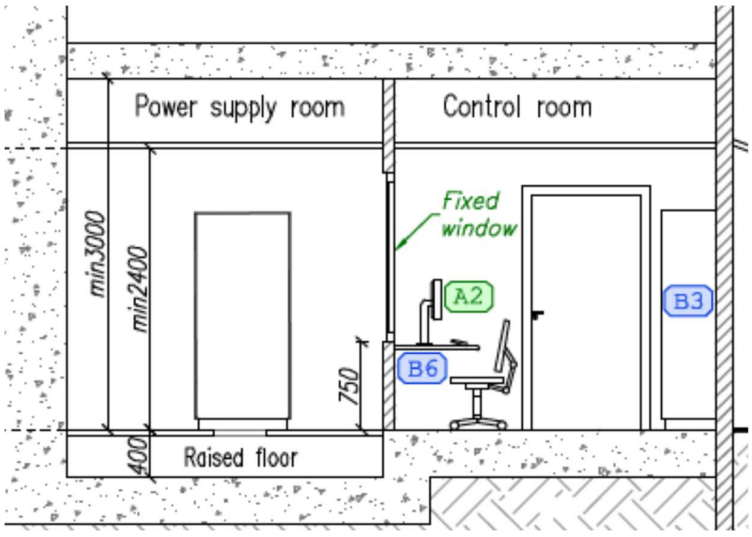


Figure 10: Power supply room cross section

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Internet access must be available in the Power Supply Room for IBA Remote Services (refer to IBA drawing **SRG-0070** and to SRG section 4.5.1): **a cable UTP Cat5-E with RJ45 connector in the false floor with internet connection** must be available for the PLC cabinet (PS4)**.**

For each control station a direct UTP cat5-E connection between PS#4 and the control station is required. It is the buyer responsibility to install these cables before start of installation.

2.3.2 FLOOR TYPE

A raised floor under the cabinets for connecting equipment is strongly recommended. The space below this floor has to be free of any device or conducts for the installation of all cables and in order to allow proper ventilation inside the power cabinets.

The floor should have a sufficient load bearing capacity even when cut-outs are made in floor tiles to allow the passage of cables.

Admissible load: 1000 kg/m², Class 5 according to DIN EN12825

During installation, the cabinet should be rolled over the entire room.



Figure 11: cut-out in floor tiles

The **recommended height** of the raised floor is **400 mm** above the concrete floor (slab). The floor pillars will be safely and permanently fixed onto the concrete floor.

The cut-out and tile arrangement are detailed in **SRG-0010**. The cut-out must have been done before cabinet installation.

Flooring does not require to be server room grade.

Flooring below the raised floor should be painted with waterproof and shock resistant paint (e.g. epoxy painting)

Alternatively, if raised floor cannot be installed, the cabinets can be mounted on a frame (optionally provided by IBA upon order: Refer to TDS 003.020 for more information).

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2.3.3 AIR CONDITIONING

The power supply room will be air-conditioned to ensure optimal operation of the electronic equipment. Heat dissipation in the power supply room, mainly due to the RF power amplifier, is very important. The hot air should be extracted from the top and the fresh air should be blown under the false floor to ensure a proper cooling of the power cabinets.

For more information, refer to section 4.4 “HVAC and Ventilation”.

2.4 HEAT EXCHANGER ROOM

2.4.1 ROOM CONFIGURATION

The room will accommodate IBA water conditioner (refer to drawing **SRG-0001**).
The equipment will be inserted through the door that should be minimum 800 mm wide.
Due to potential short half-life isotopes production during irradiation and metallic ions at the deionization cylinder, walls to accessible rooms are recommended to be made of solid concrete of minimum 100 mm thickness (2350 kg/m³) and the door shall be shielded with 2 mm of lead or equivalent.

Heat exchanger room dimensions must allow the movement of the water group to access its other side easily.

In case of water leakage, it is recommended to foresee a sump connected to a liquid waste storage if requested by local regulation.

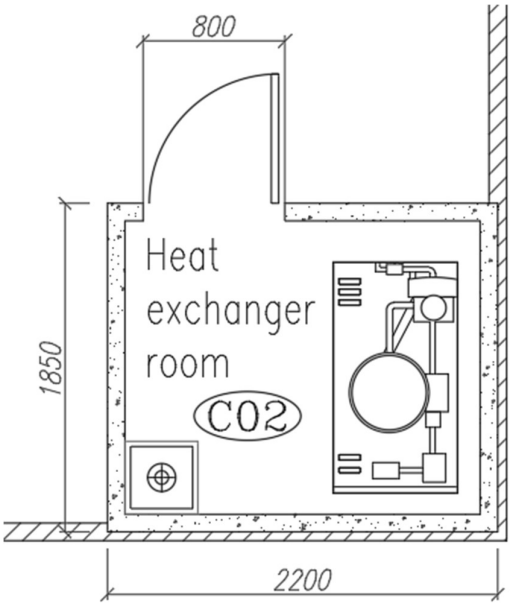


Figure 12: typical water conditioner room

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2.4.2 IBA WATER CONDITIONER

IBA water conditioner system is a closed water circuit with water tank to cool the cyclotron yoke and its sub-systems, it distributes filtered and demineralized water to the cyclotron system for cooling thanks to a variable speed pump with frequency driver commanded by the cyclotron PLC.

Heat exchanger and three-way valves are included for the secondary loop. Power supply is provided by IBA from PS1 power distribution cabinet.

The IBA heat exchanger ensures separation of the building cooling circuit and cyclotron closed-loop cooling circuit and avoids contamination risk

2.4.3 CHILLED WATER CONNECTION

The Buyer will connect the building chilled water supply to IBA heat exchanger:

For details, please refer to section 4.1 Chilled water requirements

2.4.4 RESERVATION FOR CONNECTION

The Buyer will provide reservations (empty ducts) to connect the water closed-loop pipes to the cyclotron room and power in power supply room (refer to drawing **SRG-0010**, ducts B and C). The water & electrical piping through the ducts will be installed by IBA.

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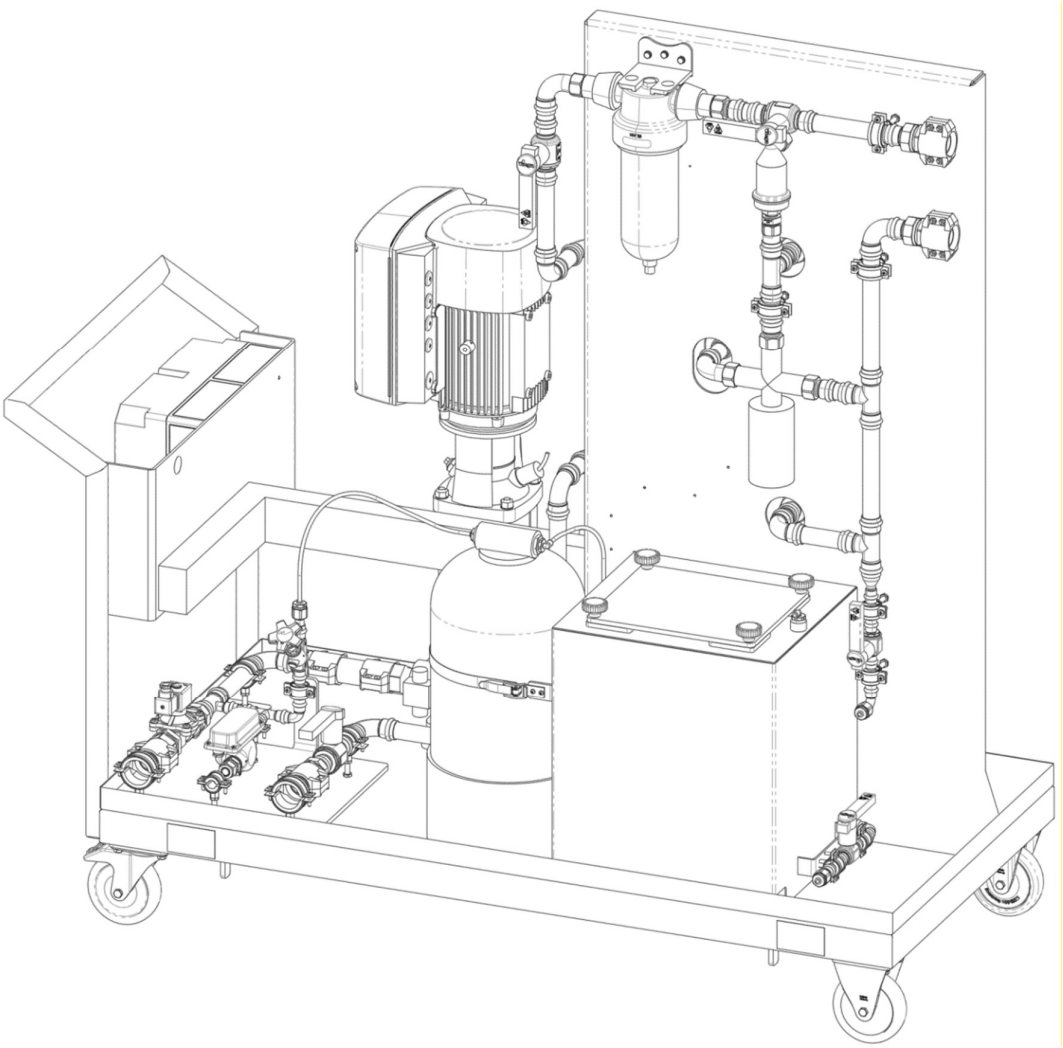


Figure 13: water conditioner

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3. INTERCONNECTION BETWEEN ROOMS AND EQUIPMENT

Refer to drawing SRG-0010 as guideline for IBA recommended interconnection.

3.1 GENERAL RECOMMENDATIONS

Cables and pipes will run from one room to another using so called 'reservation' or 'duct' usually embedded in the concrete slab and under the vault walls: ducts, trenches, cable trays or plastic pipes. Cables and pipes between IBA pieces of equipment will be provided and connected by IBA.

- All metallic ducts and cable trays should be **grounded in compliance** with applicable regulations
- Special care should be taken to **avoid any cutting edge** on ducts and trays that could damage cables and pipes
- **Obstacles** within the ducts **must be avoided**. Ducts must be seamless and designed with a bend radius at least 10 times the internal diameter of the duct
- Ducts should be designed to minimize radiation leakage from the vault. This is done by using small diameter ducts with sufficient angles between the radiation source and the outside of the vault and with ducts fully embedded in the concrete slab.

3.2 INTERCONNECTION TABLE

These minimal requirements are for IBA use only (i.e. interconnections between the cyclotron and its sub-systems, excluding utilities other than the ones distributed within IBA system). No customer nor regular building equipment may use these ducts. Otherwise, the ducts must be made larger and the design adapted. **Pulling cords must be foreseen in each duct**. For the exhaustive list of interconnections, refer to drawing SRG-0010 or SRG-0310 (Self-shielded version). **Provisions** must be made by the Buyer and his Construction Contractor **for the gas connections** (from the cold gas depot to the cyclotron vault), **all other piping and wiring** (light, power outlet, safety and emergency features, etc.) not included in IBA scope of supply and for **Ventilation ducts**. Refer to drawing SRG-0010, **ducts X, Y and Z**.

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Table 10: interconnection reservation

From / To	Power Supply room	Heat exchanger room	Hot lab
Vault	2 ducts DN 160mm (A1) 2 ducts DN 110mm (A2) 1 duct DN 110mm (E) In case of beam line: see specific drawings	1 duct DN 160mm (B)	According to configuration: for 2 targets → 2 ducts DN 110 mm (Rx) In case of solid target[1]: 1 duct DN 160 mm and 1 duct of DN 110mm (Sx)
PS room	-	1 duct DN 110mm (C)	-
Control room	1 duct DN 80mm (D)	-	-

3.3 MAXIMUM DISTANCE BETWEEN ROOMS

The maximum recommended distances to ensure optimal working conditions can be found in the table below:

	Power Supply room	Heat exchanger room	Hot Lab
Vault	< 15 m (standard RF cable length, see note 1)	< 20 m (max. tubing length)	< 20 m (see note 2)

The power supply room and control room will be no more than 80 m apart (ideally next to each other as already explained in section 2).

Note 1: When the cable length between the top of the Power Supply cabinet #2 and the cyclotron RF connection, considering the minimum bending radius, exceeds 17m, the customer must warn IBA as soon as possible for non-conventional length RF cable ordering.

Note 2: IBA guarantees activity delivery if the hot lab is less than 20 meters from the vault. From 20 meters up to 100 meters, the transfer is possible but the activity (for $^{18}F^-$) guaranteed has to be decreased by 2.5% for each additional 10 meters. For example, if the tube is 60 meters long, guaranteed activity has to be decreased by around 10%.

[1] *To be defined depending on solid target plateform*

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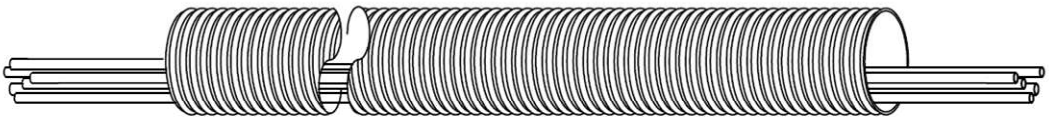
3.4 RADIOISOTOPE TRANSFER LINES SHIELDING

The design of connections / reservation from the cyclotron vault to the hotcells in the radiochemistry laboratory will be made according to hotcell type (trenches and interconnection adapted to the hotcell footprint).

The shielding thickness for radioisotope transfer lines can be achieved by embedding empty conducts in the concrete slab (usually 500 mm deep; 2350 kg/m³) or by covering open trenches with a lead plate (usually 63mm).

Total shielding thickness should be computed **according to local regulations**. The routing of the transfer lines from the cyclotron vault to the radiochemistry laboratory will be made as short as possible to reduce the shielding need but it should accommodate the GMP requirement of the radio-pharmacy and the need for technical access to the hot-cells for installation and maintenance of the transfer lines.

Each transfer line is a corrugated flexible plastic pipe of 16 mm diameter containing several 1/16" PTFE (Teflon™) lines as standard or PP (polypropylene, available upon Buyer's request, to be addressed to IBA Project Manager) for liquid target, or 1/8" stainless steel (gas target) lines for the radioisotope transfer.



The transfer lines shall be provided and installed by the Buyer in the reservation from the cyclotron vault to the hotcell inside the laboratory where activity will be measured, before IBA installation start. The lines extremities shall be hermetically caulked and with 3m free length on both sides.

3.5 VAULT VENTILATION

IBA recommends the observance of IAEA model described in TRS471 requesting 10 air renewal per hour, and thus implementation of large ducts. These ducts must comply with the radiation shielding requirement and a design proposal to reduce neutron escape out of the vault is proposed in drawing **SRG-0010**: ducts X1 and X2.

See Chapter 4.4 HVAC and ventilation

3.6 RF CABLE

The 50 ohms RF cable connecting the cyclotron to the RF power supply (in the power supply room) is a special coaxial cable for transferring RF power to the cyclotron.

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This cable has a **fixed length of twenty (20) meters**. Other lengths are available on request.

The minimal bending radius of the RF cable is 350 mm, the route between the cyclotron vault and the power supply room (up to PS2: RF amplifier cabinet) should consider this minimum-bending radius. The cable outer diameter is 36 mm and its weight is 1,55 kg/m.

After installation by IBA technicians, **the RF cable should be protected** by any suitable means against any possible mechanical damage, for example: it should not be possible to walk on the cable where it is exposed.

Caution: malfunctions or serious damage to the equipment can occur if the cable is flattened or bent further than its minimum admissible bending radius.

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4. UTILITIES

Provision of operating supplies and necessary test equipment is **Buyer's responsibility**. All utilities should be available and operational at the start of the installation and evidenced in BBOD (refer to chapter 11).

4.1 CHILLED WATER REQUIREMENT

Chilled water system will remove heat from the IBA cooling loop and will be provided by the Buyer. A dedicated water chiller system can be used or the system can be connected to general building chilled water system. Refer to drawing **SRG-0020** for location of the connection points, located close to the water conditioner at approximately 1,6m high

Table 11: chilled water requirement and connection

The chiller minimum heat removal capability	55 kW (cyclotron)
Flow	100 l/min
Buyer water-cooling temperature INLET	between min 6° - 16°C max
Maximum static pressure	10 bars
Nominal Pressure drop	1 ± 0.1 bar (100 l/min) (max 1.8 bar)
Building Interface	On IBA Side: Hose nozzle for flexible pipe internal diameter 32mm (1 ¼"); external diameter 42mm (1 5/8") 
Piping material	flexible pipe internal diameter DN32 (1 ¼")
Insulation	The pipes should be covered by suitable insulation to avoid condensation when the cooling water temperature is too low (i.e. below 13,5°C). This insulation shall be extended up to the connection to the heat exchanger of IBA water conditioner (including the flexible pipes provided by IBA between the chilled water distribution piping and the water conditioner itself).

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It is the Buyer's responsibility to install piping and flush out the system prior to IBA installation. **A by-pass** on Buyer's side should be provided (ball valve) and **a strainer** must be foreseen before the connection of IBA inlet side.

A means to display the flow (flow sight) and temperatures of the Buyer cooling water must be provided by the Buyer close to IBA water conditioner.

IBA provides a shunt regulator that is used to maintain cyclotron cooling water at an appropriate temperature by shunting the supply of chilled water. The pump located on the secondary loop can therefore operate with a constant flow. In case a variable flow pump is considered, it must be understood that the system always requires cooling for the oil diffusion pumps and that the variability of the cooling demand is very wide. In case of variable speed pump installation, the control should be linked to the return temperature with a threshold at 16°C.

A provision of 100 liters of demineralized water supply for filling the cyclotron closed-loop circuits must be made available by the customer for IBA installation.

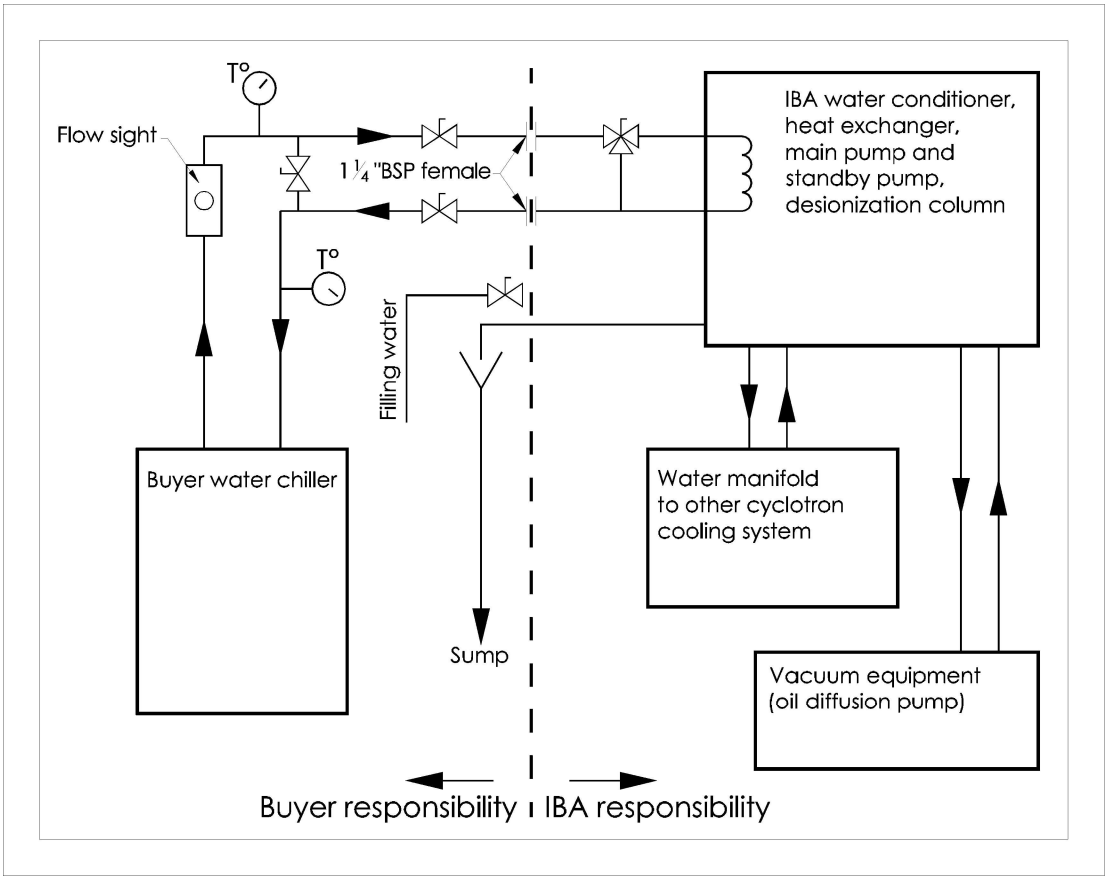


Figure 14: schematic of the chilled water supply and cyclotron cooling loops

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4.2 CYCLOTRON GAS AND COMPRESSED AIR REQUIREMENTS

The information in this section is summarized on drawing **SRG-0020** (or SRG-0320 for self-shielded version).

4.2.1 GENERAL REQUIREMENTS

Any required gases and the compressed air must be available, functional, leak free and tested on-site prior the start of the installation.

All the **gas pressure regulators** for cyclotron and targets will be available **outside the vault** (**item B7** on drawing SRG-001), safely fixed and permanently labelled (refer to next picture).



Figure 15: cyclotron gas regulators and manometers outside the vault

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4.2.2 COMPRESSED AIR FOR CYCLOTRON EQUIPMENT

Table 12: cyclotron compressed air requirements

Recommended air class	ISO8573.1 class 1-4-1
Remaining dirt particle size	< 0.1 µm
Dew-point	3°C
Remaining oil content	< 0.01 mg/m³
Pressure	7 barg @ connection point
Connection point to Cyclone® interface	12mm OD plastic connection push-on
	Average flow: < 2 l/min – peak : 70l/min @ 7 barg (0.490 Nm³/min)
Chemistry modules and hot-cells (if any)	See the Integralab laboratory design for details

The compressed air will be connected to the cyclotron gas interface on the cyclotron foot; or 3m free length hermetically caulked should be available in the cyclotron pit. The compressed air **pressure regulator** will be available **outside the vault**, safely fixed and permanently labelled (refer to picture in section 4.2.1).



Figure 16: Compressed air push-on connection on Cyclone®KIUBE foot

4.2.3 GASES FOR CYCLOTRON

These gases are **not optional**. Connections to the cyclotron system will be made inside the cyclotron vault except for the helium ultra pure (drain gas for liquid target) which should be available in Power Supply room (refer to drawing SRG-0020).

The gases will be connected to the cyclotron gas interface on the cyclotron; or 3m free length **hermetically caulked** will be made available in the cyclotron pit.

It is the customer responsibility to ensure the compliance of the gas distribution with applicable regulations, norms or standards at the place where the system will be installed.

IBA recommends the followings:

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- tubing in stainless steel (SS 316) or oxygen free copper (OFC) for gas purity (except for H_2)
- **for H_2**: SS 316 1/16" with **same diameter** and no intermediate connections (dedicated gas cylinder and supply line) from the gas cylinder up to the connection with IBA equipment.
- piping will be **as short as possible and leak-free**.
- **Proper leak test** (according to ISO 7396) **has to be performed** before the start of operation.

Table 13: cyclotron mandatory gas requirement

Gas type and purity	Dry Nitrogen[2] 99.9% (Cyclotron venting)	He 99.9999% [3] (drain gas for the liquid targets)	He 99.8% (low grade for target window cooling)	H_2 (99.9999%) (For H- source)
Piping diameter	OD 6mm (1/4") ID 4mm (0.170")	OD 1/8" ID 1/16"	OD 6mm (1/4") ID 4mm (0.170")	OD 1/16" ID 0.5mm
Piping material	SS316 or OFC	SS316 or OFC	SS316 or OFC	SS316
Bottle type	B50	B50	B50	B20
End use point	Cyclotron vault	Power supply room	Cyclotron vault	Cyclotron vault
Pressure range (barg)	0-10	0-10	0-1.5	0-3
Pressure use (barg)	1-5	~2	~0.5	~2

Be aware that any gas purity corruption will contribute to strongly reduce the ion source lifetime or pollute the gas target's specific activity.

[2] *Dry Nitrogen is recommended but not mandatory. Venting with dry nitrogen reduces the time to reach operational vacuum after cyclotron opening.*

[3] *Helium with lower purity (down to 99,995% (N45)) could be accepted but as it is used to flush carbon target, specific activity of ^{11}C could be affected by the gas impurities (lower SA).*

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Additional gas for Cyclone® KIUBE 300:
In addition to the gases listed in Table 13, the Cyclone® KIUBE 300 also requires the following:

Table 14 - additional gas for Cyclone® KIUBE 300

Gas type and purity	Oxygen ≥ 99.8%; Humidity ≤ 300ppm (ion source boost)
Piping diameter	OD 1/8“ ID 1/16”
Piping material	SS316 or OFC
Bottle type	B10
End use point	Cyclotron vault
Pressure range (barg)	0-3
Pressure use (barg)	~2

WARNING: the cyclotron system uses flammable and explosive gases. IBA recommends the use of small volume hydrogen bottle and only in well ventilated zones or cabinets to prevent reaching the explosive lower limit for a mix hydrogen/air (the hydrogen volume from the bottle should be chosen to stay under the limit of 4% of hydrogen concentration in case of bottle leakage). The choice of system, installation, operation and maintenance shall be done according to local regulations; components shall be chosen compliant with PED European Directive or equivalent safety.

4.2.4 TYPICAL CYCLOTRON GAS CONSUMPTION

Hypothesis: 2 FDG runs/day (duration of each run: 2 hours), 5 days/week

Table 15: Typical cyclotron gaz consumption

Gas type and purity	***Bottle type***	***Number of bottles per year***
Dry Nitrogen (Cyclotron venting)	B50	1
Helium (99.9999%) (drain gas for the liquid targets)	B50	2
Helium (99.8%) (low grade for target window cooling)	B50	2
H_2 (99.9999%) (For H- source)	B20	0.5

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4.2.5 GASES FOR GAS TARGETS

Depending on the isotopes you will produce (and the gas targets that will be installed on your cyclotron), additional target gases will be necessary. The gases will be piped to the cyclotron gas target accessory cabinet; or 3m free length **hermetically caulked** will be made available in the cyclotron pit.

Table 16: Nirta ^{11}C gas targets requirement

Gas type and purity	N_2 (99.9999% or 99.9995%) + 0.5% O_2 (99.99%) **For ^{11}C-CO_2**	N_2 (99.9999%) + 5% H_2 (99.99%) **For ^{11}C-CH_4**	He 99.9999% **For ^{11}C-CO_2 and ^{11}C-CH_4**
Piping diameter	OD 1/8“ ID 1/16”	OD 1/8“ ID 1/16”	OD 1/8“ ID 1/16”
Piping material	SS316	SS316	SS316
Bottle type	B50	B50	B50
End use point	Cyclotron vault	Cyclotron vault	Cyclotron vault
Pressure range (barg)	0-25	0-25	0-10
Pressure use (barg)	20	20	5

Note on Specific Activity:

Both gas targets can be flushed using the ultrapure helium system used for liquid target transfer. It is recommended to have a dedicated ultrapure helium line and pressure regulator for the Nirta 11C gas target. For carbon target, a higher target gas purity will provide higher expected ^{11}C specific activity. If highest purity (N60 99.9999%) gas cannot be provided, N55 (99.9995%) will be accepted. The specific activity depends on many process parameters and not only on the target. High specific activity of ^{11}C-CO_2 is the most difficult to achieve since stable CO_2 is all around. Gas & equipment free from hydrocarbons (CxHy) is necessary to ensure high specific activity.

Table 17: Nirta ^{15}O gas target requirement

Gas type and purity	$^{15}N_2$ (min enrichment 98%; 99% N_2 + 1% O_2)	Liquid Nitrogen	He 99.9999%
Piping diameter	OD 1/8“ ID 1/16”	Dewar in the vault to be filled before production	OD 1/8“ ID 1/16”
Piping material	SS316		SS316
Bottle type	500 ml -1000 ml		B50
End use point	Cyclotron vault	Cyclotron vault	Cyclotron vault
Pressure range (barg)	min 8	NA	0-10
Pressure use (barg)			5

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Table 18: Nirta $^{18}F_2$ gas target requirement

Gas type and purity	$^{18}O_2$ (min enrichment 96%; purity 99,99%)	Ne (99,995%)	Ne (99,995%) + 2% F2 (99,99%)	Liquid Nitrogen
Piping diameter	Bottle in the vault Fitting 1/16”	OD 1/8“ ID 1/16”	OD 1/8“ ID 1/16”	Dewar in the vault to be filled before production
Piping material		SS316	SS316	
Bottle type	B0.5 or B1[4]	B10	B10	
End use point	Cyclotron vault	Cyclotron vault	Cyclotron vault	Cyclotron vault
Pressure range (bar)	NA	0-30	0-30	NA
Pressure use (bar)		25	25	

[4] *Minimum gas volume of 5 liters*

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4.3 ELECTRICAL REQUIREMENTS

4.3.1 GENERAL RECOMMENDATIONS

The building network will supply the building equipment (light, ventilation...) and a **dedicated circuit breaker** will be provided for the cyclotron power connection. The main building distribution panel is the Buyer's responsibility.

The Buyer will provide **a hook-up (free cable) with 3m free length under PS1 distribution cabinet** in the power supply room (refer to SRG-0070). The whole cyclotron electrical power consumption will be taken from this single connection. All connections from the PS1 to IBA cyclotron related equipment will be provided and connected by IBA. Electrical cable installation and connection of IBA PS1 cabinet to mains is the Buyer's responsibility, including cable protection.

The Neutral (N) should be connected to the ground at the main building transformer only or at the step-down transformer when using 3 wires network (US only).

Earth connection (PE)

Protective Earth (PE) should not contain switch or fuse. The PE should be connected to earth with impedance less than 10 ohms. All exposed metallic components of IBA equipment will be wired to the power distribution equipotential connection (at PS1 cabinet) with a yellow/green wire provided and connected by IBA.

Local standard

The local electrical standards (standard regulations, wire color, safety issues,) will be provided by the Buyer upon one month after contract signature for information to IBA personnel. The Buyer is responsible of the compliance with local regulations and electrical norms.

4.3.2 AC POWER FOR THE CYCLOTRON

Total main power characteristics to be provided to PS1 for IBA installation (this includes power for UPS, see point 4.4.3):

Table 19: electrical power requirement and consumption

Total main power	Installed	Consumption
Cyclone® KIUBE 100	70 kVA	45 kW
Cyclone® KIUBE 150	85 kVA	55 kW
Cyclone® KIUBE 180	95 kVA	60 kW
Cyclone® KIUBE 300	100 kVA	65 kW
Vacuum stand-by (eco mode)	5 kVA	3 kW

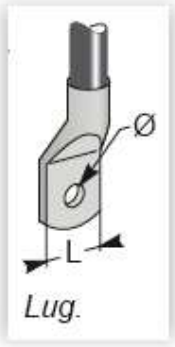
WARNING: to avoid electric shock, always switch off power before servicing the equipment.

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Optional equipment, such as external beam lines, needs additional power requirement. For details, please refer to section 8.3.

Table 20: electrical specification for connection to IBA cabinet

Voltage	3 x 400Vac +/- 5% [a]
Type	3 phases + neutral + ground PE (5 wires)
Frequency	50/60 Hz +/- 1%
Connection size	lug size L ≤ 25mm ; ϕ : 8.5mm (M8 screws); tightening torque: 15 Nm max cable 120 mm²
Max short-circuit current	Icu max ≤ 25 kA rms
Earthing system	**TN-S**
Ingress protection rating	IP 54
Additional power for beam line	(refer to chapter 7)



a. Other voltages can be accommodated; adaptation transformer shall be foreseen.

4.3.3 UPS CONNECTION

UPS dedicated connection is available on cyclotron at PS1 cabinet, for vacuum and control systems (to avoid a vacuum break in case of mains failure): three phases + neutral, 400V +/- 5%,

UPS specifications: **20 kVA, 27A nominal current**

Disclaimer: when UPS is out of IBA scope of supply, its connection is under Buyer responsibility and has to be performed by experienced personal.

4.3.4 OTHER POWER CONNECTIONS

Electrical power plugs, 230V-16A-50/60Hz, should be available in the facility: refer to **SRG-0070**:

- Inside the cyclotron vault;
- On the maintenance workbench;
- In the control room;
- In the power supply room
- In the heat exchanger room

Note: All equipements in the cyclotron vault are submitted to ageing by radiations and should be yearly checked and maintained. (Example: Inspection of insulation of lights and power outlet cables regurlarly)

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4.3.5 LIGHTING

The recommended lighting in all cyclotron areas is 500 Lux (horizontal plane at 1m high).

In cyclotron vault, 500 Lux are expected at the level of the median plane (1.12 m high, refer to drawing SRG-0070). For fluorescent lights using electronic ballasts, it is recommended to move ballasts out of the vault to protect them from radiation premature damaging. Lighting is preferably wall mounted (around 2m high) all around the vault to improve lighting in the cyclotron itself.

It is recommended that wall and floor surfaces are painted in a light color to improve visibility and promote safety.

Emergency lighting will be available in relevant area.

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4.4 HVAC AND VENTILATION

The facility HVAC performs heating, cooling, moisture control and fresh air supply.

- The system in every area should be **operational before IBA starts the installation.**
- The controlled area (rooms where possible radioactive gaseous contamination is possible) should be conditioned by a **ventilation system separated** from the general ventilation system of the building.
- IBA recommends observing the recommendations described in IAEA TRS 471 concerning the pressure cascade to be implemented in the different rooms of the controlled area. The rational behind this recommendation is the confinement of potential airborne or gaseous radioactive contamination, by reducing the pressure in rooms where the risk of such contamination is higher (higher the risk, lower the pressure). A proposal of such a pressure cascade is depicted in drawings SRG-0X30. Further to the confinement of radiactive contamination, the pressure cascade must also be implemented to protect environmental atmosphere of clean rooms from dust contamination, using then higher pressurization (higher the risk, higher the pressure).
- The **potential radioactive release is monitored** at the stack (chimney) exhaust after suitable filtration (HEPA and charcoal filters)
- Ventilation should comply with local codes, health and safety regulations concerning radioprotection and environmental release.
- Apart from the cyclotron vault and the power supply room, which will be detailed hereunder, the other areas will be ventilated and heated as necessary.

4.4.1 CYCLOTRON VAULT AND TECHNICAL ROOMS:

- A **negative relative pressure** in the cyclotron rooms versus the other rooms is recommended (refer to drawing SRG-0030).
- Moreover, IAEA recommends minimum ten air changes per hour in the vault (refer to drawing SRG-0035). (Ventilation system must ensure vault air renewal during/after production and before someone accessing the vault to remove potential air contamination)
- **Passage of ventilation ducts** in vault walls should comply with radiation regulations. A minimum of three bends in different direction (in different planes) are recommended (refer to drawing SRG-0010 – pipes X1 & X2) and straight portions should not go too deep in the shielding. Bends and connection should preferably be in the ceiling slab of the vault.

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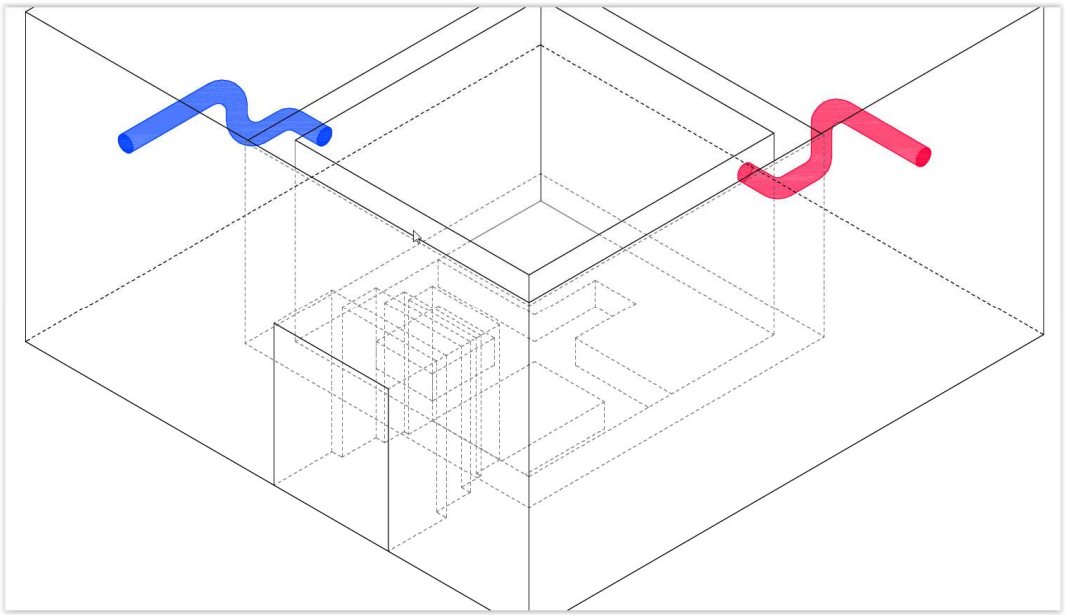


Figure 17: example of ventilation ducts in bunker walls

Table 21: ventilation requirement & temperature

Cyclotron vault and technical rooms

Temperature limit	17 - 30°C
Humidity	35 - 65% at 20°C
Power dissipated in vault	< 5kW

4.4.2 POWER SUPPLY ROOM

- The power supply room will be air-conditioned to ensure optimal operation of the electronic equipment.
- A fan coil unit (FCU) or A/C cabinet will be required for the heat dissipation removal, mainly due to / occurring in the RF power amplifier (PS2). **Heat dissipation** is approximately **15 kW for Cyclone® KIUBE 100/150/180** and **20 kW for Cyclone® KIUBE 300.** This heat dissipation is done through the ambient air. FCU sizing is the Buyer responsibility. **Approximate flow to be handled by the FCU is between 5.000m³/h and 8.000m³/h (depending on the cyclotron model and air renewal conditions).** Flow in the FCU is reverted (from top to bottom) to blow fresh air under the raised floor.
- Exhaust is ideally located above the PS2, in order to remove the warmest air out of the room. It is recommended **not to** connect the cabinet with the duct; an exhaust grid can be set at the ceiling above the cabinet. The cabinet PS2 is equipped (by IBA) with a blower to ensure a cooling flow all along the cabinet (refer to drawing SRG.0010).
- A minimum of air renewal is required for hygienic purposes according to European standards.

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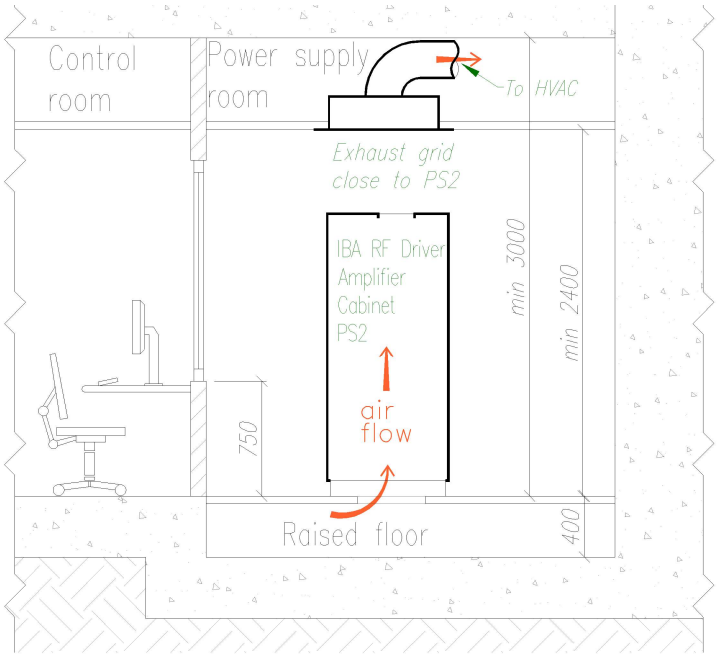


Figure 18: air flow in the power supply room

Note: The raised floor must be free of any device or conducts for the installation of the cables.

Table 22: ventilation requirement & temperature

Power supply room

Temperature	17 - 25°C
Humidity	35 - 65% at 20°C
Power dissipated in air	15 kW for Cyclone® KIUBE 100/150/180 and 20 kW for Cyclone® KIUBE 300

Caution: never obstruct the flow of cooling air entering/exiting the power supply cabinets; otherwise, equipment damage may occur.

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4.5 OTHERS

4.5.1 REMOTE ACCESS CONFIGURATION

In order to connect the machine to our remote services, a wired (RJ-45) internet connection in any other IP address range than 192.168.1.0 and 192.168.143.0 is required. The connection outlet should be situated in the power supply room (refer to drawing **SRG-0070** or SRG-0370), the next router or switch should be situated at less than 90m (cable length) to ensure proper connection.

Minimum speed required: 10 Mbits/s download speed; 5 Mbits/s upload speed.

The IP address can be either DHCP (preferred) or fixed, and can go through a proxy. In case of a fixed IP, all details must be supplied: IP address, subnet mask, gateway, DNS server address.

The firewall(s) have to let the traffic from and to the supplied address go through to and from a series of following IP/ports in TCP or UDP without restriction 24/7:

- Firewall, minimum direct output port access:
  - TCP: 80, 443
  - UDP: 1194
- Firewall, other direct output port access:
  - TCP: 25, 110, 465, 995
- Firewall, no direct input port access needed
- Whitelist:: The entire Talk2M domain should be withelisted for outbound connections. Please refer to: KB-0209-00 - Addresses and ports used by Talk2M (http://resources.ewon.biz/en/doc/support/kb-0209-00-addresses-and-ports-used-by-talk2m)

This connection must be available and working at the start of the installation. In order to test the internet connection, customer is required to install and run “Talk2M Connection Checker” tool on a laptop before the start of the cyclotron installation:
https://resources.ewon.biz/download/talk2m-connection-checker

Your Advantages

- **Improve the uptime** of the cyclotron.
- Access to **true 24/7 helpdesk**, putting you in direct contact with a cyclotron engineer who will help you solving any problem you may have (> 93 % problem are solved remotely).
- Short response time.
- **Fast diagnose**, first visit with the right part/tools.
- Secured network, no access to internet but only to a secured platform (Talk2M).

4.5.2 STANDARD EQUIPMENT REQUIRED BY IBA

IBA will require **laboratory consumables** such as gloves, demineralized water, syringes and needles for the cyclotron and target start-up. A detailed list, depending on IBA scope of supply, will be provided by IBA project manager.

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A dose calibrator is required for measurement of the produced activity during target optimization and acceptance (it could be included in IBA scope of supply, item 235.001). Calibration of this equipment is the Buyer's responsibility. The dose calibrator will be suitable for the following radio element ^{11}C, ^{15}O, ^{13}N and ^{18}F with capability **up to 20 Ci** of ^{18}F.

A lead shielded storage for cyclotron items is requested (target windows…); recommended dimensions: 200 x 200 x 200 mm; shielding: 50 mm of lead (it could be included in IBA scope of supply, item 235.020).
A dedicated room for waste storage with slightly shielded walls is a good practice in radiopharmacies.

Radiation protection equipment as described in section 2.1.7.

4.5.3 COMMUNICATION

An intercom system linking reception, vault, power supply room, control room, laboratories and packaging is nice to have in operation but not required for installation.

4.5.4 SEISMIC REQUIREMENTS

The seismic calculation to meet the site location codes is the responsibility of the Buyer and/or his contractors.

The cyclotron feet shall be anchored (Buyer's responsibility) in the floor (see Section 6.3 for more information). The power cabinets should be fixed to each other to avoid tilting.

In areas with a high level of seismic activity, the installation of the cyclotron inside a vault with a maze entrance is recommended (refer to drawing SRG-0200).

4.5.5 FIRE PROTECTION AND FIRE FIGHTING

It is the responsibility of the Buyer.

The fire protection measures should be agreed in accordance with the local fire brigade. An interlock to shut down the cyclotron in case of fire alarm is available on the IBA interlock system (see Chapter 5).

IBA recommends manual CO_2 fire extinguishers (location commonly determined with the local fire brigade). Automated water extinguishers should be avoided whenever possible in controlled areas due to significant risks associated with environmental pollution and electronics damages with extinguishing water. Fire inhibitor automated systems using argonite or FM200 should preferably selected, especially for areas where the pollution risk is high (e.g. vault, workshop, technical rooms, …) or where sensitive equipment is present (e.g. vault, power supply room, heat exchanger room, server room, …)

Fire detectors for the vault shall preferably be installed in the exhaust ducts outside of the vault to prevent their premature damage due to radiation field.

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4.5.6 NOISE LEVELS

The equipment generates noise, depending on its status. The table below summarizes the noise levels in the different room, depending on the cyclotron status.

As the noise generated by the power supplies is high, recommendation for a separating wall and window between the control room and power supply room is given in Section 2.2.

Table 23: noise levels generated by the Cyclone® KIUBE

	Power Supply room	***Heat exchanger room***	***Vault***
Standby	45 dB	60 dB	55 dB
Magnet ON	65 dB	64 dB	55 dB
Manget & RF ON	77 dB	64 dB	55 dB

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5. INTERLOCK AND CONTROL SYSTEM INTERFACE

The building interlock system is the Buyer's responsibility. This system will exchange information with IBA equipment; these signals are called building interlocks.

Cyclotron control system and Buyer building system must communicate to:

- Allow the beam production;
- Allow activity transfer from a target to a receiving station/hotcell in a laboratory;
- Ensure emergency stop of the beam whenever required.

Interlocks components, wiring and responsibilities are described in **drawing SRG-0050** and should be completed and wired before the start of Installation. Each wire must be properly and clearly labelled.

Detailed information, including the applicable and exhaustive list, about Interlock signals provided and required by IBA will be found in the standard ICD (Interface Control Document, provided by the Project Manager).

5.1 STANDARD IBA INTERLOCKS

5.1.1 BEAM PRODUCTION AND EMERGENCY STOP

Beam production and Emergency stop are controlled by IBA Redundant Safety System (RSS), which is compliant with the PL=d of the ISO 13849-1:2015 (*Safety of machinery – Safety – related parts of control systems – General principles for design*).

The RSS will ensure safe shut-down of the cyclotron's Ion Source and RF power supplies based on two redundant inputs of the two signals: "vault door closed" and "no emergency break".

5.1.1.1 VAULT DOOR CLOSED:

TWO limit switches (DS on drawing SRG-0050) must be installed on the vault door (either plug door or maze door). These switches must be compliant with at least the PL=d of the ISO 13849-1:2015 (*Safety of machinery – Safety – related parts of control systems – General principles for design*), for example they can be Schneider Electric XCS DMC switches or equivalent. They are usually provided by the vault door manufacturer.

The switches must be **independently wired** to IBA power distribution cabinet (PS1) in Power supply room. These two signals must be of the type NO (normally open) free of potential contact.

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5.1.1.2 EMERGENCY STOP

Two redundant signals for emergency stop of the cyclotron ("no emergency break") must be **independently wired** to IBA power distribution cabinet (PS1) in Power supply room. These two signals must be of the type NC (normally closed) free of potential contact.

These signals will come from Emergency Crash buttons (EB on drawing SRG-0050) connected in series and will include other redundant building signals determined by the Buyer (for example: vault ventilation OK, no fire alarm,…). These signals must be compliant with the PL=d of the ISO 13849-1:2015 (*Safety of machinery – Safety – related parts of control systems – General principles for design*), for example they can be Schneider Electric Harmony® XALK or Harmony® ZB4, or equivalent.

WARNING: the PL=d (ISO 13849-1:2015) of IBA RSS will be ensured only if Buyer respects the same safety standard for "vault door closed" and "no emergency stop" signals.

5.1.2 ACTIVITY TRANSFER AUTHORIZATION

To authorize a transfer of activity from one of the targets on the cyclotron to a receiving station/hotcell through direct connection or through liquid or gas switching boxes, the cyclotron control system requires two signals "hotcell destination ready" AND "all transfer enabled".

The "all transfer enabled" signal is a master signal that is required to authorize any transfer. It could be a master key or a general building signal (ventilation system ready in lab or other conditions to be defined by local RSO).

The "hotcell destination ready" signal should merge two pieces of information: **hotcell is closed and hotcell pressure is OK**. This signal, together with "all transfer enabled", must be ready to authorize the hot-cell destination selection and then the radioisotope transfer (DS and PS on drawing SRG-0050).

In order to avoid high exposure due to inadvertent opening of the hotcell with high level of activity inside, all the Hotcell door limit switches will comply with the ISO 13849-1:2015 and they should be checked at the end of the hotcell installation.

5.1.3 DATA FROM BUYER'S SYSTEM

IBA control system expects signals from Buyer's systems through potential free contacts as follows:

- Open contact — Low level (0V)
- Close contact — High level (24V) from IBA low voltage power supply

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Table 24: Connections to IBA safety relays (PS1) for beam production and Emergency Stop

Signal	*Contact*	*Status*	*Detail*
Vault searched and door closed (2 contacts)	Open contact	Vault open	Enable/disable ion source and RF system
	Close contact	Vault closed & Secured	
No Emergency break (2 contacts)	Open contact	No beam authorized	Enable/disable ion source and RF system
	Close contact	Beam authorized	

IBA recommends having the Interlock contacts wired with cable provision to the power supply room in order to be connected by IBA technicians to Power supply cabinet PS1 (terminals X3 3-4 & 4-5 for vault door status and X3 3-6 and 6-7 for emergency breaks loop).

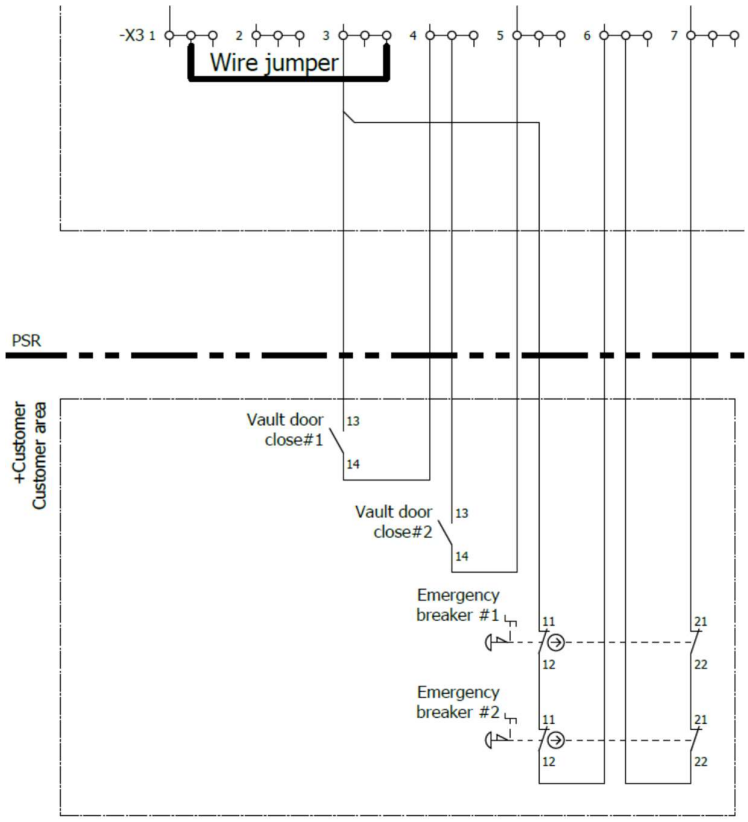


Figure 19: Redundant door and emergency wiring in IBA power distribution cabinet

Wires for door switches and emergency break are of type LIHH-OZ 2x0,5mm², refer to drawing SRG 0050.

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Table 25: Connection to IBA PLC (PS4) for Activity Transfer Authorization

Signal	*Contact*	*Status*	*Detail*
Direct transfer: Hot cell destination	Open contact	Hotcell not ready	Hotcell ready to receive radioisotopes
	Close contact	Hotcell ready	
Transfer through **liquid** switching (^{18}F, ^{68}Ga, ^{13}N): Switching destination x ready (x=1-4 or 1-8 or 1-16)	Open contact	Switching destination not ready	Switching destination ready to receive radioisotopes
	Close contact	Switching destination ready	
Transfer through **gas** switching: Switching destination x ready (x=1-4)	Open contact	Switching destination not ready	Switching destination ready to receive radioisotopes
	Close contact	Switching destination ready	
All transfers enabled	Open contact	Transfers not authorized	Master signal required to allow any transfer
	Close contact	Transfers authorized	

Maximum 3 liquid switchings and 2 gas switchings can be installed.

These inputs (one signal for each destination) must be of the type NC (normally closed) free of potential contact and must be wired from the receiving station (i.e. hotcell control box) to IBA PLC control system (PS4) in Power supply room.

5.2 DATA SUPPLIED BY CYCLOTRON CONTROL SYSTEM

5.2.1 STATUS LIGHTS

IBA will provide and install (refer to drawing SRG-0050)

- a 4-colors beacon with cyclotron status next to the vault door (BC on drawing SRG-0050)
- one flashing light in the main production laboratory (FL on drawing SRG-0050)



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5.2.2 STATUS SIGNALS FOR BUILDING INTERLOCKS

Signals for Buyer's building systems are available from IBA PLC control system (PS4) in Power supply room. The signals are providing cyclotron status and transfer warning as well as hot-cell door selection information for door opening disable system.

All signals are provided by free of potential contact from a relay in PS4. For each available signals, one inverter contact is available on the relay:

- 11: Common: max 250V/ 6A AC/DC;
- 12: NC: normally closed (recommended for fail safe operation);
- 14: NO: normally open.

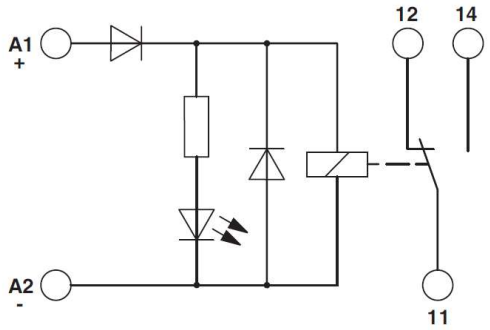


Figure 20: circuit diagram of IBA PS4 relays

5.2.2.1 STATUS SIGNALS

Table 26: status signal from the cyclotron system

Signal	*Status*	*Signification*	*Detail*
Beam ON	Active	Ion source ON; RF system ON	Beam production in cyclotron vault
RF ON	Active	RF system is ON	RF field ON
Main Coil ON	Active	Magnet power supply is ON	Magnet field ON
Transfer ON	Active	Transfer from any target is active	Activity is on its way from target to destination

Status active means that the relay is energized (11 in contact with 14)

5.2.2.2 RECEIVING STATION/HOTCELL DOOR LOCK

When destination (hotcell) is selected for a radioisotopes transfer, a signal is issued from the cyclotron Control System to disable the hot-cell door.

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A safe lock (DL in drawing SRG-0050) should be provided by the hotcells manufacturer (often standard).

Table 27: interlock signal from the cyclotron system

Signal	*Contact*	*Signification*	*Detail*
Direct transfer destination x selected (x=1-8)	Open Contact	Hotcell selected for unloading a target	Signal can be used to disable the hotcell door
^{18}F liquid Switching destination x selected (x=1-7)	Open Contact	Hotcell selected for unloading a target	Signal can be used to disable the hotcell door
Gas Switching destination selected x (x=1-7)	Open Contact	Hotcell selected for unloading a target	Signal can be used to disable the hotcell door

Warning: bypassing an interlock creates a unique situation where the control system is told that the operation of the equipment is safe for both personnel users and equipment, when it is probably not. Bypassing an interlock could therefore lead to injury, death or equipment damage.

Warning: To the exception of the Redundant Safety System (RSS, see §4.2), which is compliant with the PL=d of the ISO 13849-1:2015 (*Safety of machinery – Safety – related parts of control systems – General principles for design*), this interface is NOT designed for high level personnel safety and consequently is NOT validated as such by IBA.

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5.3 OPTIONAL INTERLOCKS

5.3.1 CUSTOMER TARGET

In case a customer target is installed on the Cyclone® KIUBE, the communication between the target and the cyclotron will be done with two signals as described in Table 28.

Table 28: interlock signals for customer target

Signal	*Contact*	*Signification*	*Detail*
Target beam ready (from target to cyclo)	Open Contact	Target ready to be irradiated	Indicates to cyclotron that target can be irradiated
Target unloading enabled (from cyclo to target)	Closed Contact	Target ready to be unloaded	Indicates to target system that target can be unloaded

5.4 IT CONFIGURATION FOR REMOTE ACCESS

Refer to Section 4.5.1.

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6. RIGGING REQUIREMENTS

IBA recommends the Buyer to contact a rigging company. Before the scheduled rigging date, in order to avoid any delay during the rigging itself with the rigging company, the buyer will communicate to IBA:

- the rigging method;
- risk assessment;
- tools and equipment planned for this operation.

6.1 UNLOADING ZONE

There should be a maneuvering area suitable for a heavy trailer and crane with adequate access to nearby road.

Typical heavy trailer for cyclotron transport is a lowboy truck.

6.2 FLOOR LOAD, RIGGING ROUTE AND OPENINGS

Two rigging methods are proposed:

- **VERTICAL**, when the cyclotron is lifted by a crane and inserted in the vault through an opening in the roof (refer to drawing SRG-0081).

Warning: in case of vertical rigging, the cyclotron could rotate. Therefore, a mean to guide the cyclotron orientation must be foreseen.

- **LATERAL** (horizontal), when the cyclotron will be rolled into the vault by an opening in one of the vault walls (refer to drawing SRG-0080).

Warning: the only authorized lifting points on the cyclotron are the eyebolt on the top of the cyclotron and the bottom of the two yoke lifting system jacks (refer to drawing **SRG-0080 and 0081**).

Table 29: rigging requirement Cyclone® KIUBE (vaulted)

Vault roof (vertical)	2400 x 2400 mm
Lateral insertion	2300 mm wide x 2100 mm high
Loading	18 000 kg on 4 pillars (220 x 220 mm)

The cyclotron rigging route will be pre-defined with the IBA representative and rigging company. Rooms and corridors will be cleared to allow maneuvering and transportation of cyclotron and related equipment. The equipment will be transported with a forklift or similar transportation vehicle from the unloading area to the place of final location.

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During cyclotron manipulation, a maximum level difference of 50 mm between feet shall be respected.

The insertion of power cabinet and water conditioner (uncrated) requires a 900 mm wide and 2000 mm high minimum opening.

In case of lateral insertion, the ground of the cyclotron vault and route should be surfaced with reinforced concrete or metal plates to withstand the cyclotron weight.

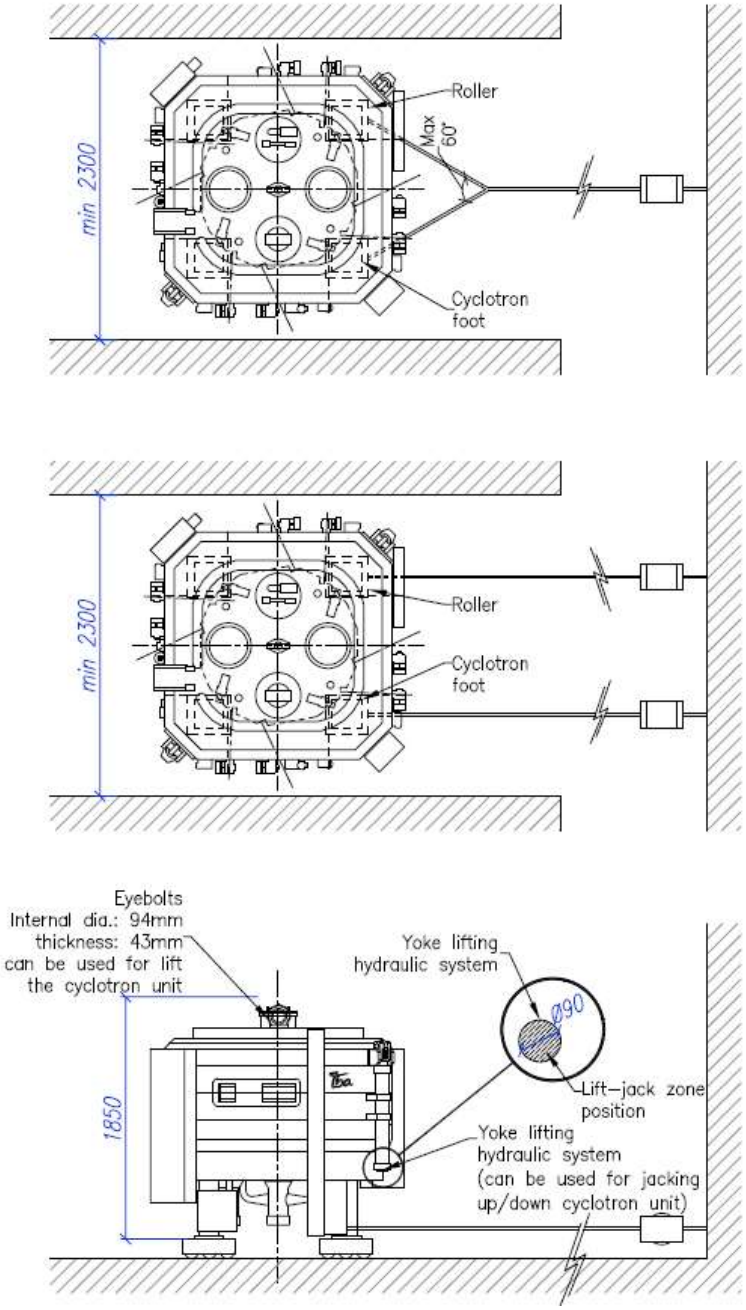


Figure 21: horizontal rigging, how to pull and lift the cyclotron

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6.3 RIGGING COMPLETION

After rigging of the cyclotron, its feet shall be anchored in the floor with the 8 clamping claws provided (two claws per foot) by the customer or rigger. The hole shall be drilled with a rotary hammer and 16mm bit around the four feet. The holders must be oriented in such a way that the cyclotron cannot move (see drawing below).

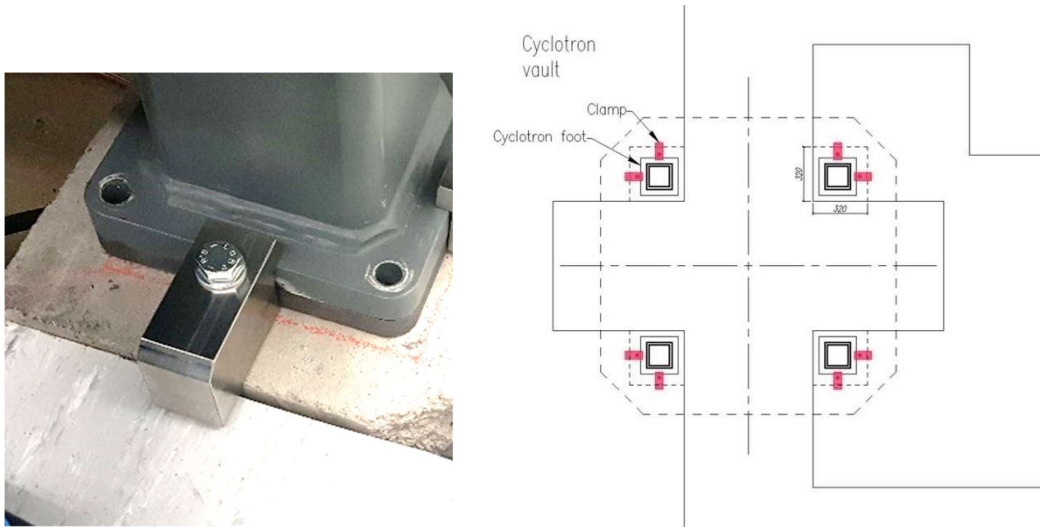


Figure 22: cyclotron anchorage

Finally, the cyclotron and equipment should be properly covered and protected against dust, humidity, temperature and shocks. The cyclotron vault should be completely closed once the cyclotron is rigged in.

6.4 RIGGER / CUSTOMER SUPPLIED EQUIPMENT

- A crane for lifting the cyclotron from delivery truck to the vault

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The Cyclone® KIUBE is equipped with one lifting point: 360° rotating lifting eyebolt M48 forged steel (Schiltz reference: 2131.11.048; vertical capacity: 32 tons). This rigging eyebolt is accessible for rigging operation without having to open the cyclotron wooden

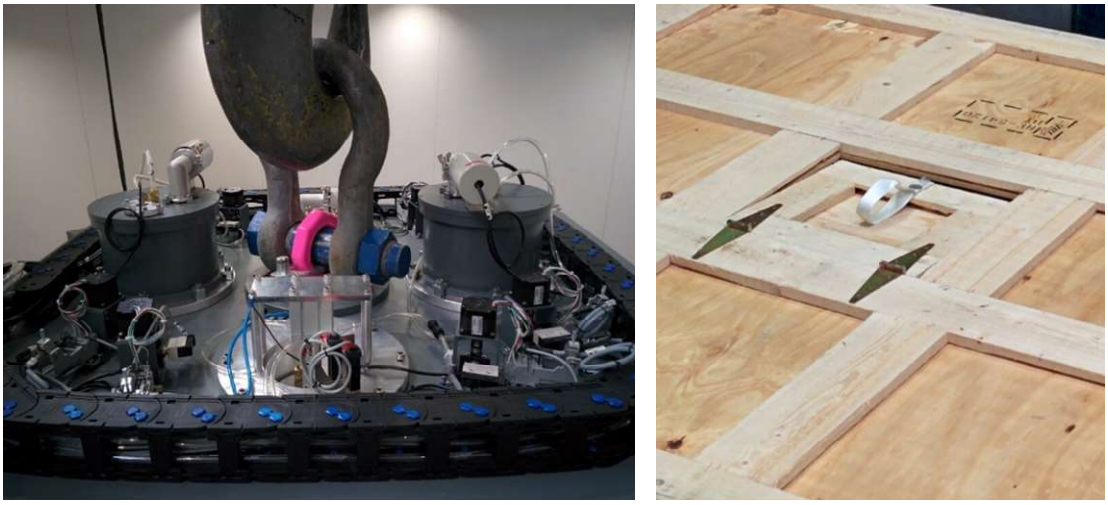


Figure 23: cyclotron lifting eyebolt

transport box.

- Steel floor plates and wood block to cover floors while transporting the cyclotron in case of lateral insertion
- Rollers in case of lateral insertion
- Wood blocks in assorted size
- 20 000 kg capacity Jacks and accessories for lifting the cyclotron
- Equipment to unload and move crates (forklift)
- Panel lifter for computer flooring/ power supply room tiles
- Ladder
- Electric screwdriver with POZIDRIVE bits to open boxes (bits PZ2)
- Set of wrenches to disassemble the equipment protection
- Crowbar
- Rotary hammer with bit of 16mm to drill in concrete

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6.5 SHIPMENT AND STORAGE

6.5.1 BOX SIZE AND WEIGHT

For a marine freight and standard configuration, typical box size and weight are:

Table 30: Cyclone® KIUBE shipment box (excluding self-shield)

Equipment	*Dimensions L x W x H (mm)*	*Total volume (m³)*	*Total gross weight (kg)*
Cyclotron	2300 x 2200 x 2100	11	18 500
Water conditioner	1200 x 800 x 1100 (x 2 boxes)	1	160
Power cabinets	1100 x 800 x 2200 (x 4 boxes)	7	1600
Other accessories cables, computer, pipes	15 boxes	11	2000

6.5.2 STORAGE

Required General Conditions for equipment storage:

The Buyer is responsible to comply with the following specifications:

- Closed and secured warehouse
- Ability to load/unload a weight of 20 000 kg from a lowboy truck
- Ability to unload a 20ft-container
- Ability to manipulate 20 000 kg
- Ability to handle boxes with a weight of 2 000 kg
- Minimal ground surface needed for standard configuration: 50 m^2 (including 25m^2 free space for manipulation)
- No stacking of boxes allowed
- Transport temperature requirements: 5 - 50°C
- Storage temperature requirements: 5 - 50°C
- Maximum humidity level: non-condensing
- Buyer liability for fire, theft and damage during storage and loading/unloading.

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7. INSTALLATION AT HIGH ALTITUDE (>2000M)

This section is applicable for all Cyclone® Kiube that aim to be installed at a higher altitude than 2000m above sea level. Because of the air rarefaction and significantly reduction of air moisture, the followings must be considered when planning a cyclotron facility:

7.1 REQUIRED ADDITIONAL INTERCONNECTIONS

Due to air dryness at high altitude, to prevent electrical short-circuit, interconnections for electrical purposes must be doubled. In paragraph 3.2 Interconnection table, table 10 Interconnection reservations must be replaced by the following information:

Table 31: interconnection reservation

From / To	*Power Supply room*	*Heat exchanger room*	*Hot lab*
Vault	**4** ducts DN 160mm (A1) **4** ducts DN 110mm (A2) **2** duct DN 110mm (E) In case of beam line: see specific drawings	1 duct DN 160mm (B)	According to configuration: for 2 targets → 2 ducts DN 110 mm (Rx) In case of solid target[5]: 1 duct DN 160 mm and 1 duct of DN 110mm (Sx)
PS room	-	**2** duct DN 110mm (C)	-
Control room	1 duct DN 80mm (D)	-	-

Provision must be made by the Buyer and his Construction Contractor for the gas connections (from the cold gas depot to the cyclotron vault), all other piping and wiring (light, power outlet, safety and emergency features, etc.) not included in IBA scope of supply and for Ventilation ducts. Refer to drawing SRG-0010, ducts X, Y and Z.

7.2 VENTILATION REQUIREMENTS FOR HEAT DISSIPATION REMOVAL

Due to air rarefaction, special care must be taken to remove the specified power out of the power supply room and cyclotron vault. The following modifications must be considered in regard with paragraph 4.4 HVAC and ventilation.

a. For the air renewal at **the cyclotron vault**, air change per hour must be adapted to ensure the removal of 5kW.

[5] *To be defined depending on solid target type*

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b. **For the power supply room**, the power to be dissipated into air remains as specified into paragraph 4.4.2 of Site Requirement Guide: 15kW for Cyclone® Kiube 100/150/180 and 20kW for Cyclone® Kiube 300. A fan coil unit (FCU) or close control unit (CCU) is recommended to remove this heat dissipation out of the room's atmosphere. For the sizing of the equipment the followings must be considered:
   - Each power supply cabinet is equipped with fans on top that drive the air across the cabinet from bottom to top.
   - Cool air must be blown below the power supply cabinets (below the recommended raised floor).
   - It is recommended not to connect the top of the cabinets with the ventilation system to prevent potential damage of the fans of the cabinets.
   - **Main heat dissipator is the power supply cabinet #2 hosting the RF amplifier**, which represents about 80% of the heat dissipation (12kW for the Kiube 100/150/180 and 16kW for Kiube 300).
   - The fan on top of the power supply cabinet #2 has a flow of 1000m³/h (air coming out of power supply cabinet #2 can easily reach a temperature above 60°C)
   - It is recommended to have the main ventilation exhaust above of the power supply cabinet #2 (but not connected) in order not to have to cool a part of this hot air flow down.
   - The flow handled by the FCU or CCU must be adapted in regard with the mix air/humidity characteristics at high altitude. For information only, the following table indicates the required flow for the FCU/CCU in order to remove 15kW with a delta temperature of 12° at different altitudes. It remains the responsibility of the buyer's engineering office to ensure an appropriate air management to ensure the heat dissipation removal in regard with site location conditions.

ALTITUDE [m]	PRESSURE [hPa]	SPECIFIC HEAT [J/kg.K]	DENSITY [kg/m³]	REQ. FLOW [m³/h]
0	1013,25	1018	1,199	**3687**
500	954,62	1018	1,129	**3915**
1000	898,76	1018	1,063	**4158**
1500	845,59	1018	1,001	**4416**
2000	794,98	1019	0,939	**4703**
2500	746,86	1019	0,882	**5007**
3000	701,12	1020	0,828	**5328**
3500	657,68	1020	0,776	**5685**
4000	616,45	1021	0,727	**6063**
4500	577,33	1021	0,681	**6472**
5000	540,25	1022	0,637	**6912**

Table A1: indicative required cooling flow in PSR in regard with the altitude

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8. BEAM LINE REQUIREMENTS (OPTIONAL)

One, two or three Beam Transport Lines (BTL) could optionnaly be installed on your Cyclone® KIUBE. The following models are available:

- Short BTL (~2m, refer to drawing **SRG-400, art. 230.030**),
- Medium BTL (~3m, refer to drawing **SRG-410, art. 230.031**)
- Long BTL (>4.5m, refer to drawing **SRG-420, for illustrion only, art. 230.032**)
- Long BTL +AC Magnet (>4.5m**, art. 230.033**)

These options are NOT compatible with a self-shielded cyclotron.

8.1 BEAM LINE COMPONENTS

Beam Transport Lines have their own vacuum pumping equipment (oil diffusion pump and primary pump) and their own cooling-water manifold. This manifold is directly connected to the cyclotron water conditioner.

At the end of the beam line, a collimator allows connection of a beam stopper or targets. An isolation valve is foreseen for target replacement without disrupting the beam line vacuum.

8.1.1 SHORT BEAM TRANSPORT LINE

Short beam transport line aims to actively control the beam shape. It is made of:

- a vacuum isolation valve,
- beam tube (approximately 2m long),
- quadrupole (2 singlets),
- and steering magnets.

8.1.2 MEDIUM BEAM TRANSPORT LINE

Medium beam transport line aims to actively control the beam shape and induce a wrobbling movement to the beam. It is made of:

- a vacuum isolation valve,
- beam tube (approx. 3m long),
- quadrupole (2 singlets),
- steering magnets

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- and AC magnet.

8.1.3 LONG BEAM TRANSPORT LINE

Long beam transport line is dedicated to transport beam into a separate vault. Long BTL is made of the following elements:

- a vacuum isolation valve,
- beam tube (min. 4,5m long, depending on options selection),
- quadrupole (4 singlets),
- steering magnets

A neutron shutter is optionnaly available to reduce neutron emission in the beam line vault while cyclotron is running.

AC magnet as an option is available.

Due to flexible design for long beam lines and their options, a preliminary design consultation must be held with IBA.

8.2 TYPICAL ROOM REQUIREMENT

A short beam line will be installed inside the cyclotron vault (refer to Figure 24). A long beam line will allow irradiating a target in a second vault (refer to Figure 26).

Table 32: vault dimensions with a beam line extension

		Short Beam Line	*Medium Beam Line*	*Long Beam Line*
Vault internal dimensions	L x W	~6 x 3,8 m	~7 x 3,8 m	~5,1 x 3,8 m
	Height	Min. 2,5 m	Min. 2,5 m	Min. 2,5 m
Second Vault internal dimensions	L x W	/	/	Min 3,15 x 3,8 m
	Height	/	/	Min. 2,5 m

Lifting eyebolts shall be installed on the roof to ease rigging and alignment of the beam line components (see drawings SRG-0400 and SRG-0410 for location and specifications).

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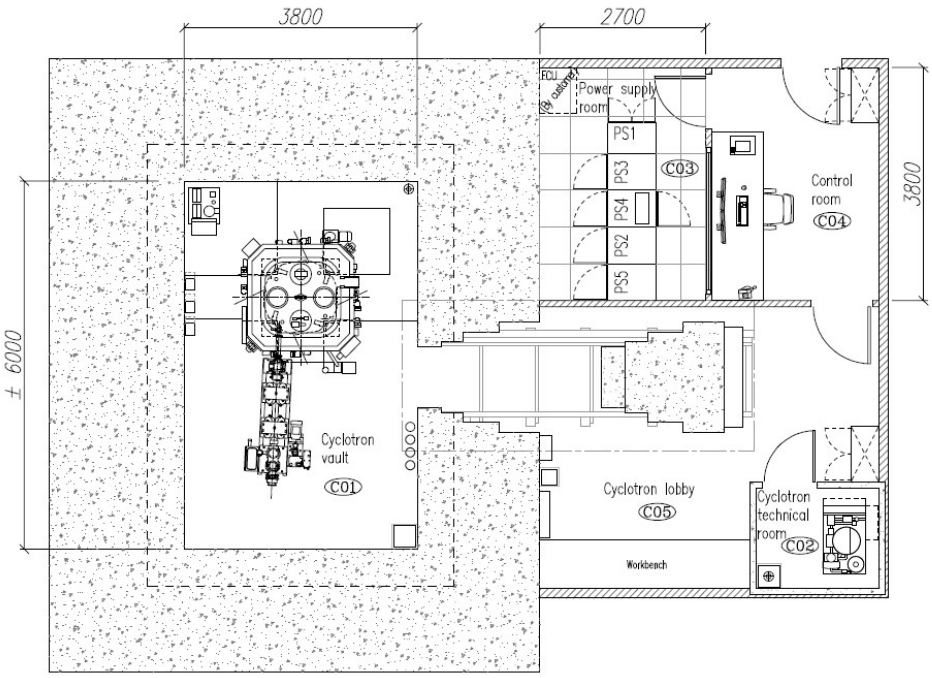


Figure 24: Cyclone® KIUBE with short beam line

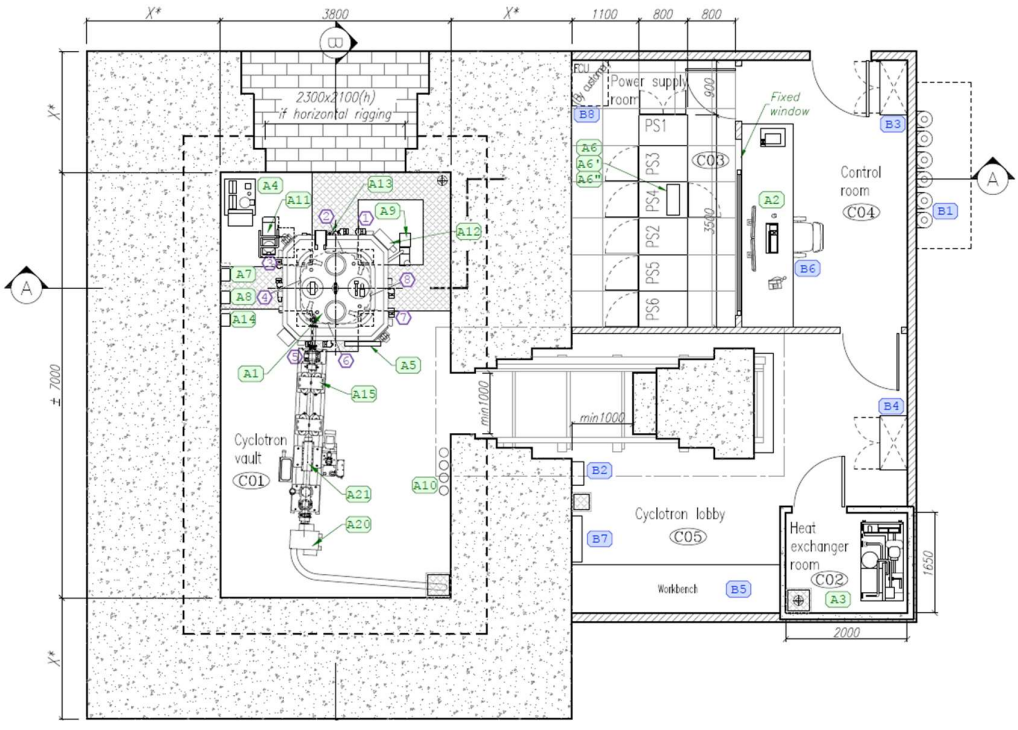


Figure 25: Cyclone® KIUBE with medium beam line

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If a second vault is built, room requirements (HVAC, safety, lights,…) are the same than for the cyclotron vault or may need to be adjusted in case of non standard project. Interconnections with other rooms shall also be adapted.

The connection for the beam line hole between the two rooms shall have a diameter of 200mm, be straight and at a central height of 1120mm (see drawings SRG-0420). After installation, customer shall provide and install polyethylene balls in order to fill the hollow space of the beam line hole.

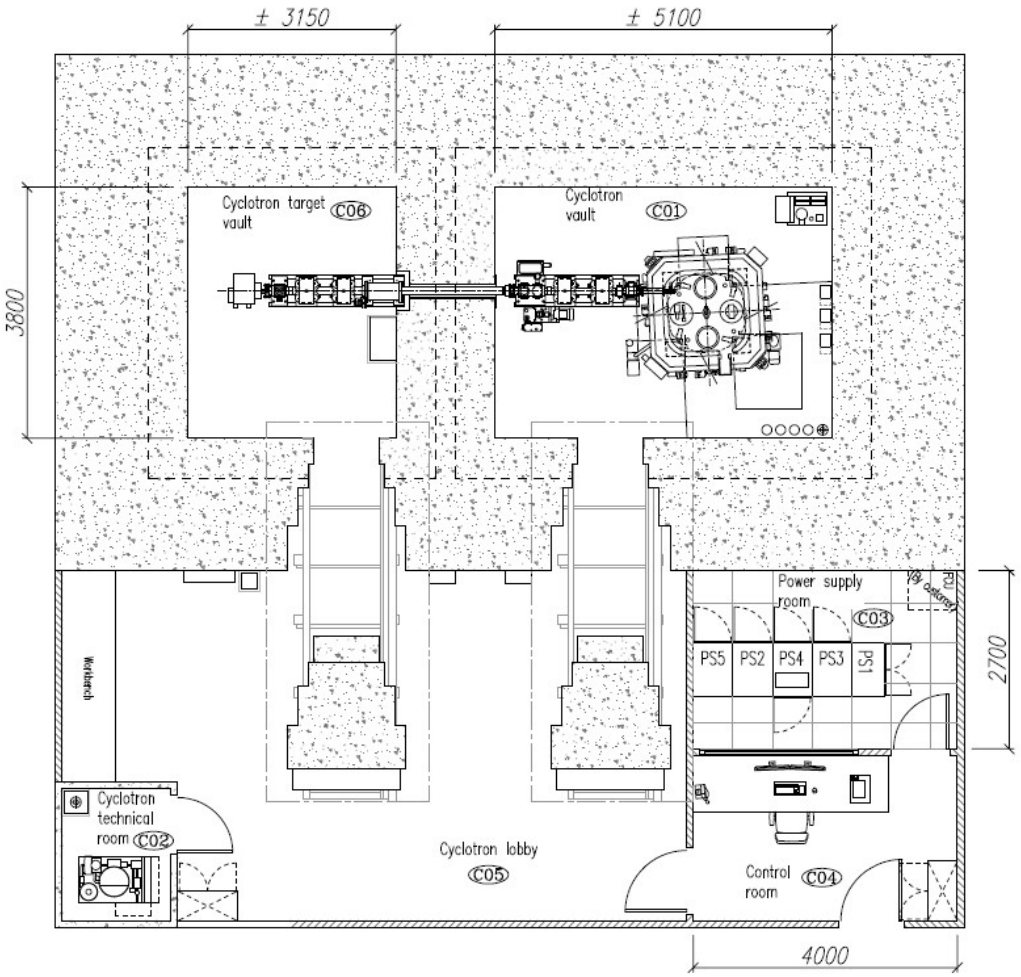


Figure 26: Cyclone® KIUBE with long beam line

8.3 CONTROL AND POWER CABINETS FOR BEAM LINE

The beam line power supply and control will require an additional cabinet in the power supply room for beam line equipment (PS5: 600 x 800 x 1880 mm; ~400kg).

For medium beam line's AC magnet power supply, an additional cabinet is required in the power supply room (PS6: 600 x 800 x 1880 mm; ~200kg).

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The beam line equipment will require to add the following installed power to the cyclotron power requirements indicated in Section 4.3.2:

Additional Power consumption:

	Additional power requirement
Short beam line	+10 kVA, 3 x 400 Vac
Medium beam line	+15 kVA, 3 x 400 Vac
Long beam line	+30 kVA, 3 x 400 Vac

8.4 RIGGING

Special attention must be paid to the levelling and orientation of the cyclotron during rigging.

In the case of a long beam line and two vaults, the alignment between the cyclotron, bunker hole and beam line is critical. IBA will provide a tool for the markup of the cyclotron footprints, so that alignment with beam line is ensured. During the rigging of the cyclotron, the cyclotron feet and orientation shall be positioned according to the footprints.

After installation and commissioning of the beam line, the cyclotron and beam line supporting feet shall be anchored to the ground to prevent any movements.

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9. SELF-SHIELDED CONFIGURATION

Please take into account that all the requirements for the facility design and the installation of the Not Self-Shielded are applicable for the Self-Shielded configuration unless differently specified in this chapter. A complete set of drawing is dedicated to self-shielded configuration (refer to drawings SRG-03XX).

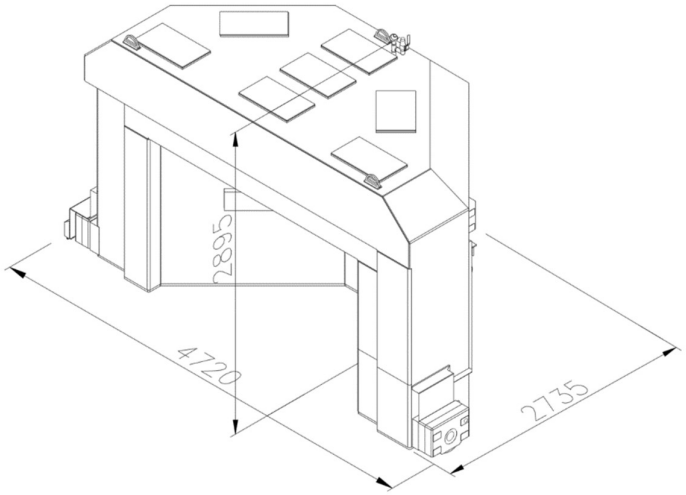
This chapter includes the general building requirements for the Cyclone® KIUBE with self-shielding. The installation procedure for the self-shielding will be provided by the IBA Project Manager (IBA ref MID73679).

The **self-shielding option is available only with Cyclone® KIUBE 100 and Cyclone® KIUBE 150**.

The self-shielding for the Cyclone® KIUBE is made of two movable steel shells constituted by an inner compartment filled with water and an outer compartment filled with magnetite aggregated drawn in water. This option allows shielding the neutrons as close as possible from the source (nuclear reaction inside the target). Shells will be delivered empty and they will be filled after installation in place. From the closed position, both halves can slide on rails to give access to the cyclotron to perform maintenance tasks.

Table 33: Cyclone® KIUBE self-shielding size and weight

Self-shielding (each piece)	Empty	Filled
Weight (kg)	6 560	48 100
Overall dimensions (L x W x H) [mm]	4720 x 2735 x 2895	



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9.1 STANDARD ROOM CONFIGURATION

Refer to drawing SRG-0300.

- **Cyclotron vault** for Cyclone® KIUBE, Self-shielding, water conditionner and auxiliary equipment;
- **Power supply room** for cyclotron power supply cabinets (RF, magnetic field, ion source, PLC, …)
- **Control room** for cyclotron control station. We recommend a separated room because of heat and noise generated by power supplies cabinets;
- **Cyclotron lobby** with access to cyclotron vault;

9.2 CYCLOTRON ROOM

Refer to drawing **SRG.0300** for the Cyclone® KIUBE self-shielded configuration.

This section describes the specific requirements for the cyclotron room equipped with self-shielding option. Recommendations about finishing. are not impacted by the installation of the self-shielding.

Table 34: Cyclone® KIUBE self-shielded room dimensions

Inner minimal dimensions (L x W x H) [mm]	6000 x 7000 x 3100
Concrete wall thickness [mm]	400 – 800

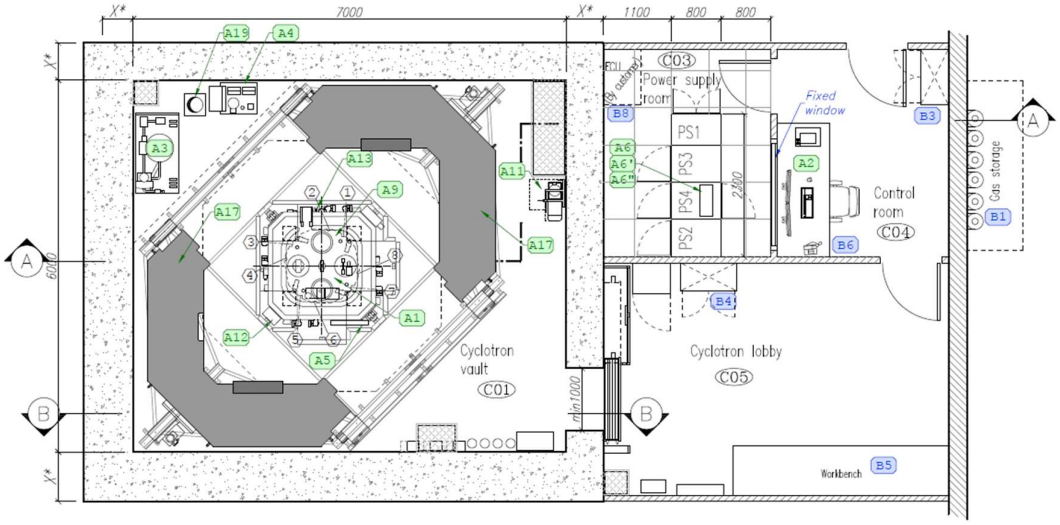


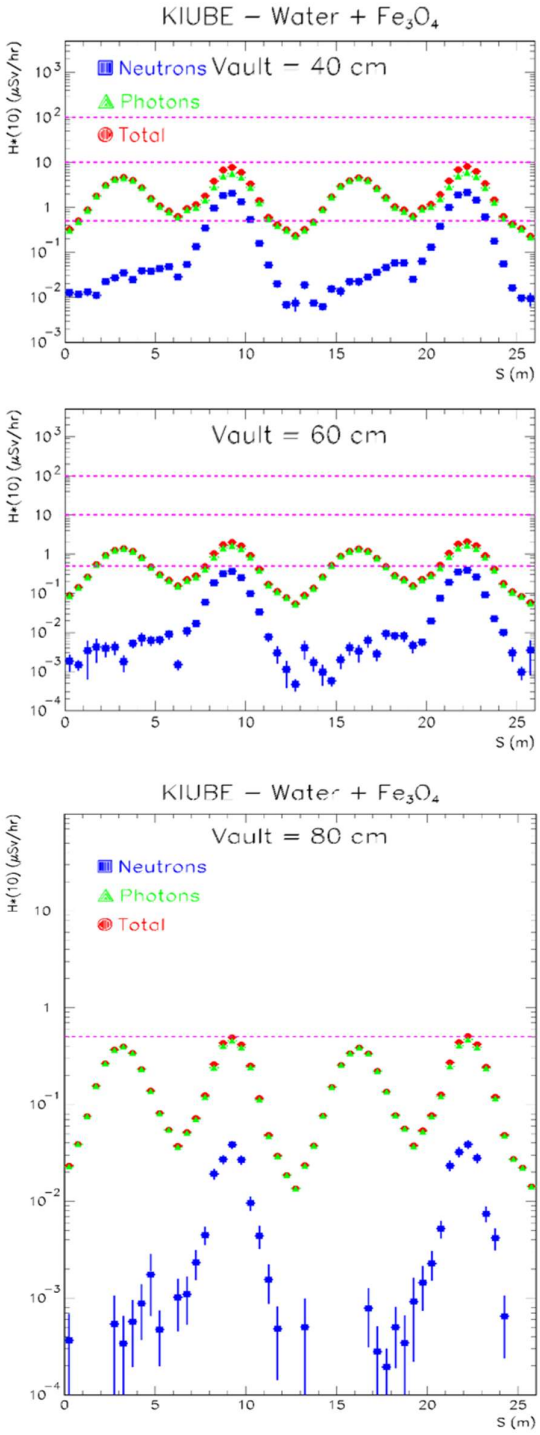
Figure 27: self-shielded configuration

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9.2.1 VAULT CONCRETE SHIELDING, ACCESS AND INTERLOCK:

Exact concrete wall thickness has to be determined according to production plan and local regulations.



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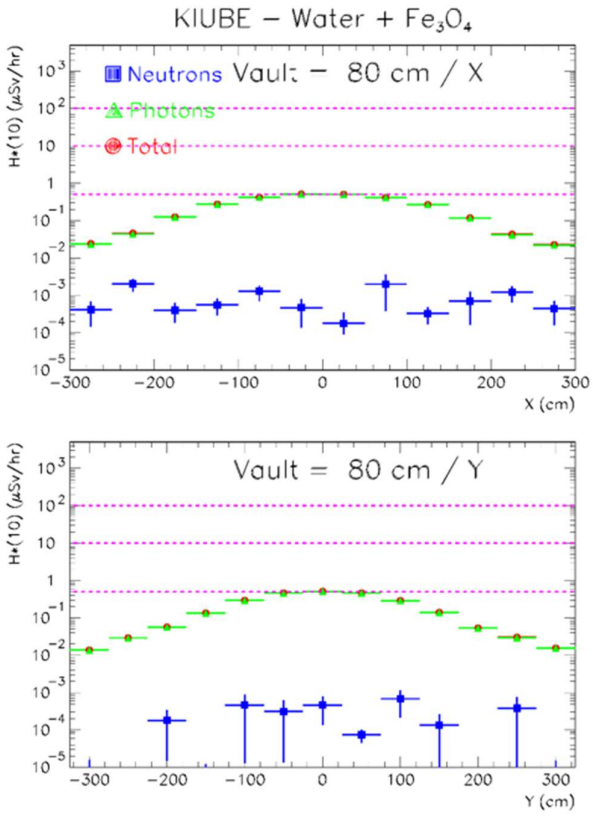


Figure 28: dose rate (KIUBE 150 µA (2x75µA on fluorine targets) self-shielded) outside the room for 40, 60 and 80cm concrete wall and 80cm concrete ceiling– dose limit 10 and 0.5 µSv/h. MID 95997

The access door must provide a shielding level equivalent to the concrete walls. A solution combining HDPE (10-20 cm) and lead (1cm) could be an option, or a short angle maze could be designed in front of the cyclotron room access. The self-shielding vault door must be equipped with emergency systems such as anti-crushing bars, emergency stop buttons, and warning lights.

An interlock system makes sure the user cannot start the cyclotron with the self-shielding open and is included with the self-shielding delivery. In addition, the cyclotron vault door must also be closed to autorise beam production and must be equipped with TWO redundant door switches as described in section 5.1.1.1 and drawing **SRG-0350**.

9.2.2 VENTILATION

A maximum of 5kW heat dissipation has to be taken into account when calculating the ventilation inside the cyclotron room. A hot air extraction fan will be installed by IBA for ventilation of the cyclotron inside the self-shield (item A19 on drawing **SRG-0300**) and this fan will be installed on top of the **duct F** (refer to drawing **SRG-0310** and **SRG-0311** if applicable), it should be a **galvanized metal spiral ducting**. The exhaust air of this fan is released inside the cyclotron room and it is recommended to place the exhaust grid of the room to the global HVAC system close to this point.

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A minimum of 10 air renewal per hour is recommended and a relative pressure of -60Pa (identical requirements in case of non-self-shielded cyclotron, refer to 4.4 and drawings **SRG-0335** and **SRG-0330**).

9.2.3 SPECIFIC ELECTRICAL REQUIREMENT

The cyclotron and its subsystems will be feeded like the not Self-Shielded configuration. Please refer to the chapter 4.3. and drawing **SRG-0370**.



The self-shielding will be powered up by the Control Box. However for the rigging and installation phases, it is required a female 400V, 16A, 3P + N + G, IP 44, power socket according to EN 60309-1 and 60309-2 located under the future location of the Control Box at 500mm above floor. A temporary power supply can also be acceptable.

A horizontal wire mesh cable tray is required around the vault to serve the different systems of the Self-Shielding like indicated in the drawing **SRG-0370**. Take into account that several vertical cable trays are also foreseen for the self-shielding fan, the self-shielding control box, the Pit 1 and also two vertical cable trays to serve the energy chain of each shell. These trays are illustrated on the drawing SRG-0370.

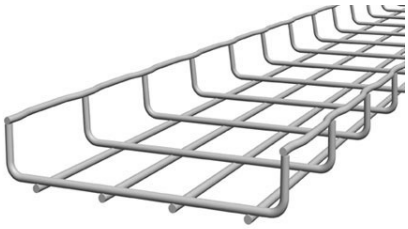


Figure 29: Example of wire mesh cable tray

9.3 INTERCONNECTION TABLE

These minimal requirements are for IBA use only. No customer equipment uses these ducts. Otherwise, the ducts must be made larger and the design adapted. **Pulling cords must be foreseen in each duct**. For the exhaustive list of interconnections, refer to drawing SRG-0310.

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Table 35: Typical interconnection ducts

From / To	*Cyclone® metallic central pit*	*Power Supply room*	*Vault pit 1*	*Hot lab*
Vault Pit 1	2 ducts DN 160mm (A1') 2 ducts DN 110mm (A2') 1 duct DN 110mm (E')	2 ducts DN 160mm (A1) 2 ducts DN 110mm (A2) 1 duct DN 110mm (E)	-	-
Vault Pit 2	1 pipe DN 110mm (B) 1 pipe DN 110mm (G)	-	1 duct DN 110mm (C)	-
Vault Pit 3	1 duct DN 110mm (R1) for radio-isotope transfer In case of solid target: 1 duct DN 110 mm (S1)	-	-	According to configuration: for 2 targets → 2 ducts DN 110 mm (Rx) In case of solid target: 1 duct DN 110 mm (Sx)
Cyclotron vault corner	1 duct DN 250 mm (F) for self-shielding exhaust air (galvanized steel)	-	-	-
Control room	-	1 duct DN 80mm (D)	-	-

Provision must be made by the Buyer and his Construction Contractor **for the gas connections** (from the cold gas depot to the cyclotron vault), **all other piping and wiring** (light, power outlet, safety and emergency features, etc.) not included in IBA scope of supply and for **Ventilation ducts**. Refer to drawing **SRG-0310**, **ducts X, Y and Z**. If the ventilation duct passes underneath the rails, a reinforced steel plate shall be installed (refer to drawing **SRG-0311**).

9.4 RIGGING

The self-shielding rails and metallic central pit should be inserted first. Note that the length of the rails, in case they will be inserted through existing corridors, could present some manipulation constraints; refer to drawing **SRG-0380** for the maximum dimensions and weight of these items.

The two shells of the self-shield will be inserted later on by either lateral or roof reservation and then the Cyclone® KIUBE will be rigged-in. For vertical rigging, the two shells will be rigged emply. For horizontal rigging, shells could be rigged after being filled with magnetite (total weight: 48,1 tons per shell). Each shell has 5 lifting eyes of 5,000 kg for its rigging (refer to drawing **SRG-0380**). Each motor assembly of 214 kg has one lifting bolt.

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The Cyclone® KIUBE with self-shield has larger rigging requirement due to the size of the self-shield shells.

Table 36: rigging requirement for self-shield (drawing SRG-0380)

Vault roof (vertical)	2600 x 5000 mm
Lateral insertion	3100 mm wide x 3050 mm high
Biggest shell dimensions (L x l x h) w/o accessories	4255 x 2200 x 2680 mm

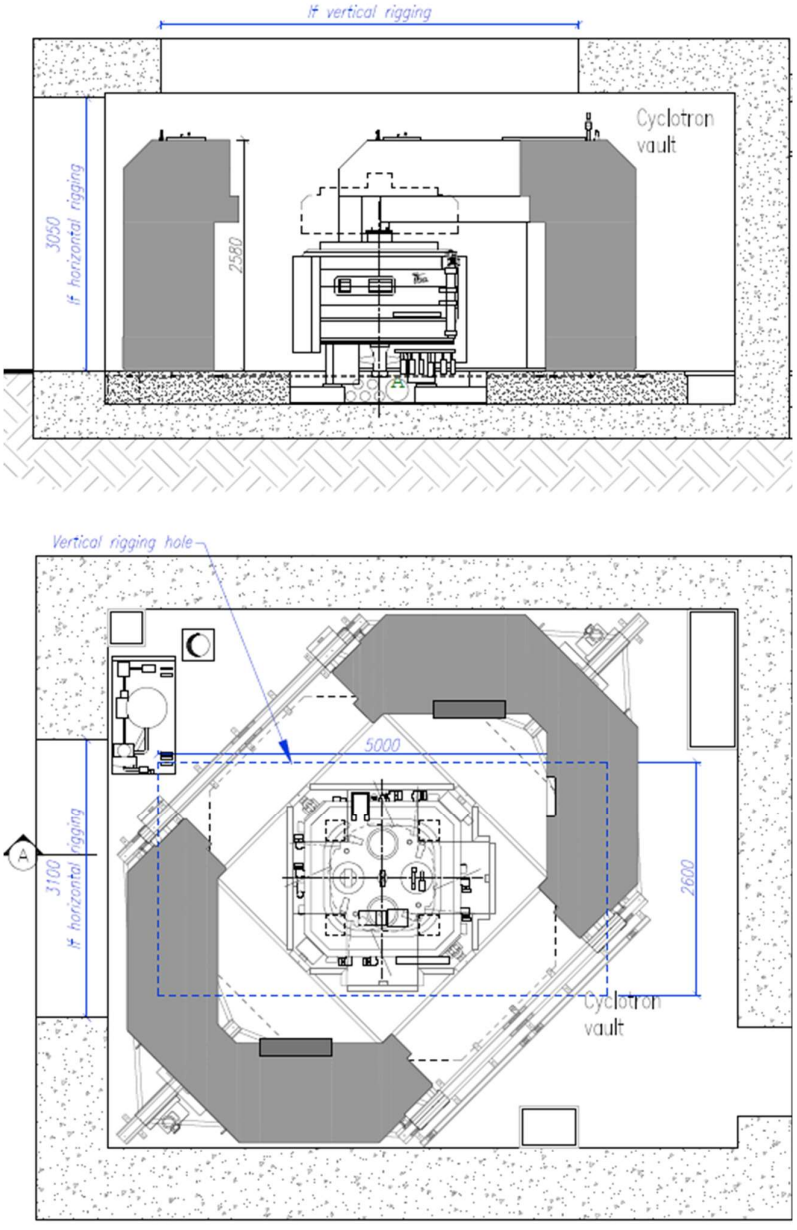


Figure 30: rigging insertion for self-shield (Drawing SRG-0380)

All pictures used in this document are non-contractual and for illustrational purpose onluy

9.5 INSTALLATION & FILLING OF THE SELF-SHIELDING

The installation has to take place in several steps. Together with the self-shielding, IBA will deliver a metallic central pit for the cyclotron and the structure for the rails (refer to drawing **SRG-0310, SRG-0315 and SRG-0380**).

In the following description, the vertical reference level will be the cyclotron vault finished floor level.

1. The bottom of the metallic central pit will be positioned at a level - 350mm compared to the reference level.
2. A first layer of the concrete will be poured to reach the level – 75mm compared to the reference level.
3. The rail structure will be positioned and levelled with screws provided with the rails.
4. The second concrete layer will be poured until the reference level with a tolerance of ± 1mm including the floor finishing. It is recommended that this finishing is hard, washable, and smooth. For instance, it could be used an epoxy coating, so that there will be a minimum of dust collected and contamination can be removed.
5. The two empty shells and the cyclotron will be rigged and positioned.
6. The self-shielding empty shells will be filled:
    a. Filling with agregates
    b. Filling with demineralized water
7. The vault will be closed.

Table 37: Cyclone® KIUBE self-shielding empty and filled weight

Self-shielding (each piece)	*Approx. weight (kg)*
Empty Shell	6 560
Filling material	+/- 34 000
Water	+/- 7 500
Filled Shell	48 100

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10. REGULATORY

10.1 APPLICABLE SAFETY REGULATIONS

IBA is committed to the protection of the Environment and the Occupational Health and Safety (EHS) of our employees, contractors, visitors and others affected by our operations. In order to demonstrate this commitment, a set of IBA EHS Policies has been established and define the requirements that need to be respected by IBA Employees, Contractors and Buyers during both Installation and Servicing/Maintenance activities. These Policies are gathered in the "*IBA EHS requirements for Contracts – RadioPharma Solutions*" document and shall be acknowledged by the Buyer prior to project start. The requirements are effective when IBA Employees or one of his contractors are on Buyer site and is restricted to the IBA working areas and related IBA activities.

Applicable Safety Regulations may vary significantly according to the country. IBA Employee is foreseen to follow Buyer EHS requirements and IBA EHS requirements.

Some important topics related to EHS policies for installation phase are highlighted in this section.

10.1.1 BUYER SAFETY OFFICER

In any country, the Buyer or his Representative is requested to designate a competent Site Manager or Project Supervisor and a competent Safety Officer (also referred to as the Safety Supervisor or Safety Coordinator).

These individuals should interface with the IBA Project Manager regarding any safety issues during the whole installation process including:

- Transport of the cyclotron and related equipment
- Unloading of the cyclotron and peripherals on the Buyer's site
- Rigging operations and related risk assessment
- Installation and connection of the various equipment and utilities
- Control and testing before operation up to (and including) the commissioning of the equipment.

For EU countries specifically, the 89/391/EEC directive on "The introduction of measures to encourage improvements in the safety and health of workers at work" are applicable.

In particular, the Buyer or the Project Supervisor should ensure that prior to the setting up of a construction site, a Safety and Health Plan is drawn up in accordance with Article 5 (b) of the Directive.

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10.1.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

IBA has established a Personal Protective Equipment (PPE) Policy. This policy describes which type of PPE needs to be worn during which occasion/ in which location.

When within the contract the Buyer receives or is requested to acquire a set of PPE to perform maintenance on the IBA equipment, it is the responsibility of the Buyer to ensure proper storage of these PPE and to ensure that they are readily available in case an intervention is requested towards IBA. Not having the necessary PPE available before the intervention will lead to stopping the intervention and this at cost of the Buyer.

10.2 EU DIRECTIVES

The Cyclone® KIUBE system was developed and CE marked according to following EU Directives:

- Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC relating to electromagnetic compatibility (2014/30/EU)
- Directive 2014/35/EU of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of electrical equipment designed for use within certain voltage limits
- Directive 2014/30/EU of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to electromagnetic compatibility

10.3 APPLICABLE STANDARDS

The standards under which the Cyclone® KIUBE has been developed are listed in the Declaration of Conformity of the equipment and in the Product Description.

The Cyclone® KIUBE is a permanently installed equipment of Group 1 – Class A.

All precautions described in this Site Requirements Guide related to layout and electrical requirements shall be respected to ensure Electromagnetic compatibility.

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11. INSTALLATION PROCESS AND TYPICAL INSTALLATION PLANNING

11.1 CYCLOTRON FACILITY DESIGN

The design of the rooms dedicated to the installation of the cyclotron and auxiliary equipment must be done following the recommendations and drawings of this Site Requirement Guide. Final design of this zone must be shared with IBA Project Manager before the construction works begin.

In case Building design services are included in your Scope of Supply, IBA Integralab team will help in customizing the facility design and integration of equipment according to the specific constraints of your site.

11.2 BUILDING COMPLETION REQUIREMENT

The building completion will be evaluated by IBA representative. The evaluation will be done using the two BBOD General Acceptance Document (Building Beneficiary Occupancy Document (BBOD)) as part of the contractual document of the cyclotron system.

11.2.1 RIGGING & BBOD RIGGING

This BBOD Document is related to the requirements that should be completed before the rigging of the cyclotron in the vault and the other equipment. IBA will authorize shipment of the equipment only if these requirements are met.

A rigging company should be contracted to unload equipment and to place it in the proper location. A representant of the rigging/transport company should be present at the moment of BBOD rigging inspection to check access and rigging specificities. The rigging should include unpacking and cleaning of site. IBA representative can be present as advisor during unloading and rigging.

Qualified personnel from IBA will start the preparation of installation and put equipment in place.

11.2.2 INSTALLATION & BBOD INSTALLATION

This second BBOD Document is related to the site completion before installation. The installation will not start until the site is accepted and utilities are in working conditions for the IBA installation team.

Qualified personnel from IBA will install all cables and pipes between IBA equipment (except main power which is the Buyer's responsibility) and make the connections and perform testing.

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The water-cooling loop, ventilation and compressed air equipment must be totally operational at this time and available 24/24 and 7/7. The Buyer should make the gas connections to the cyclotron and start-up the gas system with enough provision of gas bottle.

11.3 SYSTEM START-UP

Qualified engineers will perform the start-up and preliminary testing of the equipment.

Ion beam at full power and full intensity can be produced; the cyclotron control parameters will be optimized. All the tests are recorded in a logbook.

The equipment and chemical products and laboratories equipment needed during start-up are the responsibility of the Buyer and must be ready on site.

After the first week of installation, IBA can produce beam with full power and full energy. The Buyer should have obtained the operating permits for such production before installation starts.

The beam tests are performed and the IBA specialist will begin radioisotopes production for optimization of targets.

11.4 ACCEPTANCE TESTS

When all the systems are working properly, the Acceptance Tests will be performed according to the contractual Site Acceptance Test document (SAT).

A Buyer representative must attend the Acceptance tests performed by an IBA engineer and sign all designated forms in the acceptance document, as each is successful.

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12. LIST OF RELATED DOCUMENTS

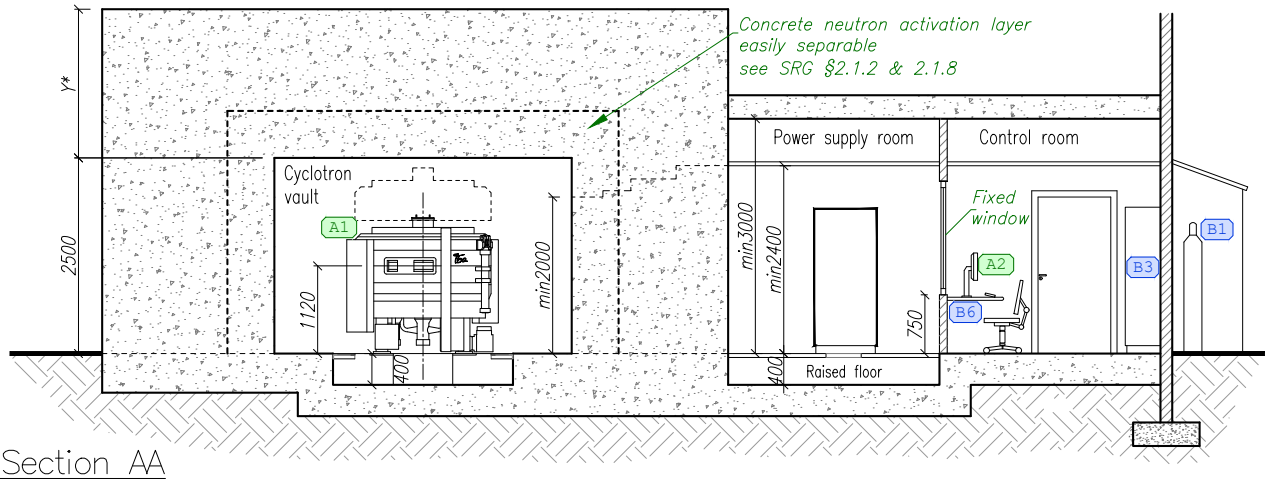
- IBA Cyclone® KIUBE System Description: IBA ref 58027
- IBA Cyclone® KIUBE Maintenance Guide: IBA ref 62133
- IBA Cyclone® KIUBE Supervisor Guide: IBA ref 66607
- IBA Cyclone® KIUBE Manual Mode User Guide: IBA ref 62272
- IBA Cyclone® KIUBE Production Mode User Guide: IBA ref 67061
- IBA Cyclone® KIUBE Product description: IBA ref 51819
- BBOD: Building Beneficiary Occupancy Date (contractual document): IBA ref 60116
- ICD: Interface Control Document: IBA ref 57421
- SAT: Site Acceptance Document related to your project based on IBA template ref 58361 (refer to contract appendixes)
- Cyclone® KIUBE Remote Access System Description: IBA ref 61735

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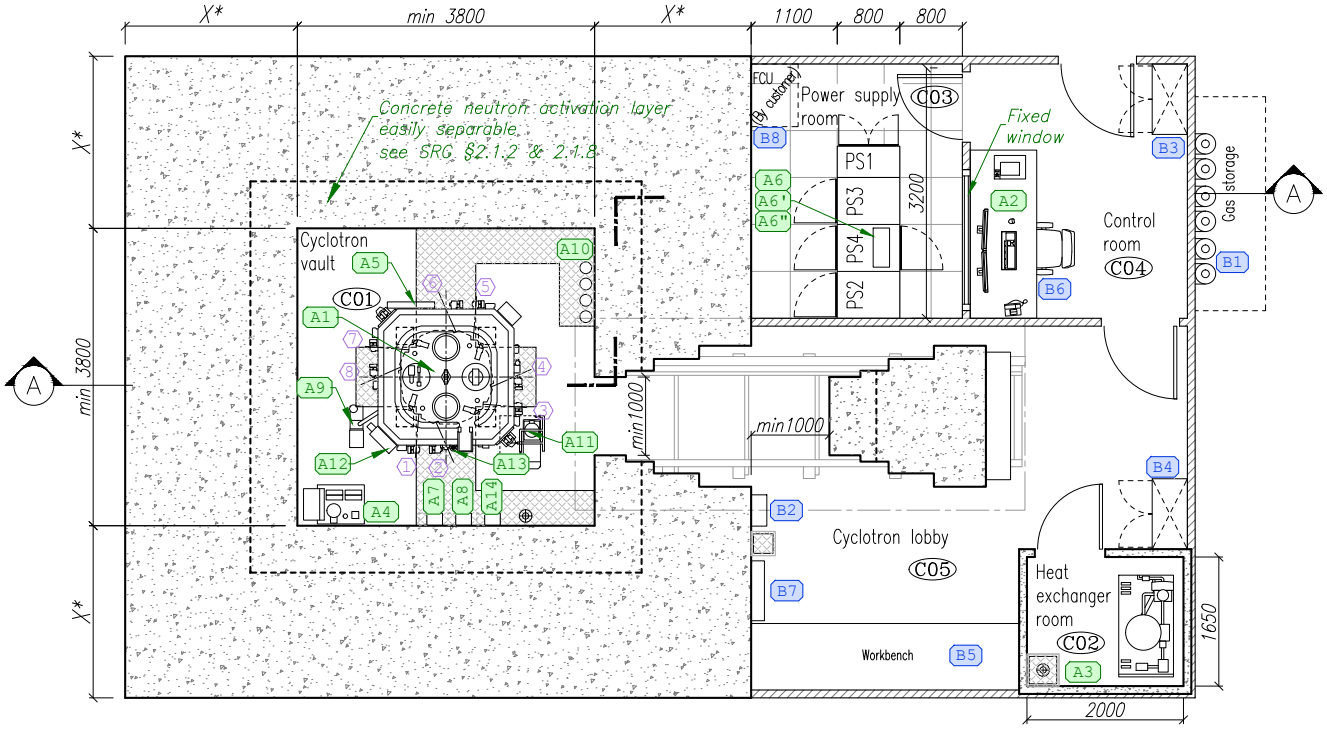
13. LIST OF ANNEXED DRAWINGS

Drawing	Reference	Contents
SRG-0001	07.46.00.0001	Site configuration Cyclone® KIUBE
SRG-0010	07.46.00.0010	Trenches and connections
SRG-0020	07.46.00.0020	Utilities and Sewage
SRG-0030	07.46.00.0030	Recommended pressure cascade
SRG-0035	07.46.00.0035	Minimum recommended air change rate
SRG-0050	07.46.00.0050	Cyclotron interlocks & control system interface
SRG-0070	07.46.00.0070	Electricity & data
SRG-0080	07.46.00.0080	Equipment dimensions & Rigging route. Cyclotron horizontal rigging
SRG-0081	07.46.00.0081	Equipment dimensions & Rigging route. Cyclotron vertical rigging
SRG-0200	07.46.00.0200	Site configuration with maze
SRG-0300	07.46.00.0300	Site configuration Cyclone® KIUBE with self-shielding
SRG-0310	07.46.00.0310	Self-shielding Trenches and connections
SRG-0311	07.46.00.0311	Self-shielding reinforcement plate
SRG-0315	07.46.00.0315	Self-shielding installation set
SRG-0320	07.46.00.0320	Self-shielding Utilities and Sewage
SRG-0330	07.46.00.0330	Self-shielding Recommended pressure cascade
SRG-0335	07.46.00.0335	Self-shielding Minimum recommended air change rate
SRG-0350	07.46.00.0350	Self-shielding Cyclotron interlocks & control system interface
SRG-0370	07.46.00.0370	Self-shielding Electricity and data
SRG-0380	07.46.00.0380	Self-shielding rigging holes dimensions
SRG-0400	07.46.00.0400	Site configuration Cyclone® KIUBE & short BTL
SRG-0410	07.46.00.0410	Site configuration Cyclone® KIUBE & medium BTL
SRG-0420	07.46.00.0420	Site configuration Cyclone® KIUBE & long BTL w/o AC magnet

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Section AA



Cyclotron Equipment List		
Item	Name	Supplied/installed by
A1	Cyclone® KIUBE	IBA
A2	Control st. Thin Customer+printer+2 screens	IBA
A3	Water conditioner	IBA
A4	Hydraulic Yoke lifting system	IBA
A5	Water manifold	IBA
A6	18F Syringe pumps cabinet (in PS4)	IBA
A6'	18O Rinsing Syringe pumps cabinet (in PS4)	IBA (optional)
A6''	Other Syringe pumps cabinet (13N/68Ga)	IBA (optional)
A7	Target switching input module (optional)	IBA (optional)
A8	Target switching output module (optional)	IBA (optional)
A9	Primary pump	IBA
A10	Safety pillars	IBA
A11	He compressor	IBA
A12	Plug to plug	IBA
A13	RF connection	IBA
A14	Target Gas distribution cabinet (optional)	IBA (optional)
PS1	Power Distribution Cabinet	IBA
PS2	R.F. Driver Amplifier Cabinet	IBA
PS3	Main Coils Ion Source P.S. Cabinet	IBA
PS4	Siemens P.L.C. Cabinet	IBA
B1	Gas cylinders supply	Customer
B2	Vault door control box	Customer
B3	Documentation storage cabinet	Customer
B4	Spare parts storage cabinet	Customer
B5	Workbench	Customer
B6	Desk + chair	Customer
B7	Cyclotron gas regulator panel	Customer
B8	Fan coil unit	Customer
1→8	Exit port number	IBA

All displayed dimensions are minimal values.

X* = 2000 to 2400mm
Y* = 1700 to 2300mm

The shielding thickness shown on this layout is indicative only. It is the responsibility of the buyer to determine the appropriate level of shielding and to execute the construction to fulfill the applicable local legislation.

0 1 2 3meter
0 5' 10'

S.A. Ion Beam Applications

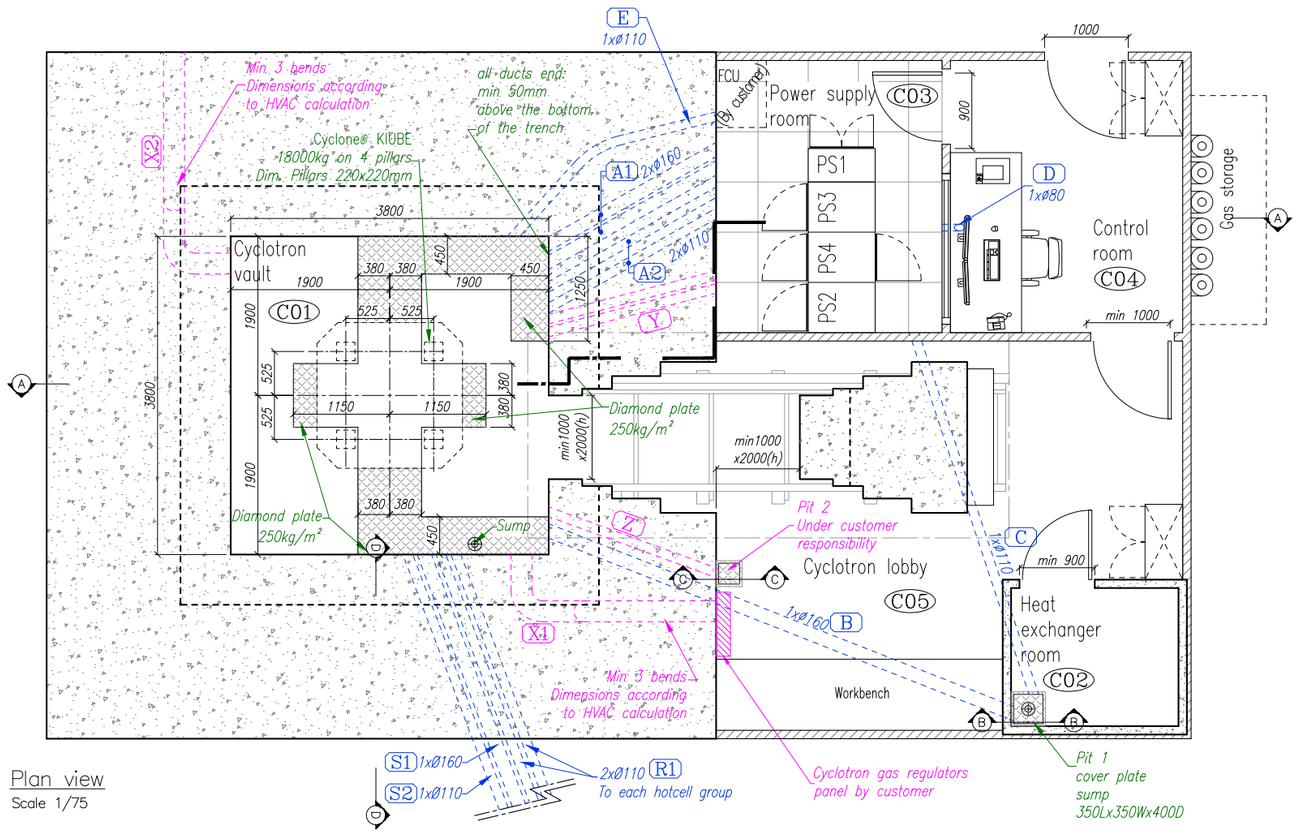
Cyclone® KIUBE

Contents:
Site configuration

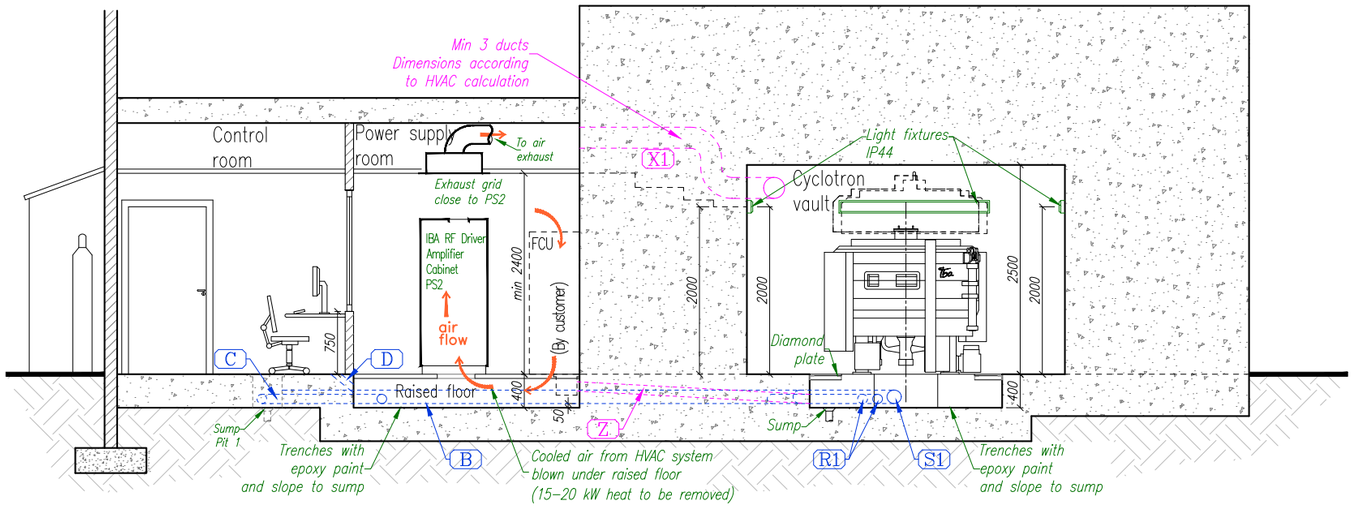
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Drawing SRG-0001
Ref.: 07.46.00.0001-G

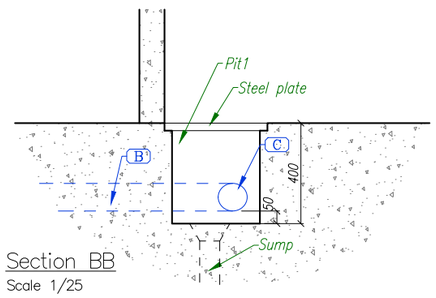
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Scale: 1/100



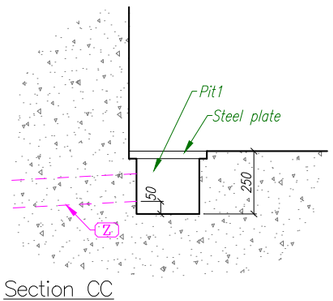
Plan view
Scale 1/75



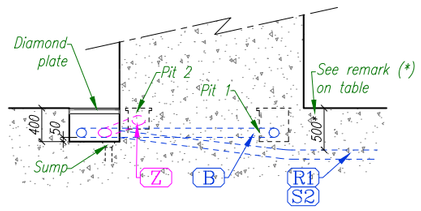
Section AA
Scale 1/75



Section BB
Scale 1/25



Section CC
Scale 1/25



Section DD
Scale 1/75

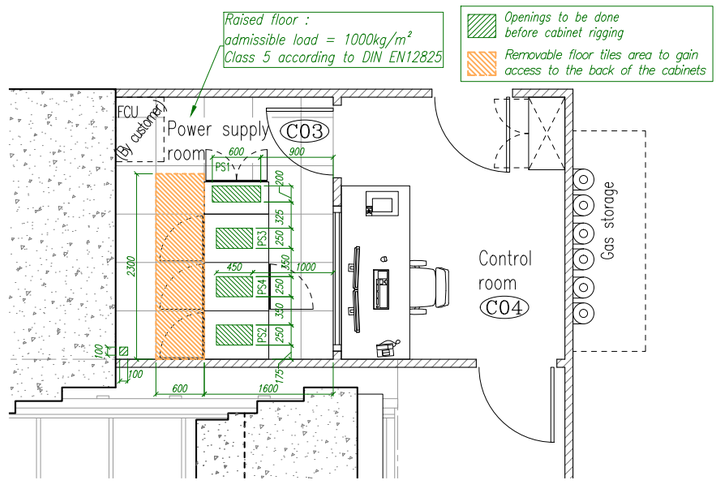
No.	Duct size (DN)	From	To	Use
A1	2xø160	Cyclotron Vault	Power Supply Room	Electrical cables
A2	2xø110	Cyclotron Vault	Power Supply Room	Electrical cables
B	1xø160	Cyclotron Vault	Pit1 (Technical room)	Cooling water connections
C	1xø110	Power Supply Room	Pit1 (Technical room)	Electrical cables
D	1xø80	Power Supply Room	Control room	Electrical cables
E	1xø110	Cyclotron Vault	Power Supply Room	Syringe driver connection
R1	2xø110*	Cyclotron Vault	To each Hotcell group	Radio-isotope transfer lines
S1	1xø160*	Cyclotron Vault	To Hotcell (Solid target)	Solid target transfer lines
S2	1xø110*	Cyclotron Vault	To Hotcell (Solid target)	Solid target blower pipe

Only for IBA equipments

X1	**	Cyclotron Vault	HVAC system	HVAC duct IN
X2	**	Cyclotron Vault	HVAC system	HVAC duct OUT
Y	**	Cyclotron Vault	Power Supply Room	Customer Electrical cables (sockets, lighting, safety, probe,...)
Z	**	Cyclotron Vault	Pit2 (Cyclotron lobby)	Gas ducts

Under Customer responsibility

	Pull cords are required in every PVC ducts.
	Conduit ducts shape shall have a minimum radius of 10x the inner diameter (excepted specific information mentioned)
*	• Radiation protection rules requires transfer lines embedded into : concrete (recommended 500mm concrete : 2350kg/m³) or lead (recommended 63 mm lead) in all directions. • It is the customer's responsibility to define the shielding, compliant with local radiation safety regulations
**	Notes: These connections are indicative only. It is Customer responsibility to calculate and validate, dimension, position. This list is not exhaustive.



Cut-out in raised floor
Scale 1/75

0 1 2 3meter
0 5' 10'

S.A. Ion Beam Applications

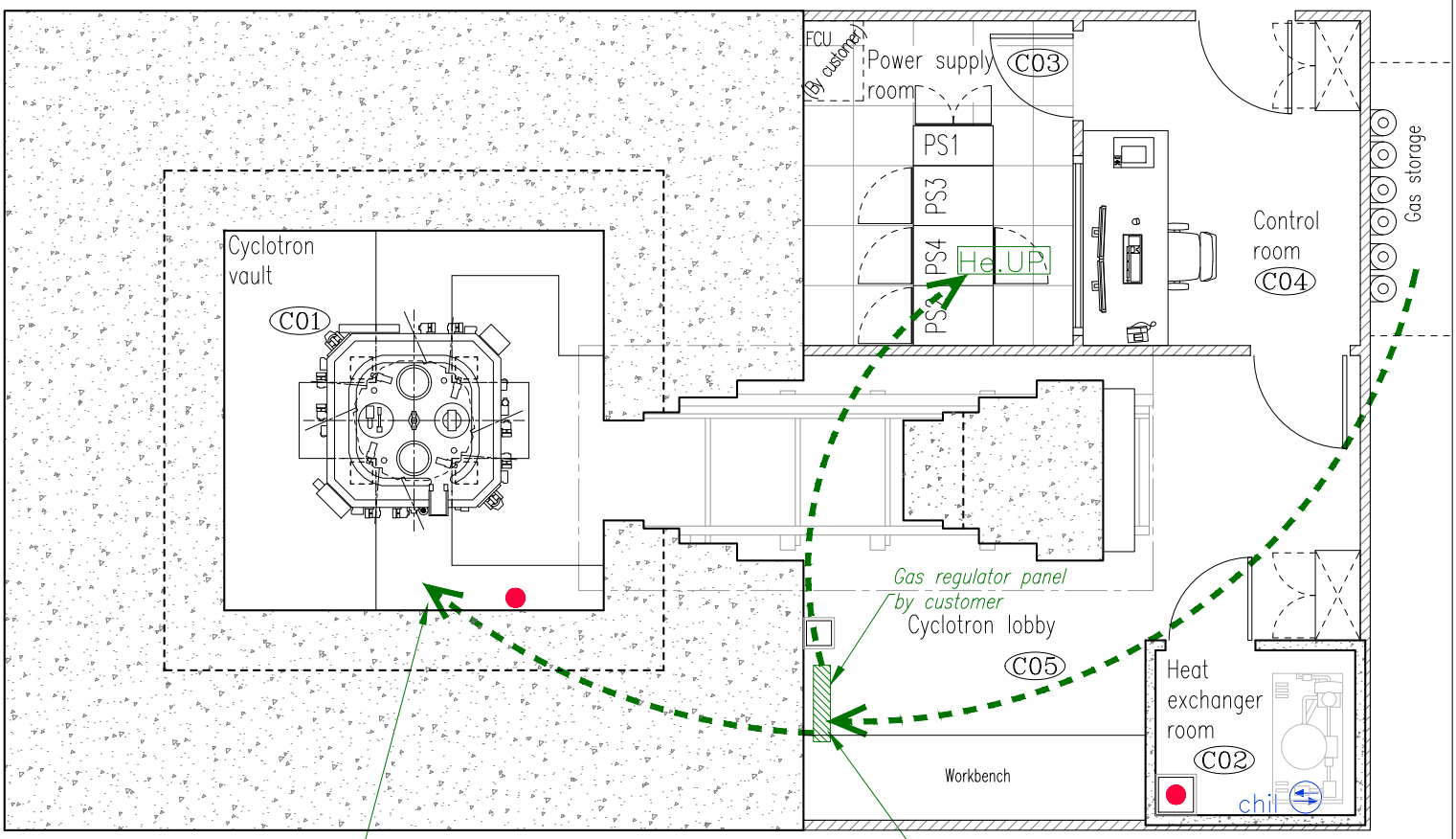
Cyclone® KIUBE

Contents:
Trenches and connections

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Drawing SRG-0010
Ref.: 07.46.00.0010-G

Dimensions: mm
Scale: 1/75 & 1/25



Gas type and purity	Dry Nitrogen (99.9%) —Optional— D.N2	Helium ultra pure (99.9999%) He.UP	Helium pure (99.8%) He.p	Hydrogen (99.9999%) H2	KIUBE 300 Reserve gas line (optional) Res.
Piping diameter	OD 6mm (¼") ID 4mm (0.170")	OD ⅛" ID 1/16"	OD 6mm (¼") ID 4mm (0.170")	OD 1/16" ID 0.5mm	OD ⅛" ID 1/16"
Piping material	SS316 or OFC	SS316 or OFC	SS316 or OFC	SS316	SS316
Bottle type	B50	B50	B50	B20	B50
Pressure range (bar)	0–10	0–10	0–1.5	0–3	0–3
Pressure use (bar)	1–5	~2	~0.5	~2	~2

Compressed air C.air	
Recommended air class	ISO8573.1 class 1–4–1
Remaining dirt particle size	<0.1 µm
Dew–point	3°C
Remaining oil content	<0.01 mg/m³
Pressure	7 bar @ connection point
Connection point to Cyclone® interface	12mm OD plastic connection push–on
	Average flow: < 2 l/min peak: 70 l/min @ 7bar (0.490 Nm³/min)

Gases: only for Cyclone® operation
Refer to SRG for specific targets gases related to project scope

→ Gas connection schematic route

chil	Chilled water IN and OUT			
	Flow	100l/min	Power to be removed	55kW
	Temperature	6–16°C	Pressure drop in the heat exchanger	1 bar
	ΔT	10–15°C	Pressure max.	8 bar
	+ Flow IN display			
	+ Temp IN & OUT display			
●	Accidental liquid waste connection point			

0 1 2 3meter
0 5' 10'

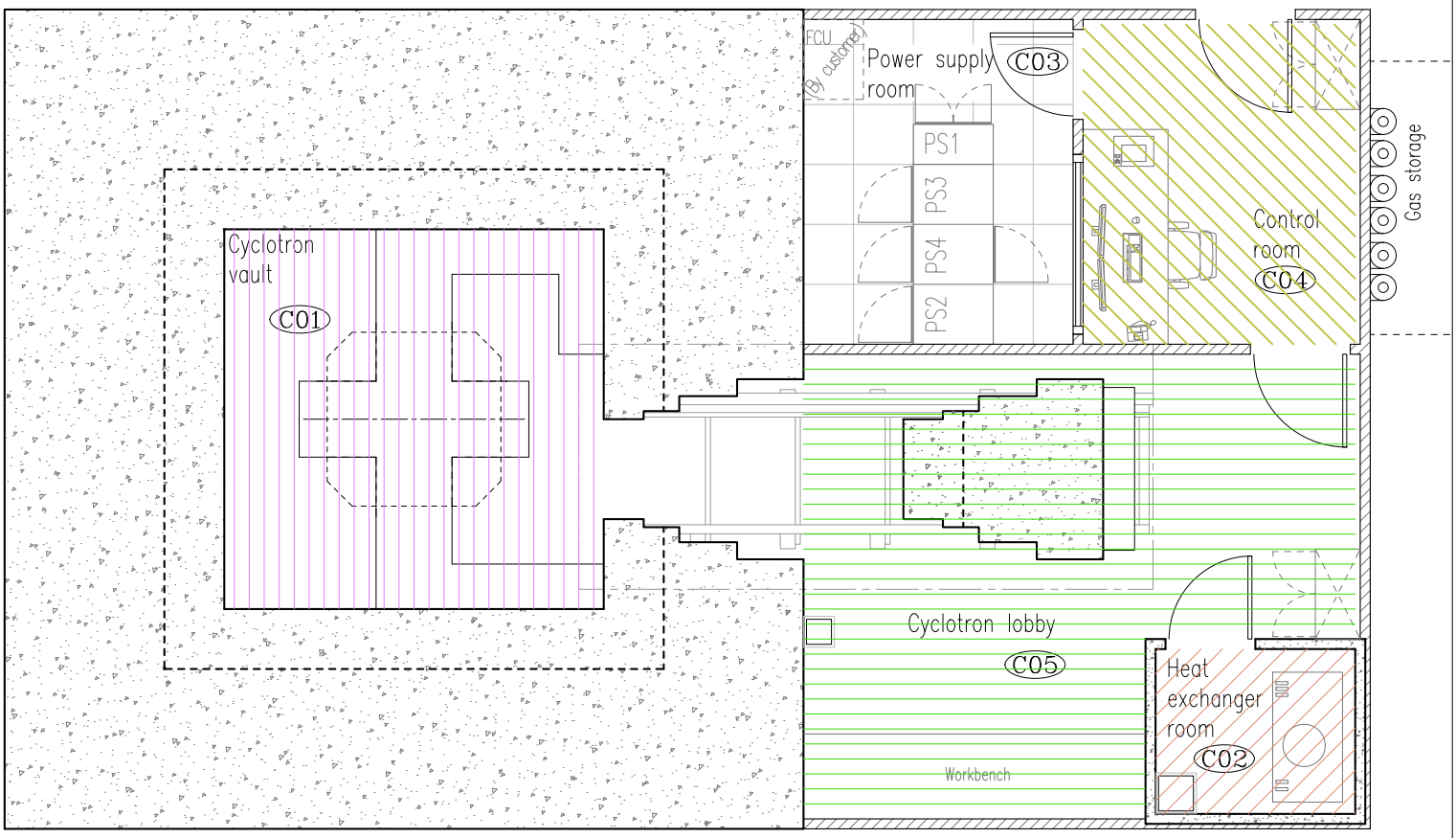
S.A. Ion Beam Applications

Cyclone® KIUBE

Contents:
Utilities and Sewage

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Drawing SRG–0020 | Dimensions: mm
Ref.: 07.46.00.0020–G | Scale: 1/75



Legend

- 0Pa
- −15Pa
- −30Pa
- −60Pa
- No requirement

based on:
- IAEA TRS 471

0 1 2 3meter
0 5' 10'

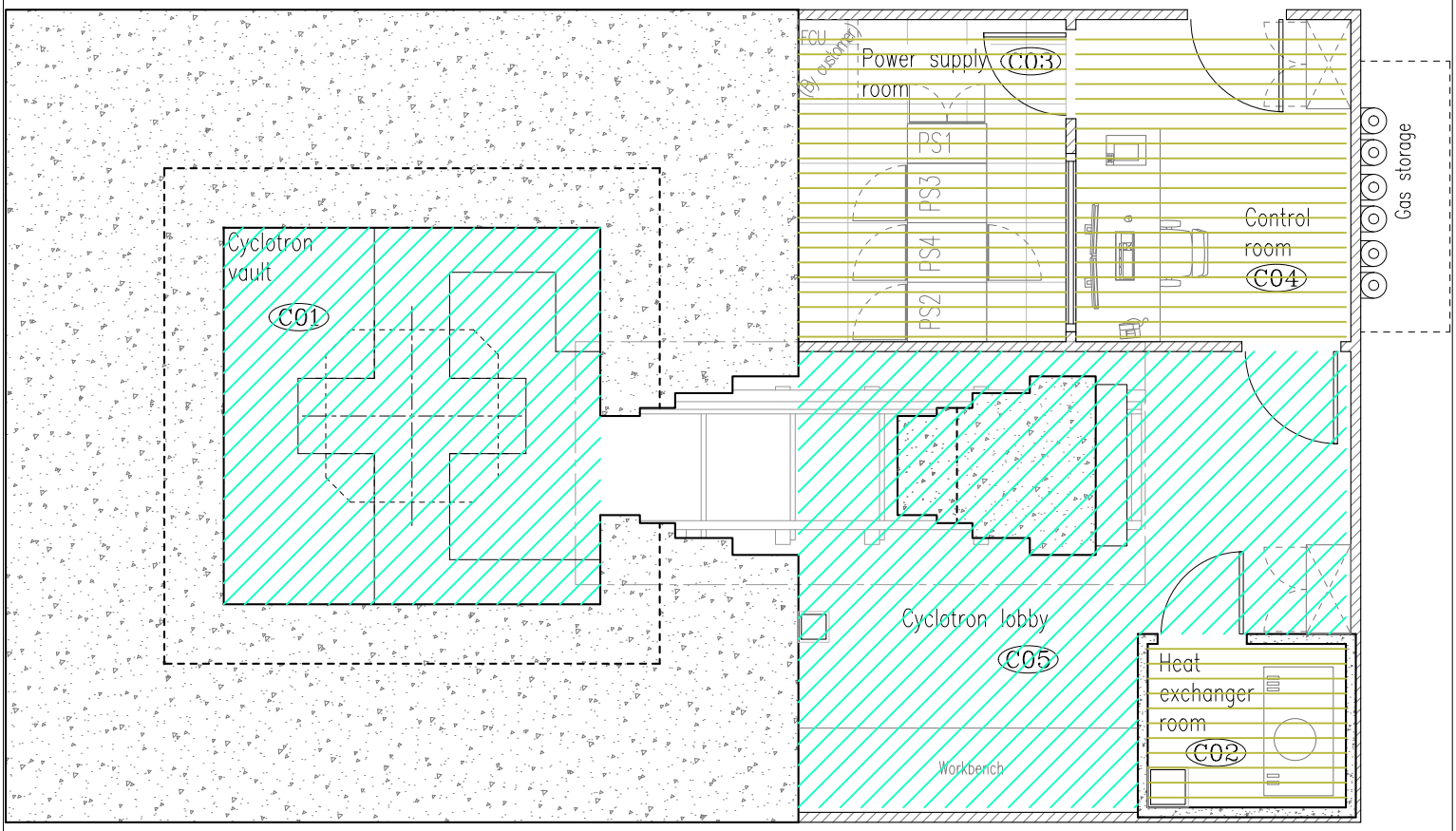
S.A. Ion Beam Applications

Cyclone® KIUBE

Contents:
Recommended pressure cascade

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Drawing SRG-0030
Ref.: 07.46.00.0030-G
Dimensions: mm
Scale: 1/75



Legend

Minimum Air change per hour

5x

10x

based on:
- IAEA TRS 471

0 1 2 3meter

0 5' 10'

S.A. Ion Beam Applications

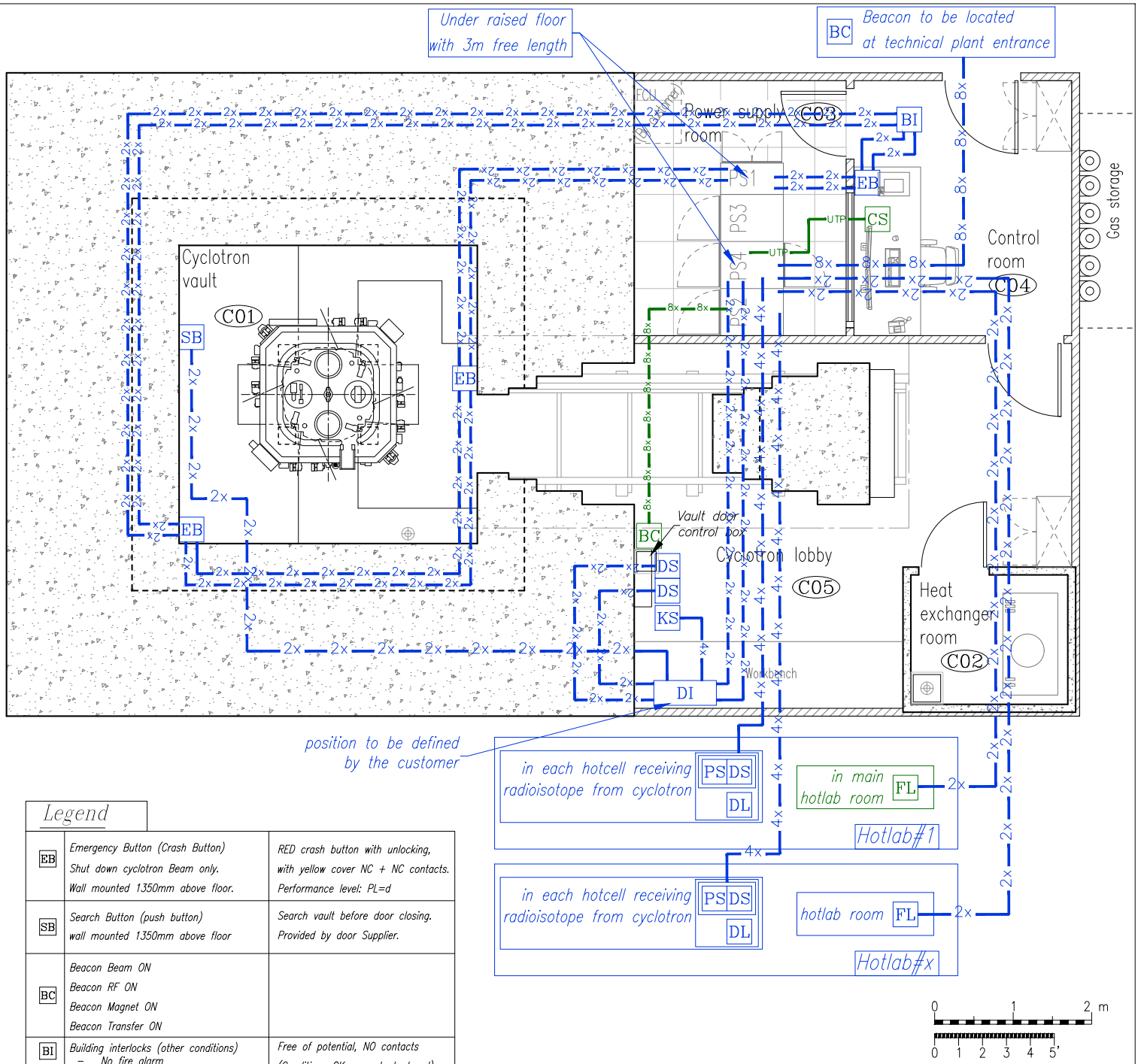
Cyclone® KIUBE

Contents:
Minimum recommended air change rate

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Drawing SRG-0035 | Dimensions: mm

Ref.: 07.46.00.0035-G | Scale: 1/75



Legend

EB	Emergency Button (Crash Button) Shut down cyclotron Beam only. Wall mounted 1350mm above floor.	RED crash button with unlocking, with yellow cover NC + NC contacts. Performance level: PL=d
SB	Search Button (push button) wall mounted 1350mm above floor	Search vault before door closing. Provided by door Supplier.
BC	Beacon Beam ON Beacon RF ON Beacon Magnet ON Beacon Transfer ON	
BI	Building interlocks (other conditions) – No fire alarm – Vault ventilation OK – additional EB in series – + other conditions to Customer request	Free of potential, NO contacts (Conditions OK = contact closed) Performance level: PL=d Provided by Customer
CS	Cyclotron control station	Provided by IBA (To 3xRJ45 Plugs)
FL	Flash light (Transfer ON)	24 Vdc BLUE 500mA
DL	Hot cell door signal Disable door opening during transfer	Free of potential NC contact (disable = contact open) Provided by Equipment supplier
DS	Door switch hotcell and vault door	Free of potential, NC contacts (door closed = contact closed) Performance level: PL=d Provided by Equipment supplier
PS	Pressure switch hotcell	Free of potential, NC contacts (pressure OK = contact closed) Performance level: PL=d Provided by Equipment supplier
DI	Door Interlock – SB = OK – Vault door close = OK	Free of potential, NO contacts (door closed = contact closed) Provided by Equipment supplier
KS	Key switch (optional)	Free of potential, NO contacts Mono stable key switch
2x	LIYY , 2 x 0.5 mm², pour signal 24 Vdc et contacts redondants	
4x	LIYY , 4 x 0.5 mm², pour signal 24 Vdc et contacts redondants	
8x	LIYY , 8 x 0.5 mm², pour signal 24 Vdc et contacts redondants	
UTP	UTP Cat. 5	

Blue = Provided & installed by Customer
Green = Provided by IBA

Remark:
Customer has to take care of the interlock between the bunker door and the controller of the Gamma Detector in the cyclotron vault. The door cannot be opened if the dose rate in the vault is too high.

The position and the number of the safety devices are indicative. They need to be adapted to the local safety regulations and specific site operators procedures.

Customer must provide :
– Terminal for connections
– Proper labeling on cables
– Schematic wiring

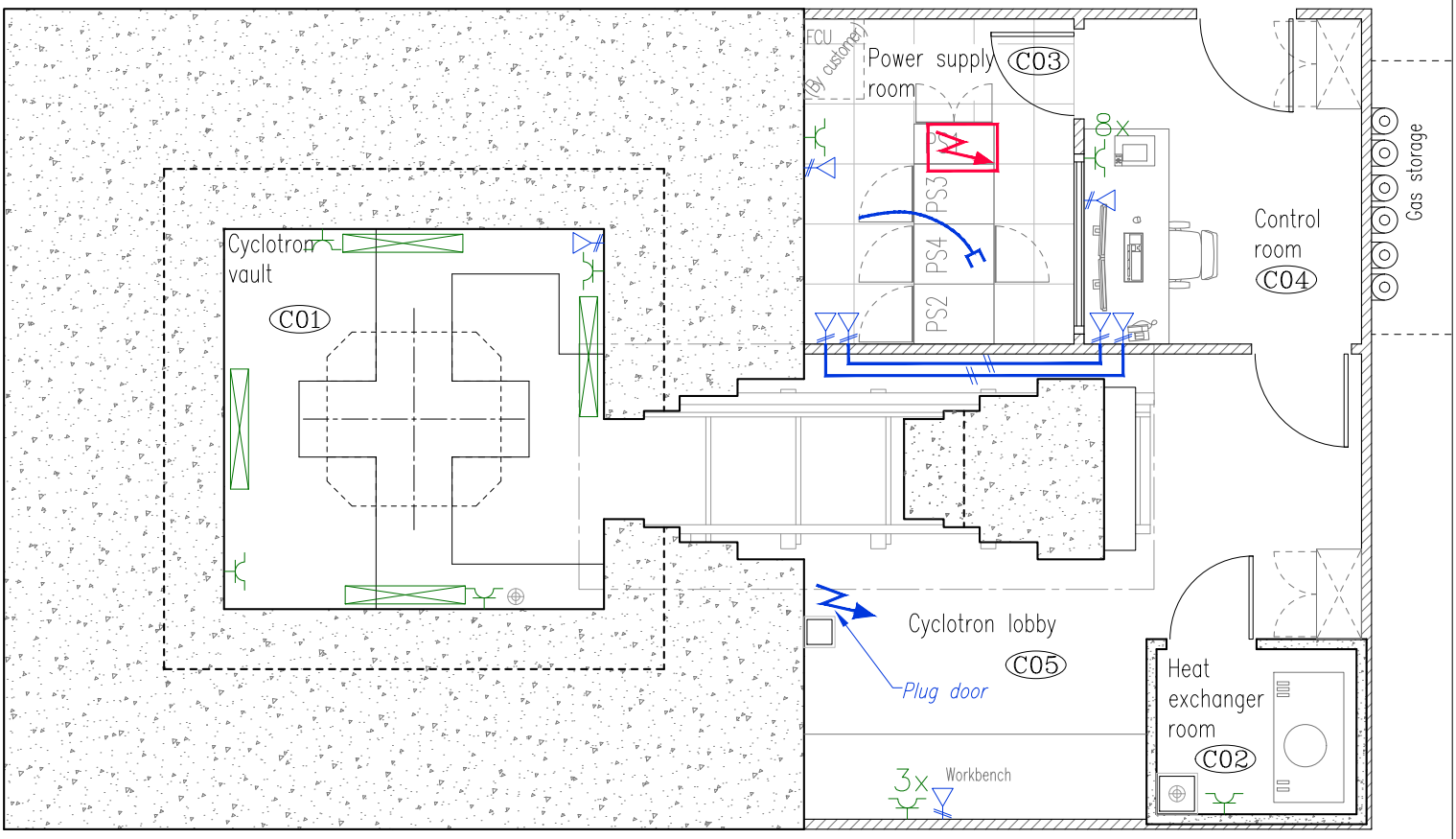
S.A. Ion Beam Applications

Cyclone® KIUBE

Contents:
Cyclotron interlocks & control system interface: minimum requirements

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Drawing SRG–0050 | Dimensions: mm
Ref.: 07.46.00.0050–G | Scale: 1/75



	Cyclotron feed (cable) – 3x400V (± 5%), AC, 50/60 Hz (± 1%) – 70 to 100 kVA : cyclotron alone (Cyclone®KIUBE 100/150/180/300) with transformer to supply 400V, 3 phases – 3 phases + neutral + ground PE (5wires) – Free cable length 3m under raised floor and PS1 cabinet – 15kVA on UPS recommended for vacuum + PLC Max. short-circuit : Isc max. ≤ 25kA rms
	Electrical feed (cable) – 230V – 16A – Arrive in false ceiling + 3m free cable length. – Following Buyer's door manufacturer requirement
	Cable UTP cat. 5-E + RJ45 connector to Internet network Available in the raised floor Free cable length : 3m under raised floor and PS4 cabinet
	Fluorescent light fixture (IP44) at 2000mm high

nx	Power outlet – Plug 230V, 50/60 Hz, 16A – Quantity minimum only for IBA equipment – Plug type E.
nx	Twin socket RJ45

Note: *
- All plugs near workbench/table shall be installed above the table.
- The plugs in Cyclotron vault are at 1000mm above the floor
- All others are at 180mm above floor

This layout shows only the requirements for specific equipment related to IBA scope.
→ To be completed by local engineering

0 1 2 3meter
0 5' 10'

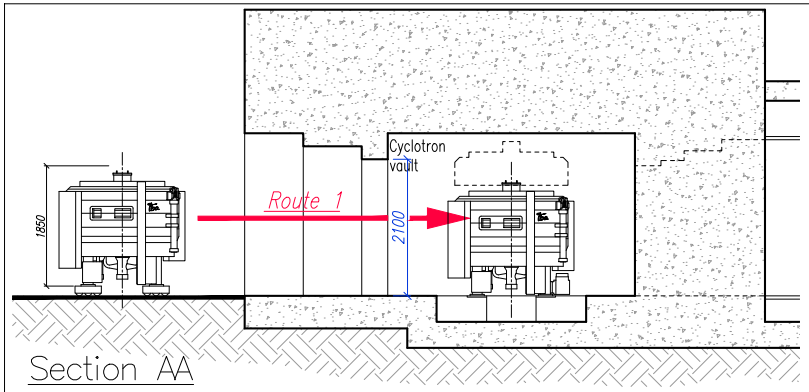
S.A. Ion Beam Applications

Cyclone® KIUBE

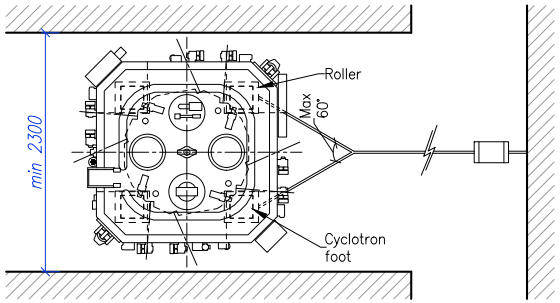
Contents:
Electricity & data

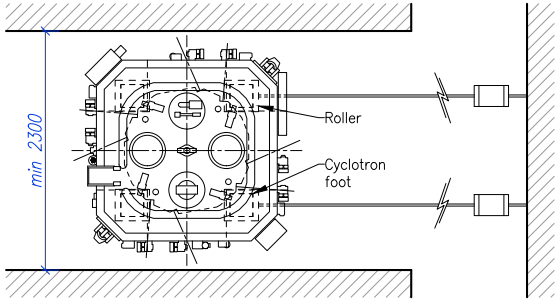
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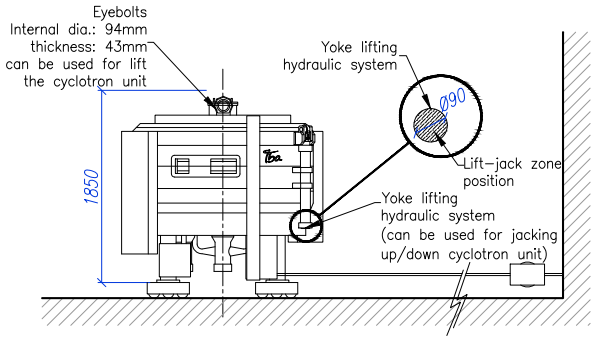
Drawing SRG-0070 | Dimensions: mm
Ref.: 07.46.00.0070-G | Scale: 1/75

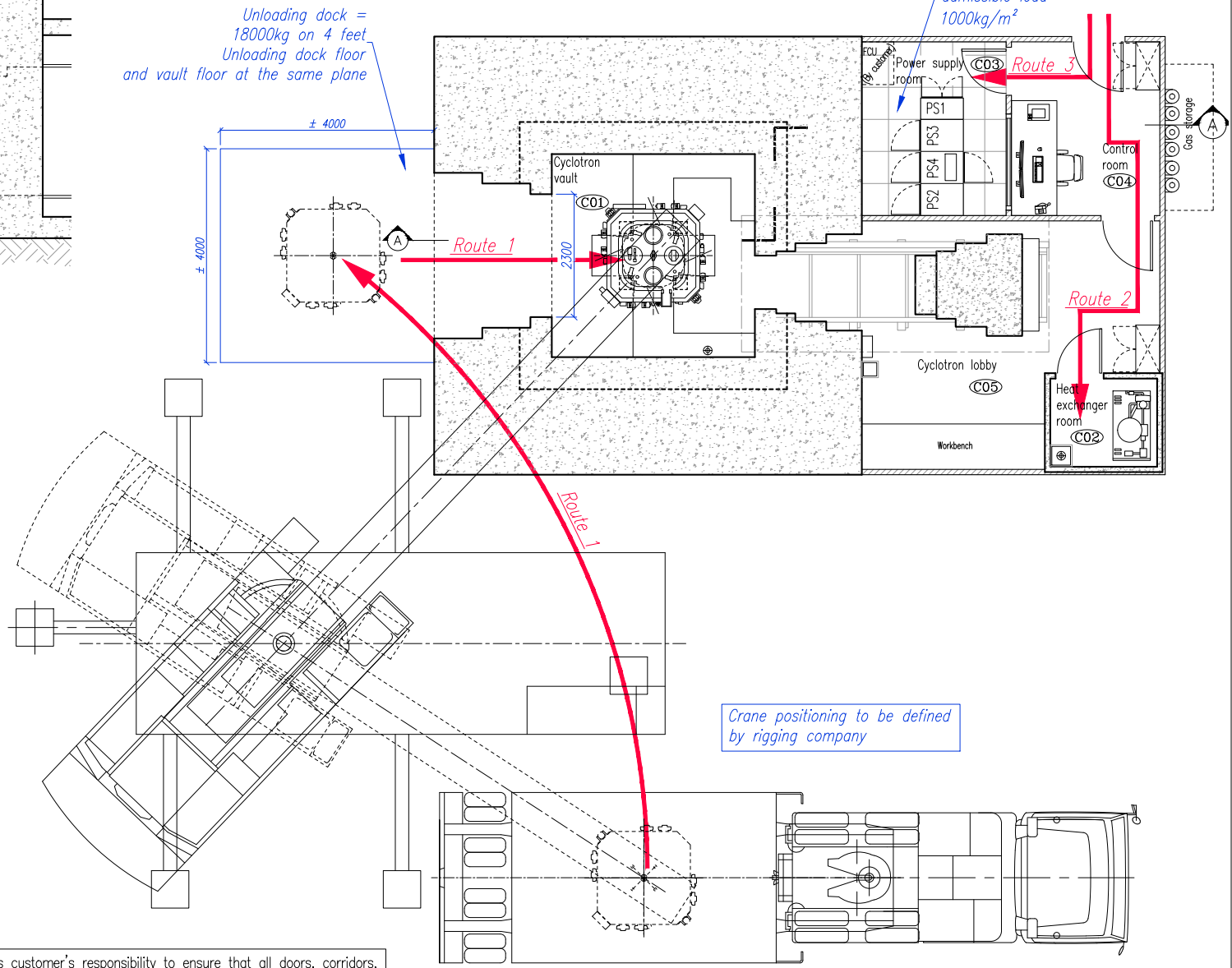


Section AA

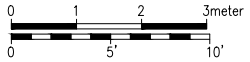








It is customer's responsibility to ensure that all doors, corridors, shafts,... allow biggest component's rigging. Clearances must be foreseen for biggest component (mentioned dimensions) plus handling tools. Ground must be flat, smooth, without any step and with appropriate strength.



Rigging route: biggest dimensions

Route	equip. type	equipment dimension L	W	H	Max.weight
1	Cyclotron	2050	2050	1850	18000kg
2	Water conditioner	1300	700	1610	220kg
3	Electrical cabinets	600	800	1880	600kg

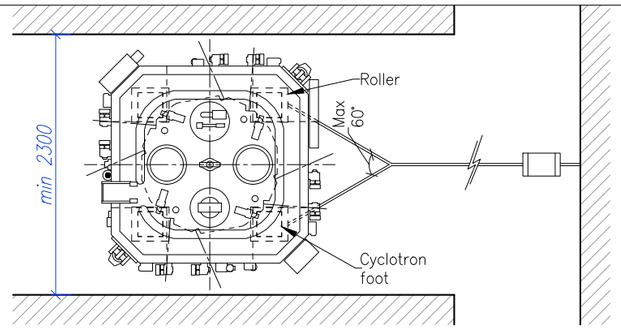
S.A. Ion Beam Applications

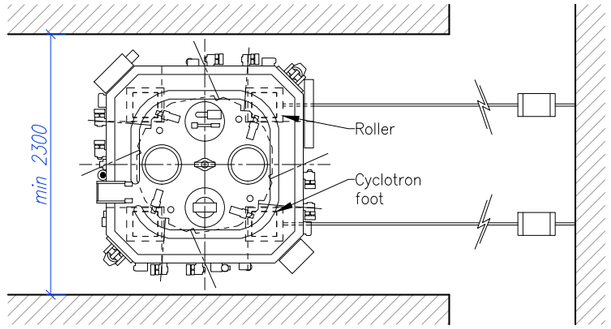
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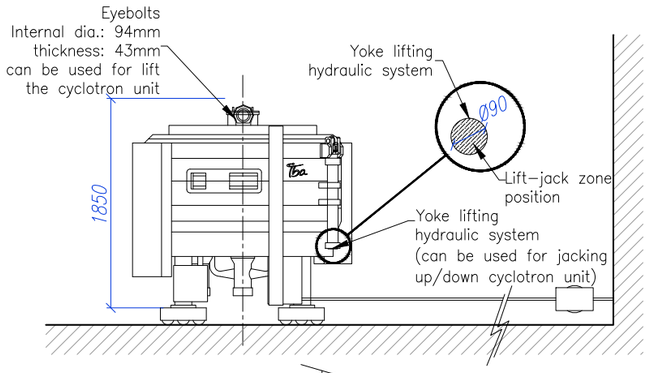
Cyclone® KIUBE

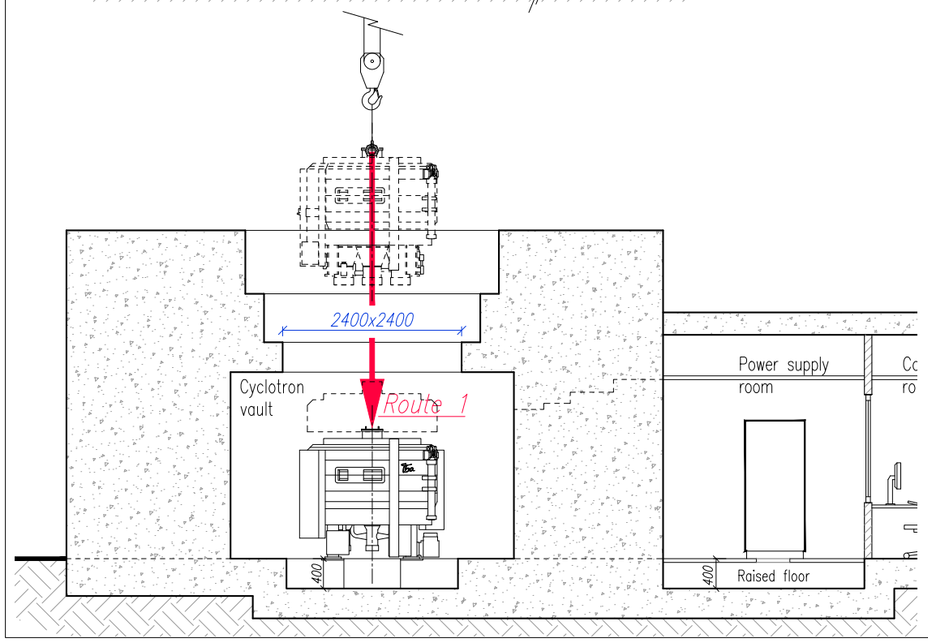
Contents:
Equipment dimensions & Rigging routes
Cyclotron horizontal rigging

Drawing SRG-0080 | Dimensions: mm
Ref.: 07.46.00.0080-G | Scale: –

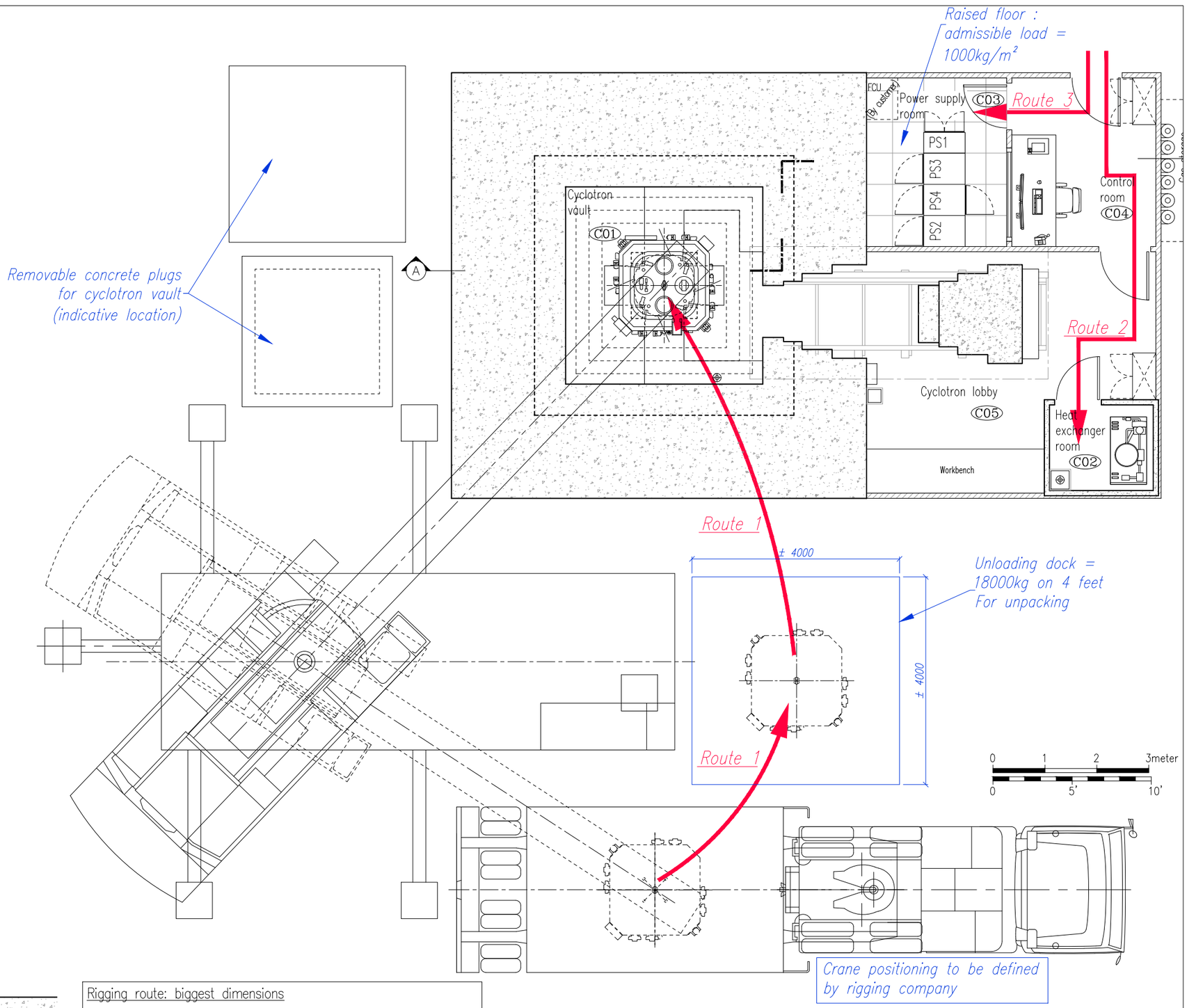








Section AA



Rigging route: biggest dimensions

Route	equip. type	equipment dimension L	W	H	Max.weight
1	Cyclotron	2050	2050	1850	18000kg
2	Water conditioner	1300	700	1610	220kg
3	Electrical cabinets	600	800	1880	600kg

It is customer's responsibility to ensure that all doors, corridors, shafts,... allow biggest component's rigging. Clearances must be foreseen for biggest component (mentioned dimensions) plus handling tools. Ground must be flat, smooth, without any step and with appropriate strength.

S.A. Ion Beam Applications

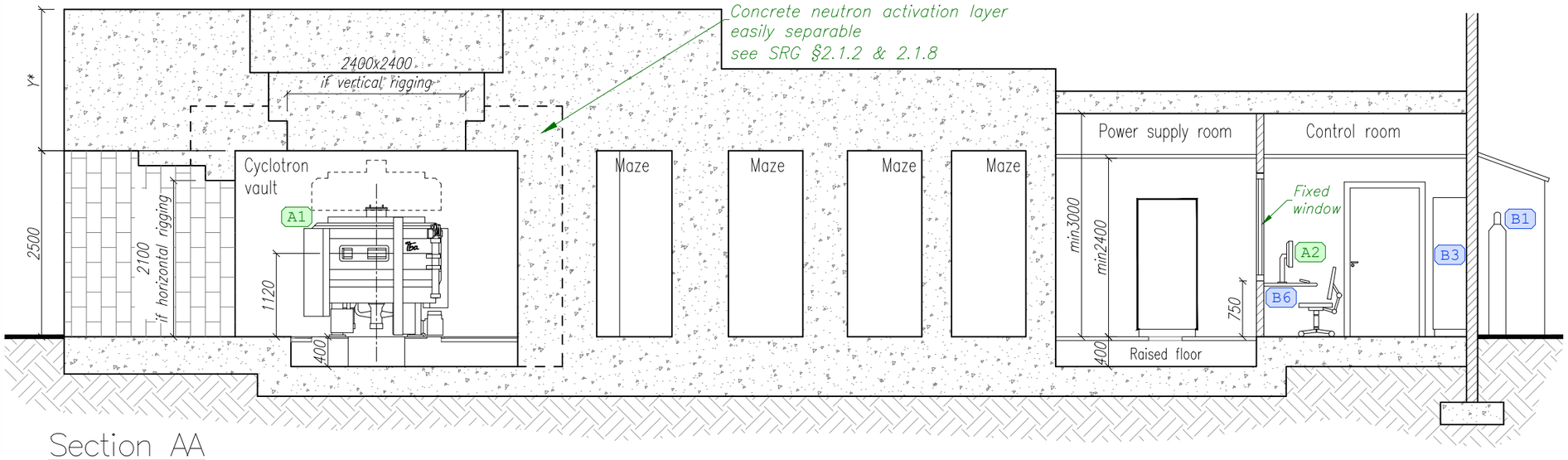
Cyclone® KIUBE

Contents:
Equipment dimensions &
Rigging routes
Cyclotron vertical rigging

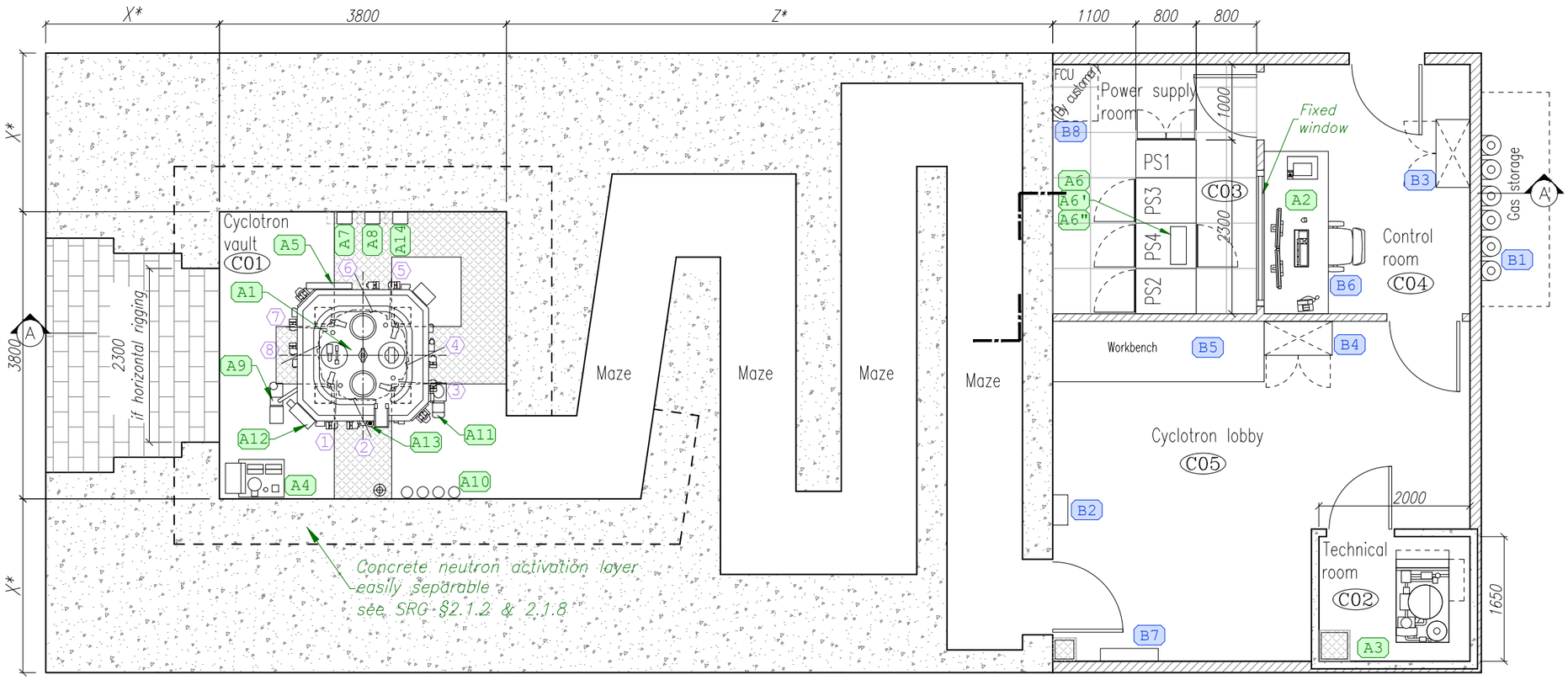
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Drawing SRG-0081 | Dimensions: mm

Ref.: 07.46.00.0081-G | Scale: –



Section AA



Cyclotron Equipment List		
Item	Name	Supplied/installed by
A1	Cyclone® KIUBE	IBA
A2	Control st. Thin Customer+printer+2 screens	IBA
A3	Water conditioner	IBA
A4	Hydraulic Yoke lifting system	IBA
A5	Water manifold	IBA
A6	18F Syringe driver cabinet (in PS4)	IBA
A6'	160 Rinsing Syringe driver cabinet (in PS4)	IBA (optional)
A6''	Other Syringe driver cabinet (13N/68Ga)	IBA (optional)
A7	Target switching input module (optional)	IBA (optional)
A8	Target switching output module (optional)	IBA (optional)
A9	Primary pump	IBA
A10	Safety pillars	IBA
A11	He compressor	IBA
A12	Plug to plug	IBA
A13	RF connection	IBA
A14	Target Gas distribution cabinet (optional)	IBA (optional)
PS1	Power Distribution Cabinet	IBA
PS2	R.F. Driver Amplifier Cabinet	IBA
PS3	Main Coils Ion Source P.S. Cabinet	IBA
PS4	Siemens P.L.C. Cabinet	IBA
B1	Gas cylinders supply	Customer
B2	Vault door control box	Customer
B3	Documentation storage cabinet	Customer
B4	Spare parts storage cabinet	Customer
B5	Workbench	Customer
B6	Desk + chair	Customer
B7	Cyclotron gas regulator panel	Customer
B8	Fan coil unit (optional)	Customer
1→8	Exit port number	IBA

X* = 2000 to 2400mm
Y* = 1700 to 2300mm

The shielding thickness shown on this layout is indicative only. It is the responsibility of the buyer to determine the appropriate level of shielding and to execute the construction to fulfill the applicable local legislation.

All displayed dimensions are minimal values.

0 1 2 3meter
0 5' 10'

S.A. Ion Beam Applications

Cyclone® KIUBE

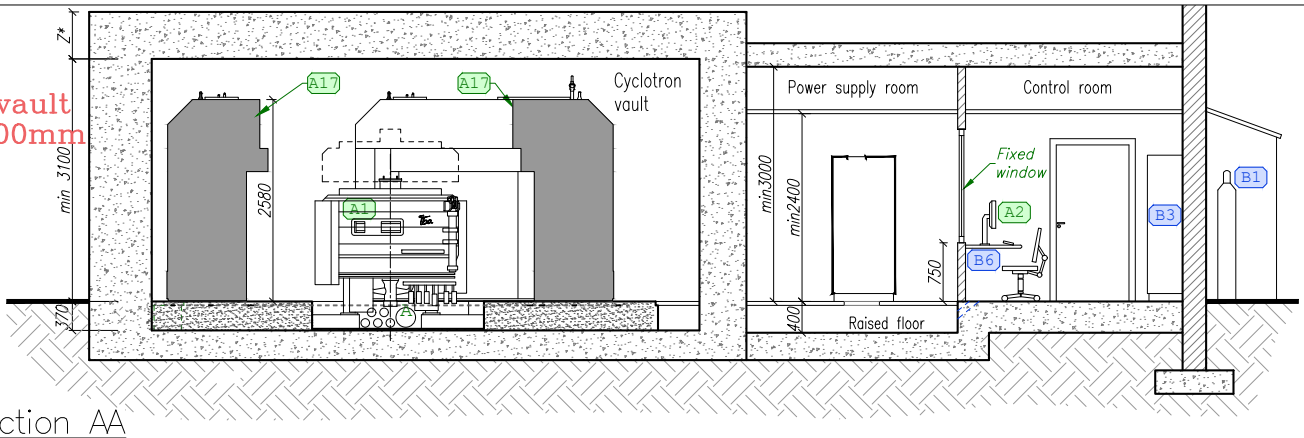
Contents:
Site configuration
With maze + regular door

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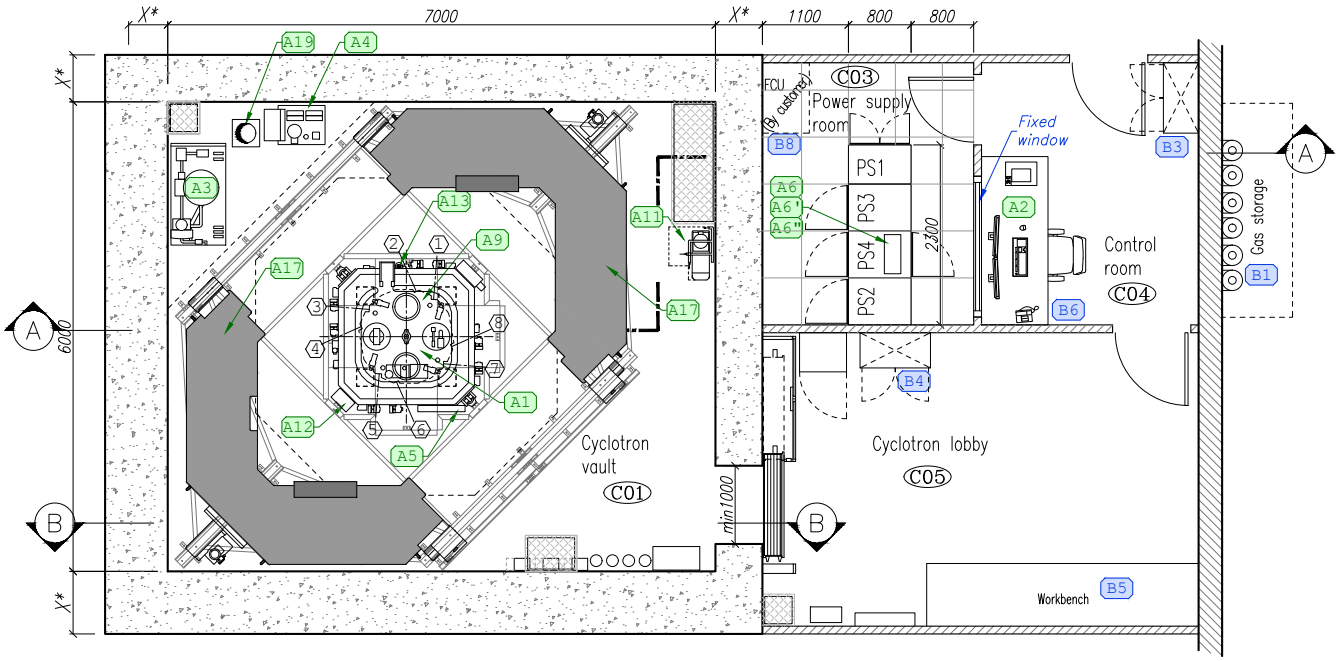
Drawing SRG-0200
Ref.: 07.46.00.0200-F

Dimensions: mm
Scale: 1/100

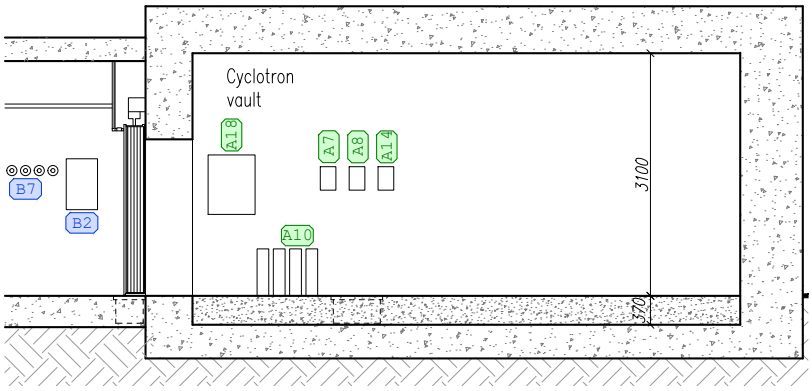
Cyclo vault
H= 3100mm



Section AA



Cyclotron Equipment List		
Item	Name	Supplied/installed by
A1	Cyclone® KIUBE	IBA
A2	Control st. Thin Customer+printer+2 screens	IBA
A3	Water conditioner	IBA
A4	Hydraulic Yoke lifting system	IBA
A5	Water manifold	IBA
A6	18F Syringe pumps cabinet (in PS4)	IBA
A6'	18O Rinsing Syringe pumps cabinet (in PS4)	IBA (optional)
A6''	Other Syringe pumps cabinet (13N/68Ga)	IBA (optional)
A7	Target switching input module (optional)	IBA (optional)
A8	Target switching output module (optional)	IBA (optional)
A9	Primary pump	IBA
A10	Safety pillars	IBA
A11	He compressor	IBA
A12	Plug to plug	IBA
A13	RF connection	IBA
A14	Target Gas distribution cabinet (optional)	IBA (optional)
A17	Self shielding	IBA
A18	Self shielding control box	IBA
A19	Self shielding Fan	IBA
PS1	Power Distribution Cabinet	IBA
PS2	R.F. Driver Amplifier Cabinet	IBA
PS3	Main Coils Ion Source P.S. Cabinet	IBA
PS4	Siemens P.L.C. Cabinet	IBA
B1	Gas cylinders supply	Customer
B2	Vault door control box	Customer
B3	Documentation storage cabinet	Customer
B4	Spare parts storage cabinet	Customer
B5	Workbench	Customer
B6	Desk + chair	Customer
B7	Cyclotron gas regulator panel	Customer
B8	Fan coil unit	Customer
1→8	Exit port number	IBA

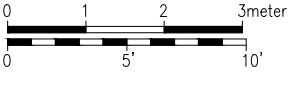


Section BB

All displayed dimensions are minimal values.

X* = 400 to 800mm
Z* = 400 to 600mm

The shielding thickness shown on this layout is indicative only. It is the responsibility of the buyer to determine the appropriate level of shielding and to execute the construction to fulfill the applicable local legislation.



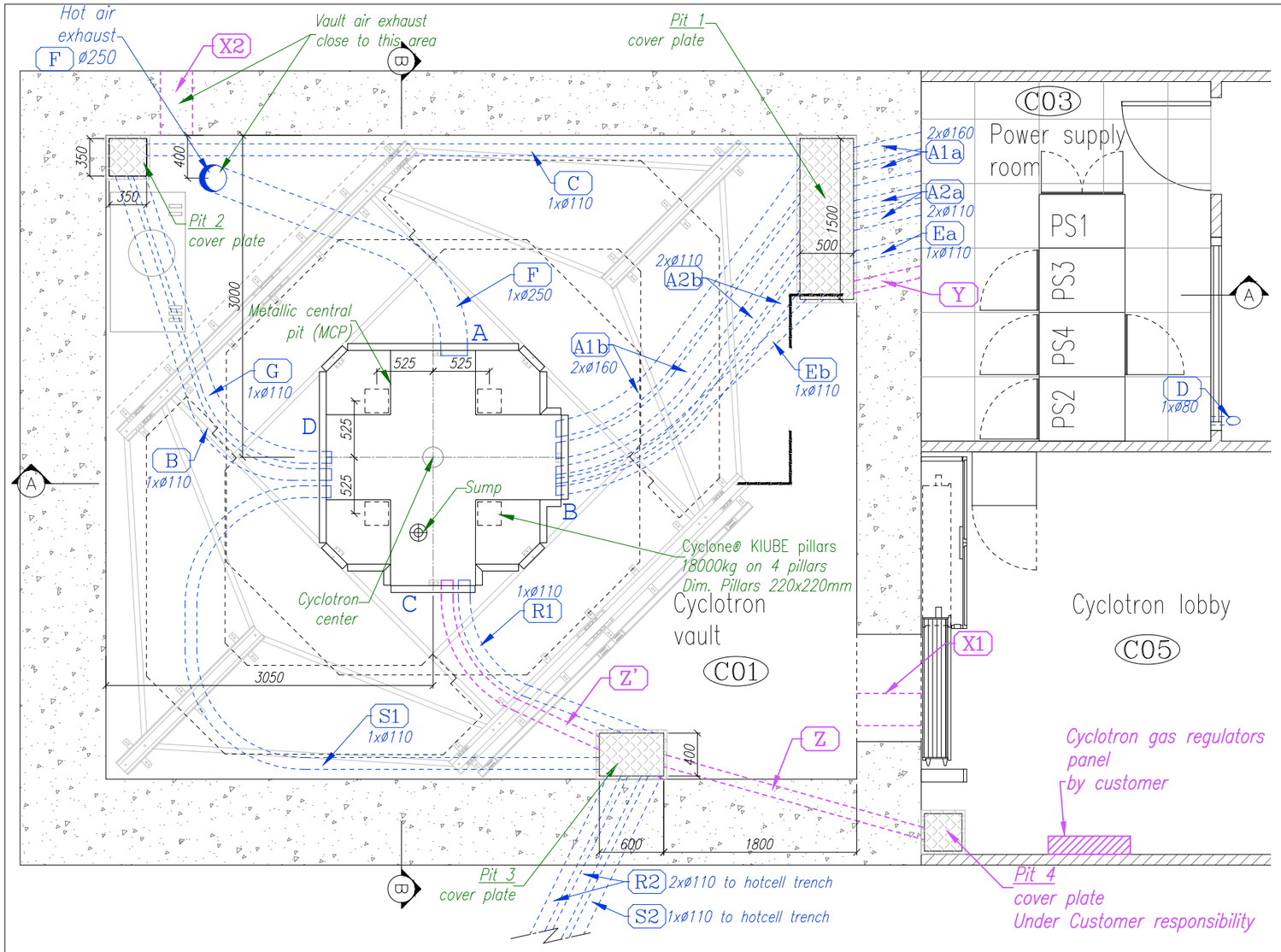
S.A. Ion Beam Applications

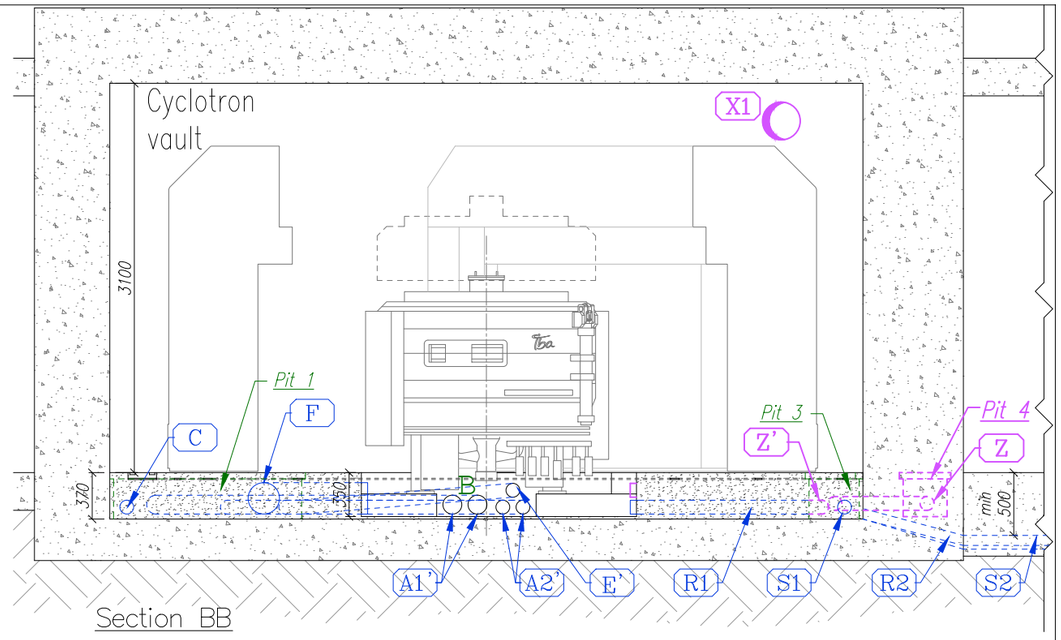
Cyclone® KIUBE

Contents:
Site configuration
Self-shielding

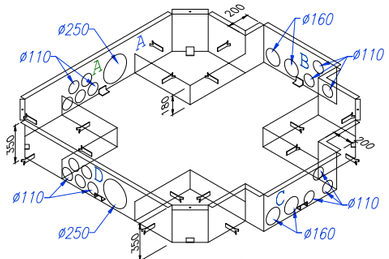
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Drawing SRG-0300
Ref.: 07.46.00.0300-G
Dimensions: mm
Scale: 1/100





Section BB



Metallic central pit
Isometric view

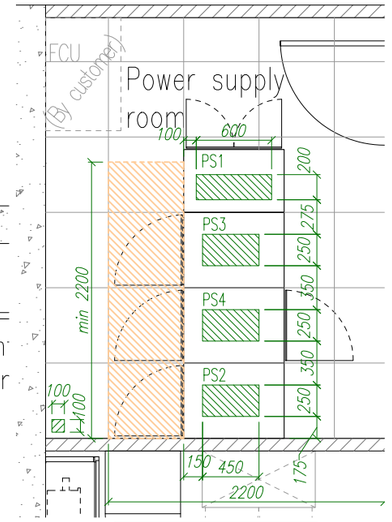
No.	Duct size (DN)	From	To	Use
A1a	2xø160	Cyclotron Vault (pit1)	Power Supply Room	Electrical cables
A1b	2xø160	Cyclotron Vault (pit1)	Metallic central pit	Electrical cables
A2a	2xø110	Cyclotron Vault (pit1)	Power Supply Room	Electrical cables
A2b	2xø110	Cyclotron Vault (pit1)	Metallic central pit	Electrical cables + He cooling
B	1xø110	Cyclotron Vault (pit2)	Metallic central pit	Cooling water connections
C	1xø110	Cyclotron Vault (pit1)	Cyclotron Vault (pit2)	Electrical cables
D	1xø80	Power Supply Room	Control room	Electrical cables
Ea	1xø110	Cyclotron Vault (pit1)	Power Supply Room	Syringe driver connection
Eb	1xø110	Cyclotron Vault (pit1)	Metallic central pit	Syringe driver connection
F	1xø250	Cyclotron Vault	Metallic central pit	Air (Galvanized steel)
G	1xø110	Cyclotron Vault (pit2)	Metallic central pit	Yoke lifting system
R1	1xø110	Cyclotron Vault (pit3)	Metallic central pit	Radio-isotope transfer lines
R2	2xø110*	Cyclotron Vault (pit3)	To Hotcell trench	Radio-isotope transfer lines
S1	1xø110	Cyclotron Vault (pit3)	Metallic central pit	Solid target transfer lines
S2	1xø110*	Cyclotron Vault (pit3)	To Hotcell (Solid target)	Solid target transfer lines

Only for IBA equipments

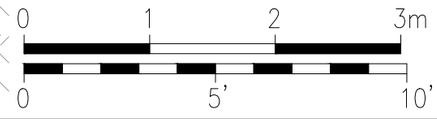
X1	**	Cyclotron Vault	HVAC system	HVAC duct IN
X2	**	Cyclotron Vault	HVAC system	HVAC duct OUT
Y	**	Cyclotron Vault (pit1)	Power Supply Room	Customer Electrical cables (sockets, lighting, safety, probe,...)
Z	**	Cyclotron Vault (pit3)	Cyclotron lobby (pit4)	Gas pipes
Z'	**	Cyclotron Vault (pit3)	Cyclone pit	Gas pipes

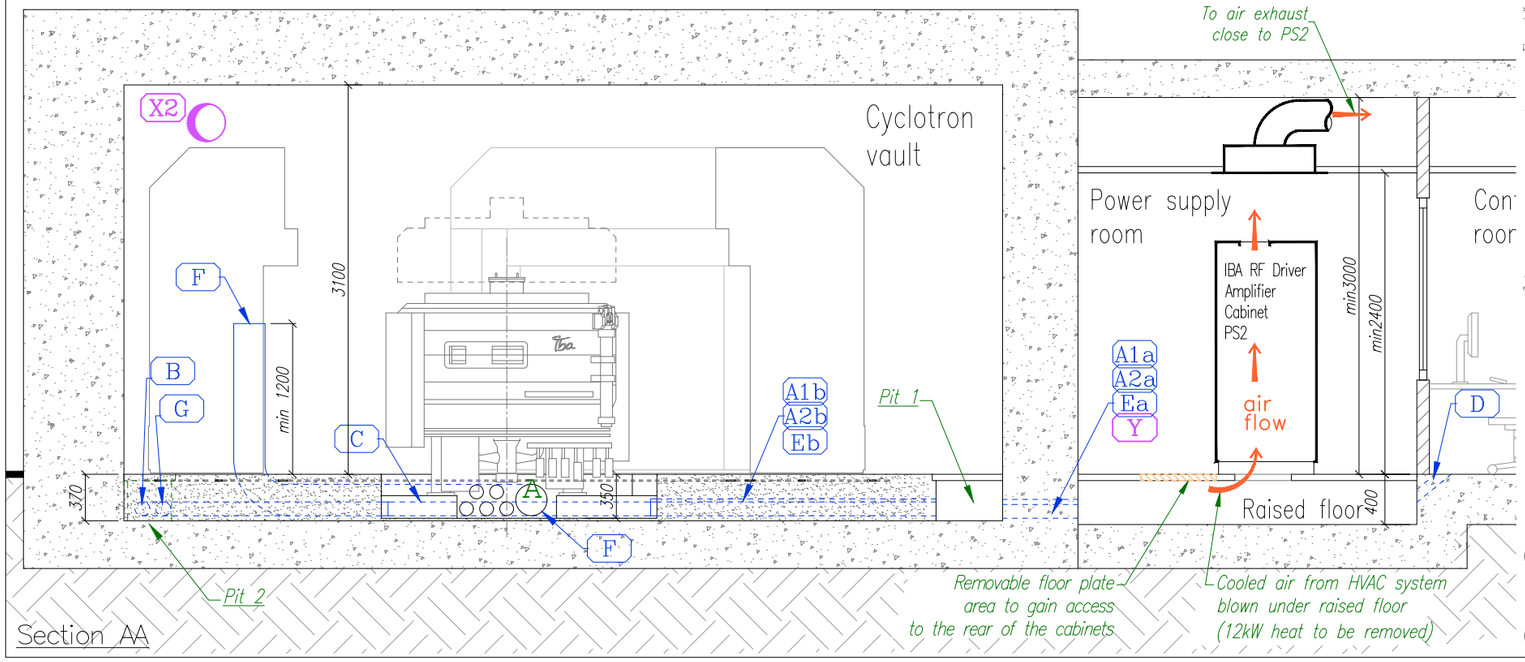
Under Customer responsibility

	Pull cords are required in every PVC ducts.
	Conduit ducts shape shall have a minimum radius of 10x the inner diameter (excepted specific information mentioned)
*	• Radiation protection rules requires transfer lines embedded into : concrete (recommended 400mm concrete : 2350kg/m³) or lead (recommended 50 mm lead) in all directions. • It is the customer's responsibility to define the shielding, compliant with local radiation safety regulations
**	**Notes:** **These connections are indicative only.** **It is Customer responsibility to calculate and validate, dimension, position.** **This list is not exhaustive.**



Cut-out in raised floor





Section AA

S.A. Ion Beam Applications

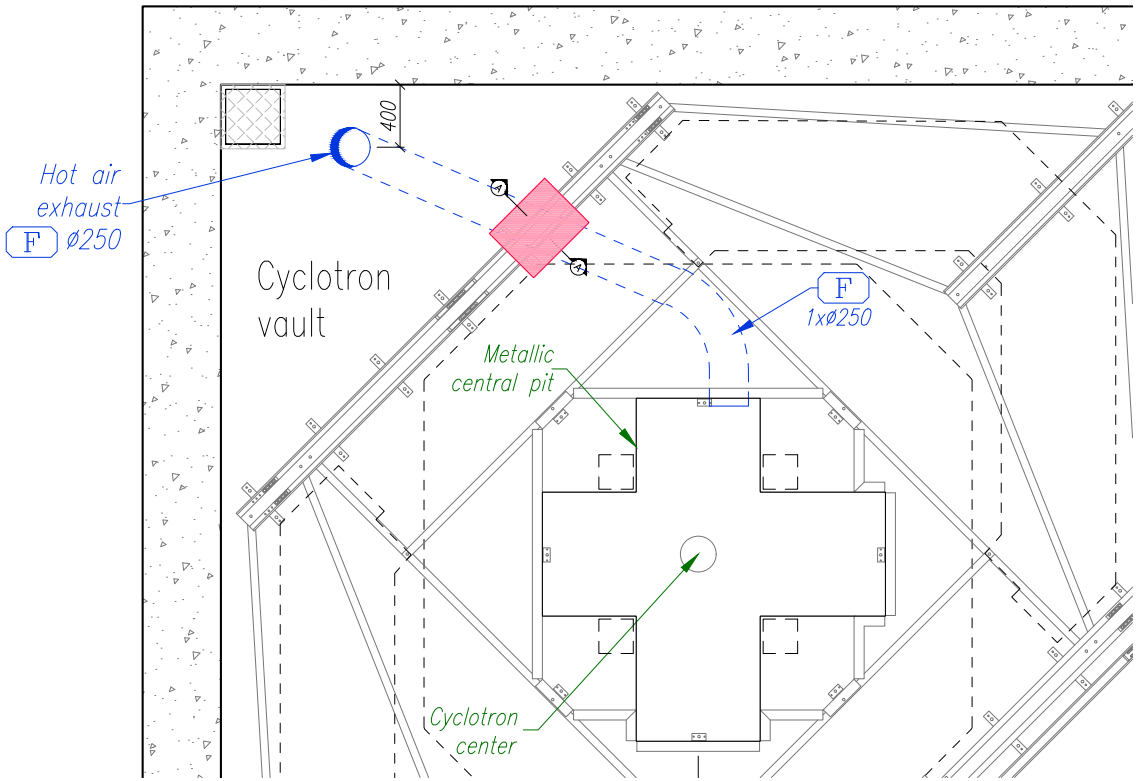
Cyclone® KIUBE

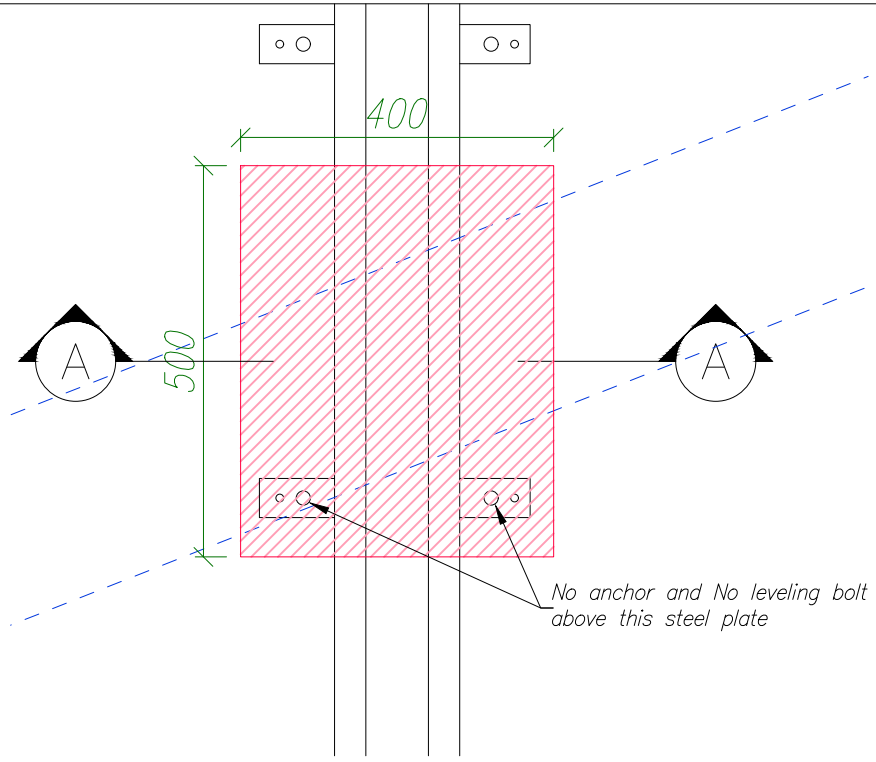
Contents:
Self-shielding
Trenches & connections
Utilities

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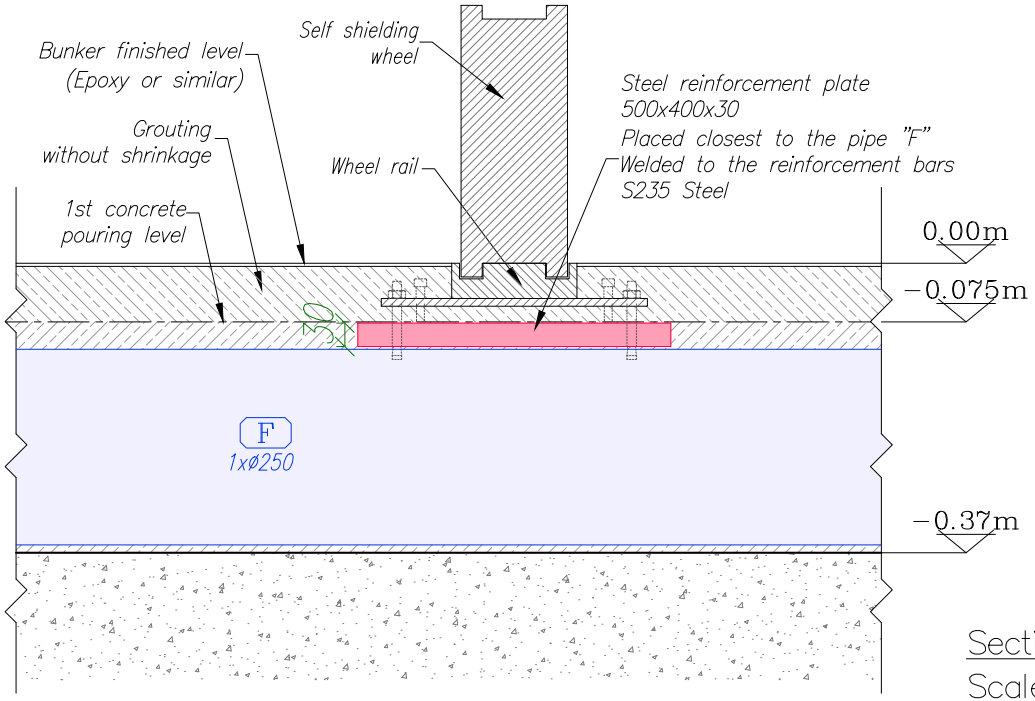
Drawing SRG-0310
Ref.: 07.46.00.0310-G

Dimensions: mm
Scale: 1/50





Detail
scale : 1/10



Section AA
Scale : 1/10

Note that this reinforcement is recommended if "F" pipe cross under the Self-sheilding rails

S.A. Ion Beam Applications

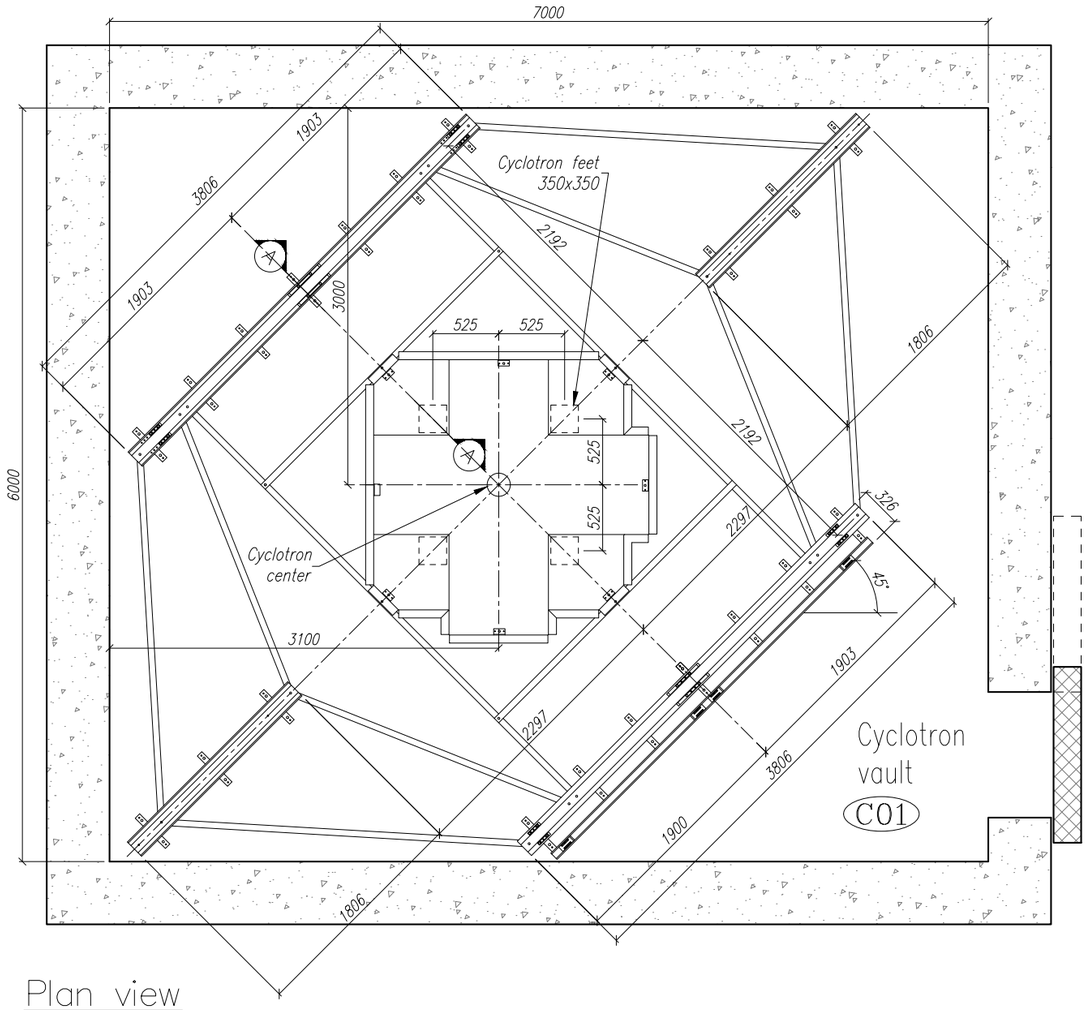
Cyclone® KIUBE

Contents:
Steel reinforcement plate

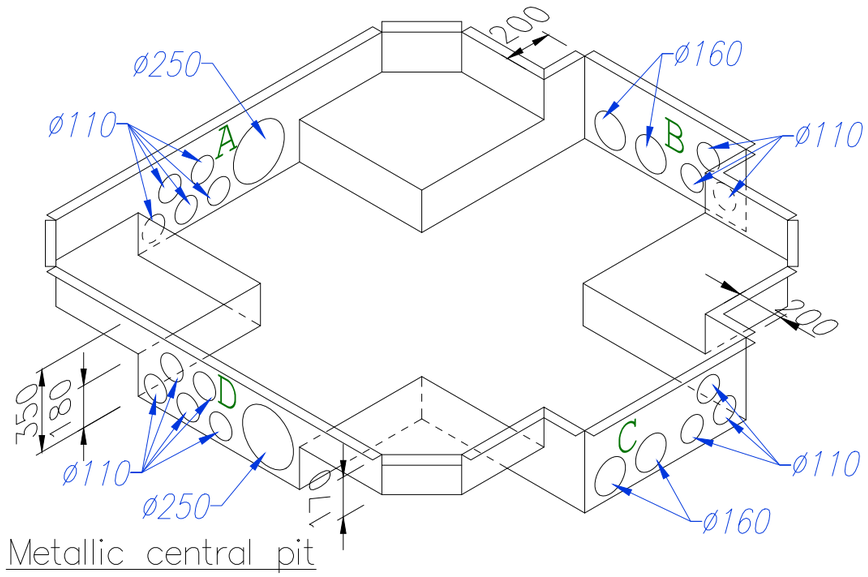
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Drawing SRG-0311
Ref.: 07.46.00.0311-G

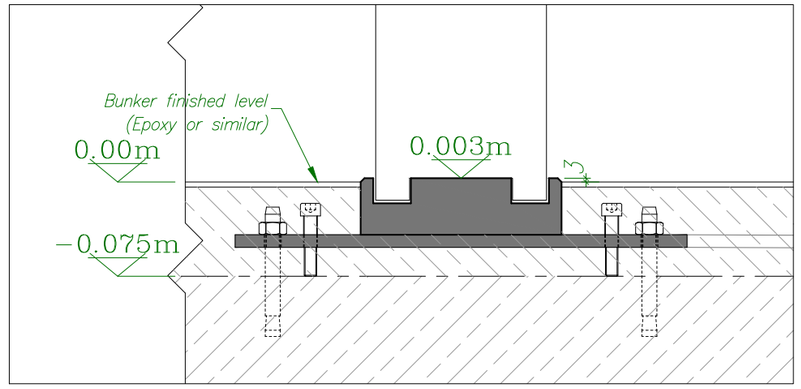
Dimensions: mm
Scale: 1/100-1/25



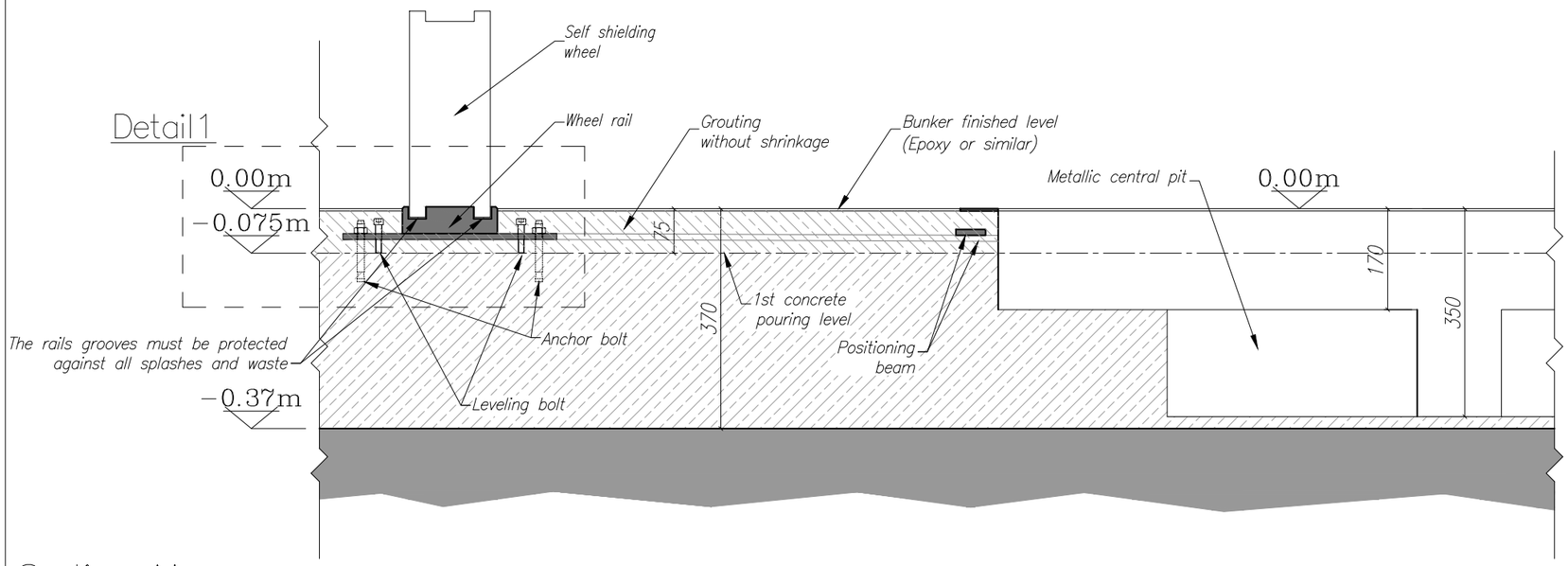
Plan view
scale: 1/50



Metallic central pit



Detail 1
scale: 1/5



Section AA
scale: 1/10

S.A. Ion Beam Applications

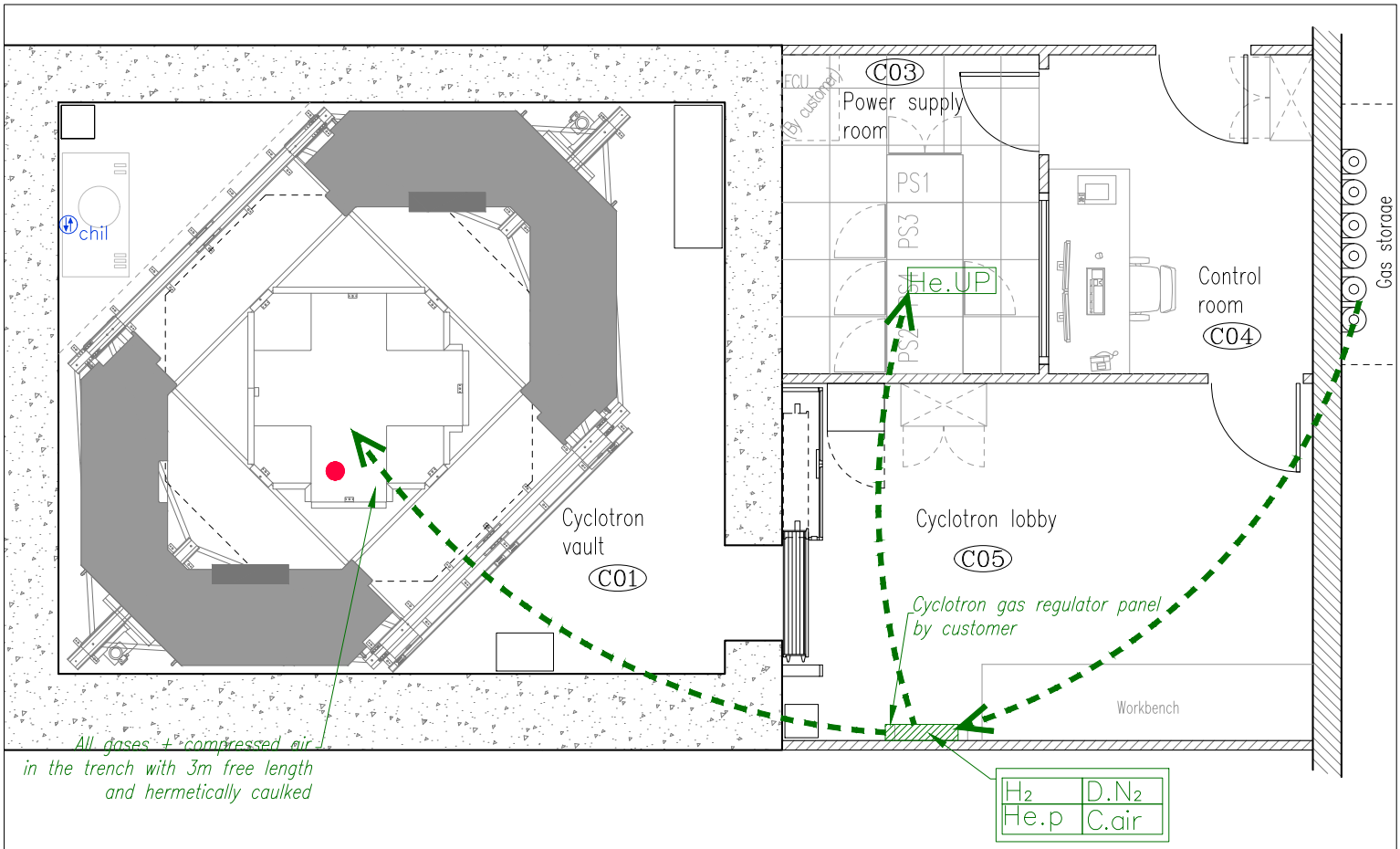
Cyclone® KIUBE

Contents:
Self−shielding
Installation set

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Drawing SRG−0315
Ref.: 07.46.00.0315−G

Dimensions: mm
Scale: 1/50−1/10−1/5



Gas type and purity	Dry Nitrogen (99.9%) –Optional– D.N_2	Helium ultra pure (99.9999%) He.UP	Helium pure (99.8%) He.p	Hydrogen (99.9999%) H_2
Piping diameter	OD 6mm (¼") ID 4mm (0.170")	OD ⅛" ID 1/16"	OD 6mm (¼") ID 4mm (0.170")	OD 1/16" ID 0.5mm
Piping material	SS316 or OFC	SS316 or OFC	SS316 or OFC	SS316
Bottle type	B50	B50	B50	B20
Pressure range (bar)	0–10	0–10	0–1.5	0–3
Pressure use (bar)	1–5	~2	~0.5	~2

Compressed air C.air	
Recommended air class	ISO8573.1 class 1–4–1
Remaining dirt particle size	<0.1 µm
Dew–point	3°C
Remaining oil content	<0.01 mg/m³
Pressure	7 bar @ connection point
Connection point to Cyclone® interface	12mm OD plastic connection push–on
	Average flow: < 2 l/min peak: 70 l/min @ 7bar (0.490 Nm³/min)

Gas connection schem

chil	Chilled water IN and OUT			
	Flow	100l/min	Power to be removed	55kW
	Temperature	6–16°C	Pressure drop in the heat exchanger	1 bar
	ΔT	10–15°C	Pressure max.	8 bar
	+ Flow IN display			
	+ Temp IN & OUT display			
●	Accidental liquid waste connection point			

0 1 2 3m
0 5' 10'

S.A. Ion Beam Applications

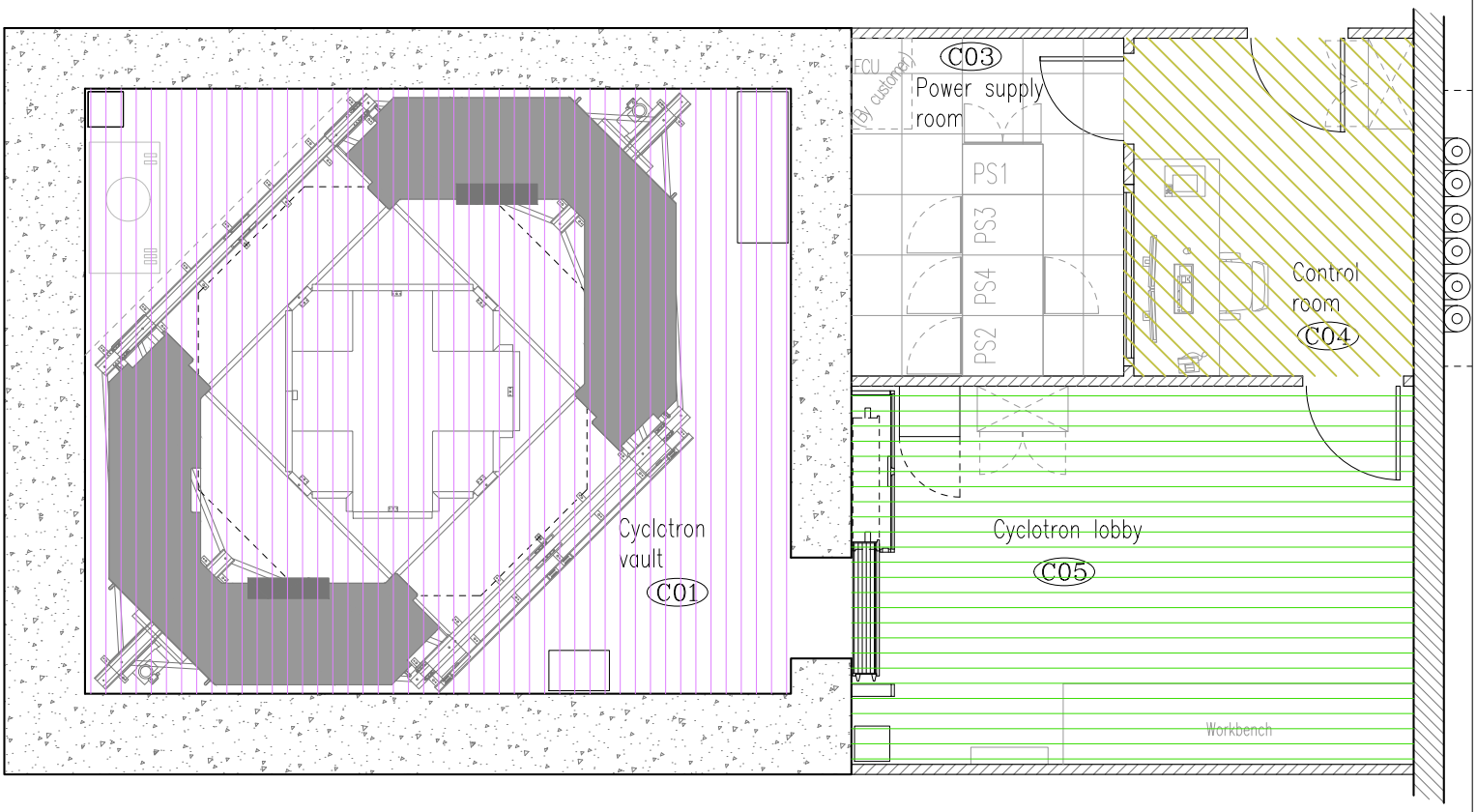
Cyclone® KIUBE

Contents: Self–shielding Utilities and Sewage

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Drawing SRG–0320 | Dimensions: mm

Ref.: 07.46.00.0320–G | Scale: 1/75



Legend

- 0Pa
- −15Pa
- −30Pa
- −60Pa
- No requirement

based on:
- IAEA TRS 471

0 1 2 3m
0 5' 10'

S.A. Ion Beam Applications

Cyclone® KIUBE

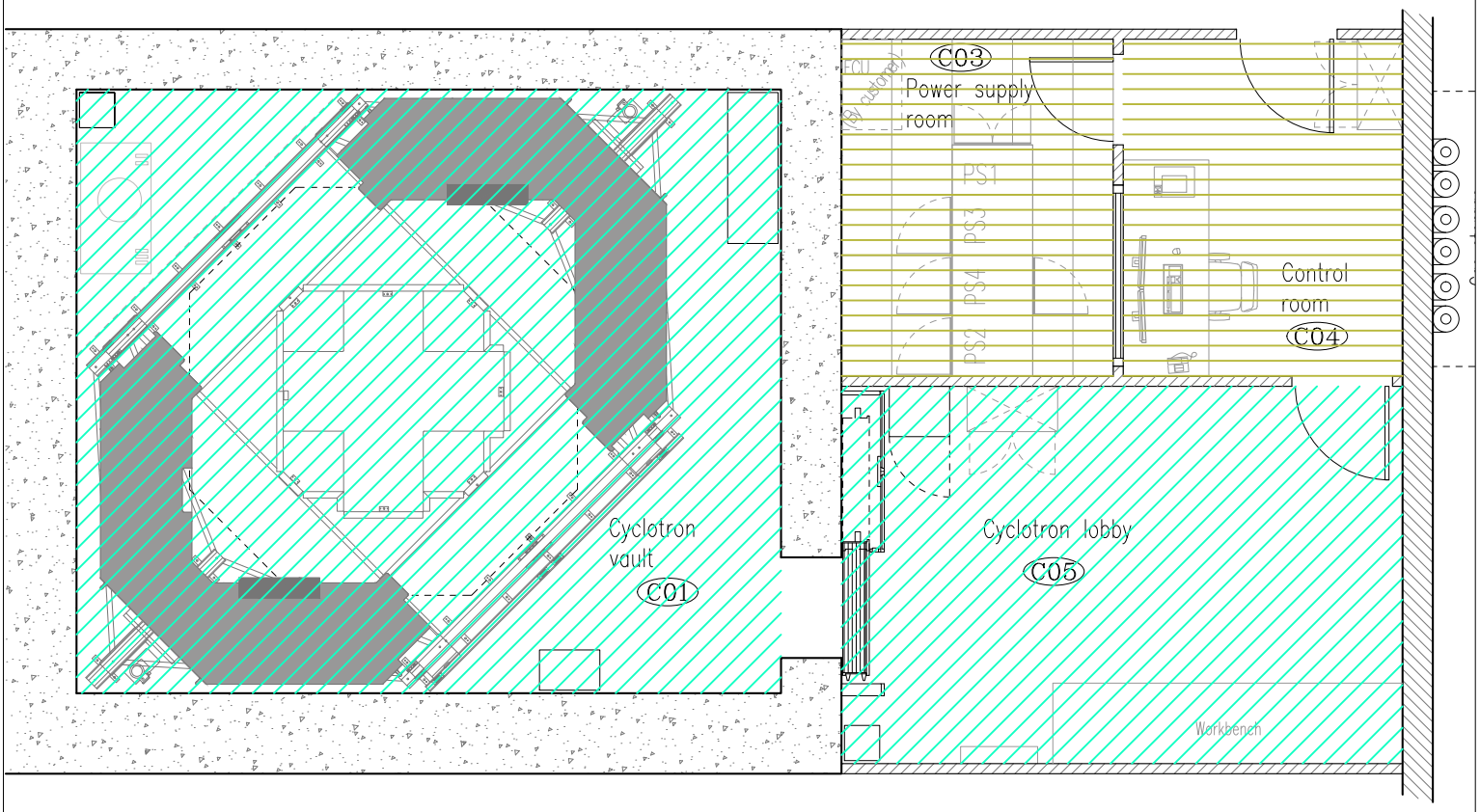
Contents: Self−shielding

Recommended pressure cascade

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Drawing SRG−0330 | Dimensions: mm

Ref.: 07.46.00.0330−G | Scale: 1/75



Legend

Minimum Air change per hour

5x

10x

based on:
- IAEA TRS 471

0 1 2 3m

0 5' 10'

S.A. Ion Beam Applications

Cyclone® KIUBE

Contents: Self-shielding

Minimum recommended air change rate

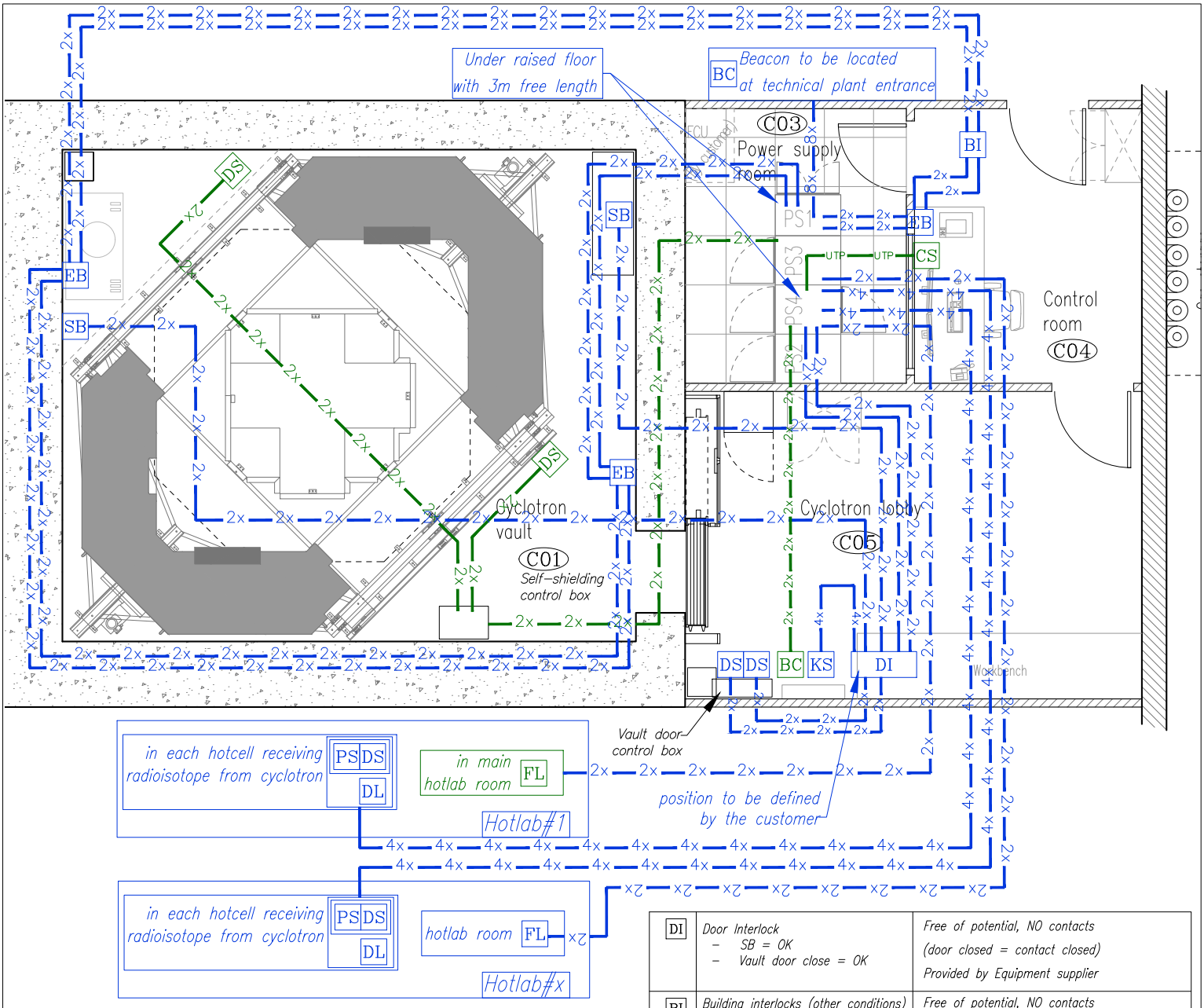
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Drawing SRG-0335

Ref.: 07.46.00.0335-G

Dimensions: mm

Scale: 1/75



Legend

EB	Emergency Button (Crash Button) Shut down cyclotron Beam only. Wall mounted 1350mm above floor.	RED crash button with unlocking, with yellow cover NC + NC contacts. Performance level: PL=d
SB	Search Button (push button) Minimum requirement wall mounted 1350mm above floor	Search vault before door closing. Provided by door Supplier.
BC	Beacon Beam ON Beacon RF ON Beacon Magnet ON Beacon Transfer ON	Provided by IBA
CS	Cyclotron control station	Provided by IBA
FL	Flash light (Transfer ON)	24 Vdc BLUE 500mA
DL	Hot cell Disable door opening during transfer	Free of potential NC contact (disable = contact open) Provided by Equipment supplier

DS	Door switch hotcell and vault door	Free of potential, NO contacts (door closed = activated) Performance level: PL=d Provided by Equipment supplier
PS	Pressure switch hotcell	Free of potential, NO contacts (door closed = activated) Performance level: PL=d Provided by Equipment supplier

DI	Door Interlock – SB = OK – Vault door close = OK	Free of potential, NO contacts (door closed = contact closed) Provided by Equipment supplier
BI	Building interlocks (other conditions) – No fire alarm – Vault ventilation OK – additional EB in series – + other conditions to Customer request	Free of potential, NO contacts (Conditions OK = contact closed) Performance level: PL=d Provided by Customer
KS	Key switch (optional)	Free of potential, NO contacts Mono stable key switch

Blue = Provided by Customer/Equipment supplier

Green = Provided by IBA

Remark:
Customer has to take care of the interlock between the bunker door and the controller of the Gamma Detector in the cyclotron vault. The door cannot be opened if the dose rate in the vault is too high.

The position and the number of the safety devices are indicative. They need to be adapted to the local safety regulations and specific site operators procedures.

Customer must provide :
- Terminal for connections
- Proper labeling on cables
- Schematic wiring

2x	LIYY , 2 x 0.5 mm², pour signal 24 Vdc et contacts redondants
4x	LIYY , 4 x 0.5 mm², pour signal 24 Vdc et contacts redondants
8x	LIYY , 8 x 0.5 mm², pour signal 24 Vdc et contacts redondants
UTP	UTP Cat. 5

0 1 2 3m
0 5' 10'

S.A. Ion Beam Applications

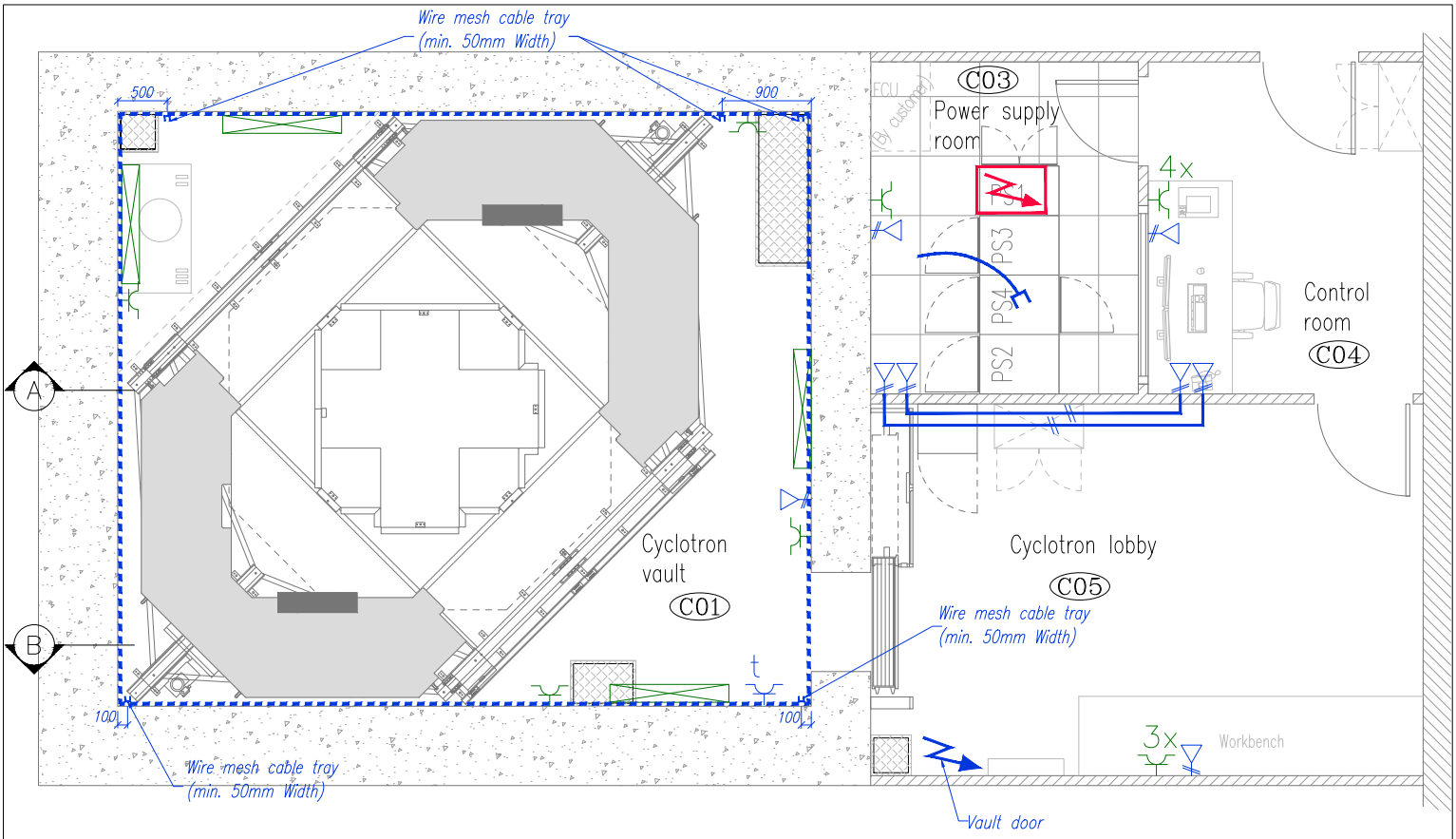
Cyclone® KIUBE

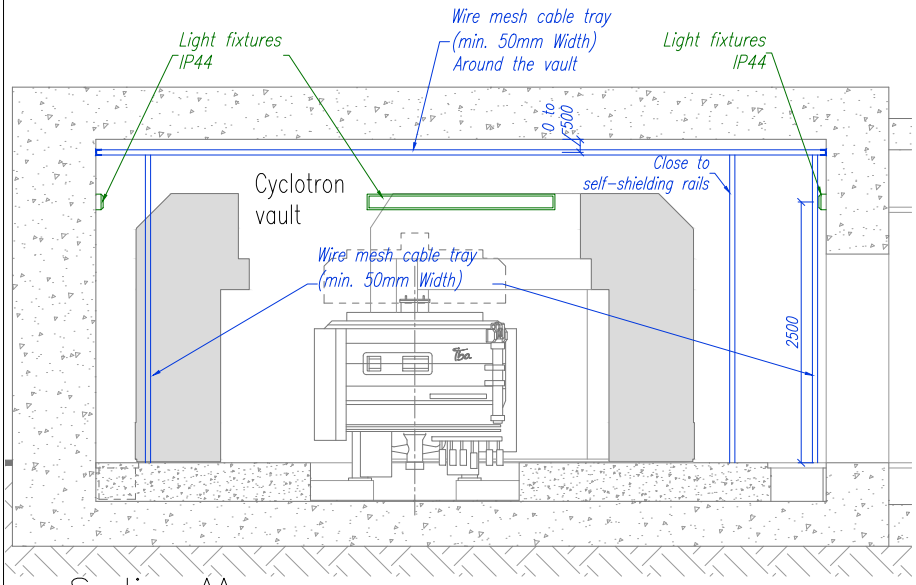
Contents: Self-shielding Cyclotron interlocks & control system interface: minimum requirements

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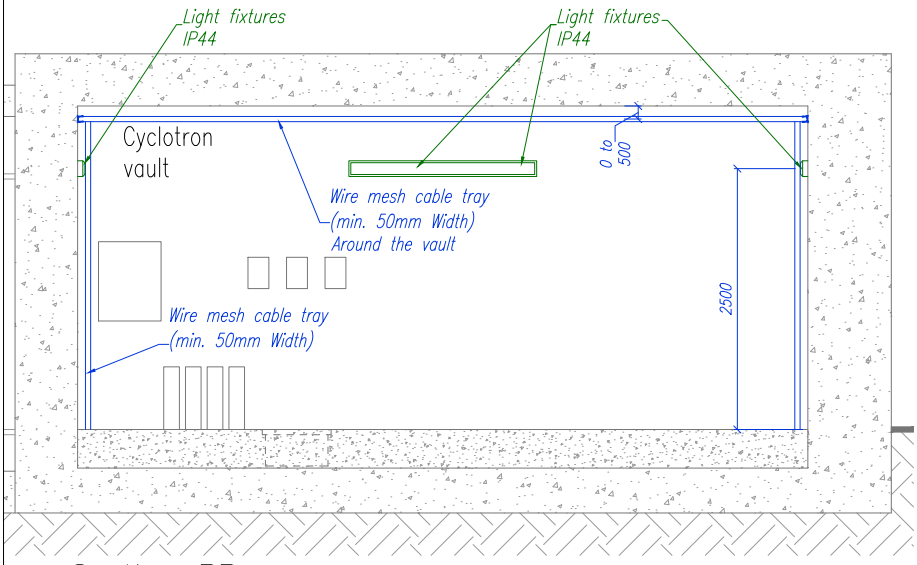
Drawing SRG-0350 | Dimensions: mm

Ref.: 07.46.00.0350-G | Scale: 1/100





Section AA



Section BB

0 1 2 3m

0 5' 10'

	Cyclotron feed (cable) – 3x400V (± 5%), AC, 50/60 Hz (± 1%) – 70 to 100 kVA : cyclotron alone (Cyclone®KIUBE 100/150/180/300) with transformer to supply 400V, 3 phases – 3 phases,+ neutral+ ground PE (5wires) – Free cable length 3m under raised floor and PS1 cabinet – 15kVA on UPS recommended for vacuum + PLC Max. short-circuit : Isc max. ≤ 25kA rms
	Electrical feed (cable) – 230V – 16A – Arrive in false ceiling + 3m free cable length. – Following Buyer's door manufacturer requirement
	Cable UTP cat. 5-E + RJ45 connector to Internet network Available in the raised floor Free cable length : 3m under raised floor and PS4 cabinet
	Fluorescent light fixture (IP44) at 2000mm high

nx	Power outlet – Plug 230V, 50/60 Hz, 16A – Quantity minimum only for IBA equipment – Plug type E.
nx	Twin socket RJ45
t	Female 400V, 16A, 3P + N + T, IP 44 Power socket according to EN 60309-1 and 60309-2 At 500mm above floor (Or temporary connection during rigging : see SRG paragraph 9.2.3)

Note: *
- All plugs near workbench/table shall installed be above the table.
- The plugs in Cyclotron vault are at 1000mm above the floor
- All others are at 180mm above floor

This layout shows only the requirements for specific equipments related to IBA scope.
→ To be completed by local engineering

S.A. Ion Beam Applications

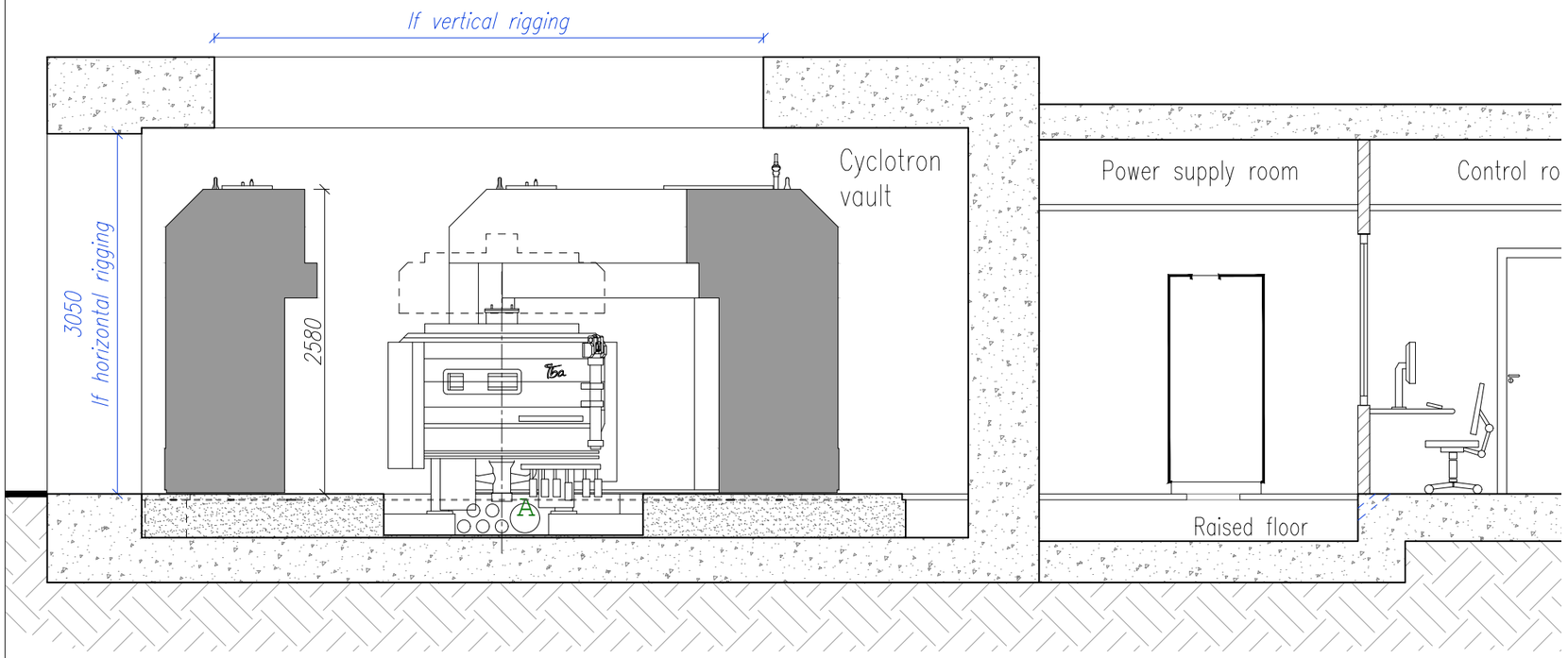
Cyclone® KIUBE

Contents: Self-shielding Electricity & data

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Drawing SRG-0370 | Dimensions: mm

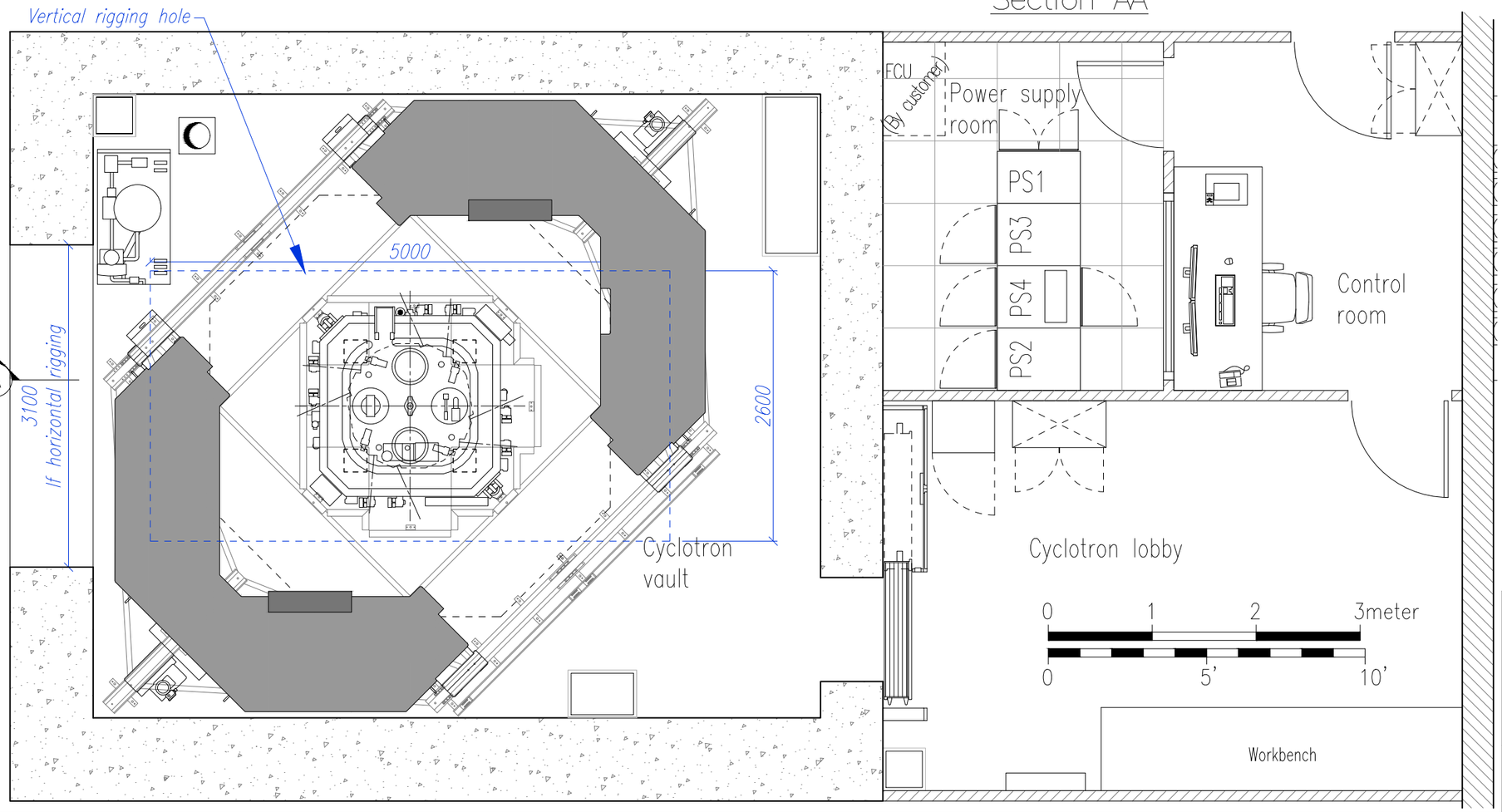
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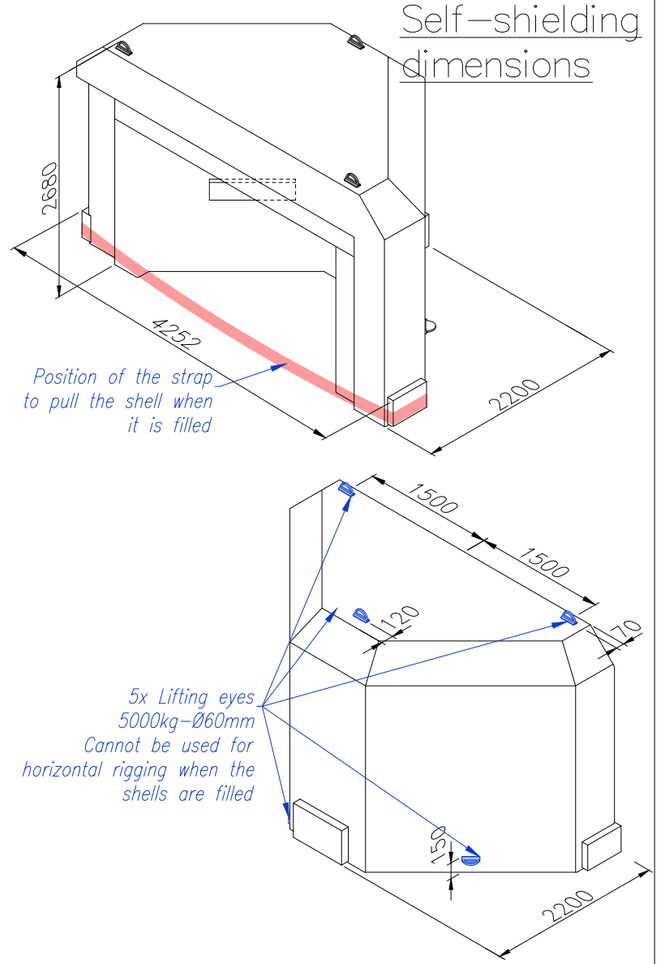
Rigging route: biggest dimensions

	L	W	H	Max. weight
Cyclotron	2050	2050	1850	18000kg
Water conditioner	1300	700	1610	220kg
Electrical cabinets	600	800	1880	600kg
Self-shielding (each piece)				
Empty Shell	4252	2200	2680	6560kg
Filled Shell+wheels & engine	4690	2652	2680	48100kg
Self-shielding Engine	680	550	907	210kg
Central pit (½ piece)	2320	1260	350	80kg
Longest rail	4450	50	10	18kg
Heaviest rail	3800	416	55	260kg

Section AA



Self-shielding dimensions



S.A. Ion Beam Applications

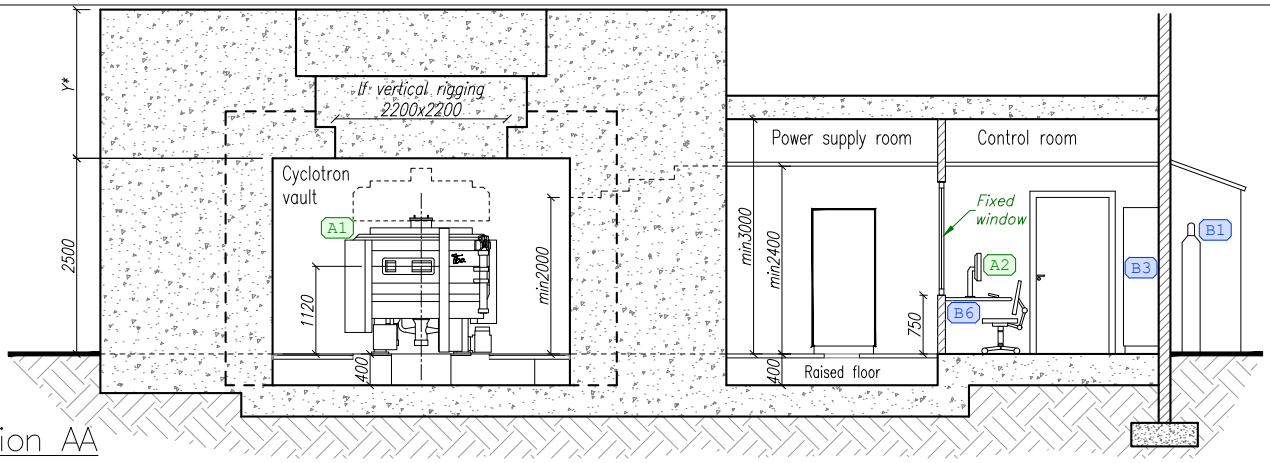
Cyclone® KIUBE

Contents: Self-shielding Rigging holes dimensions

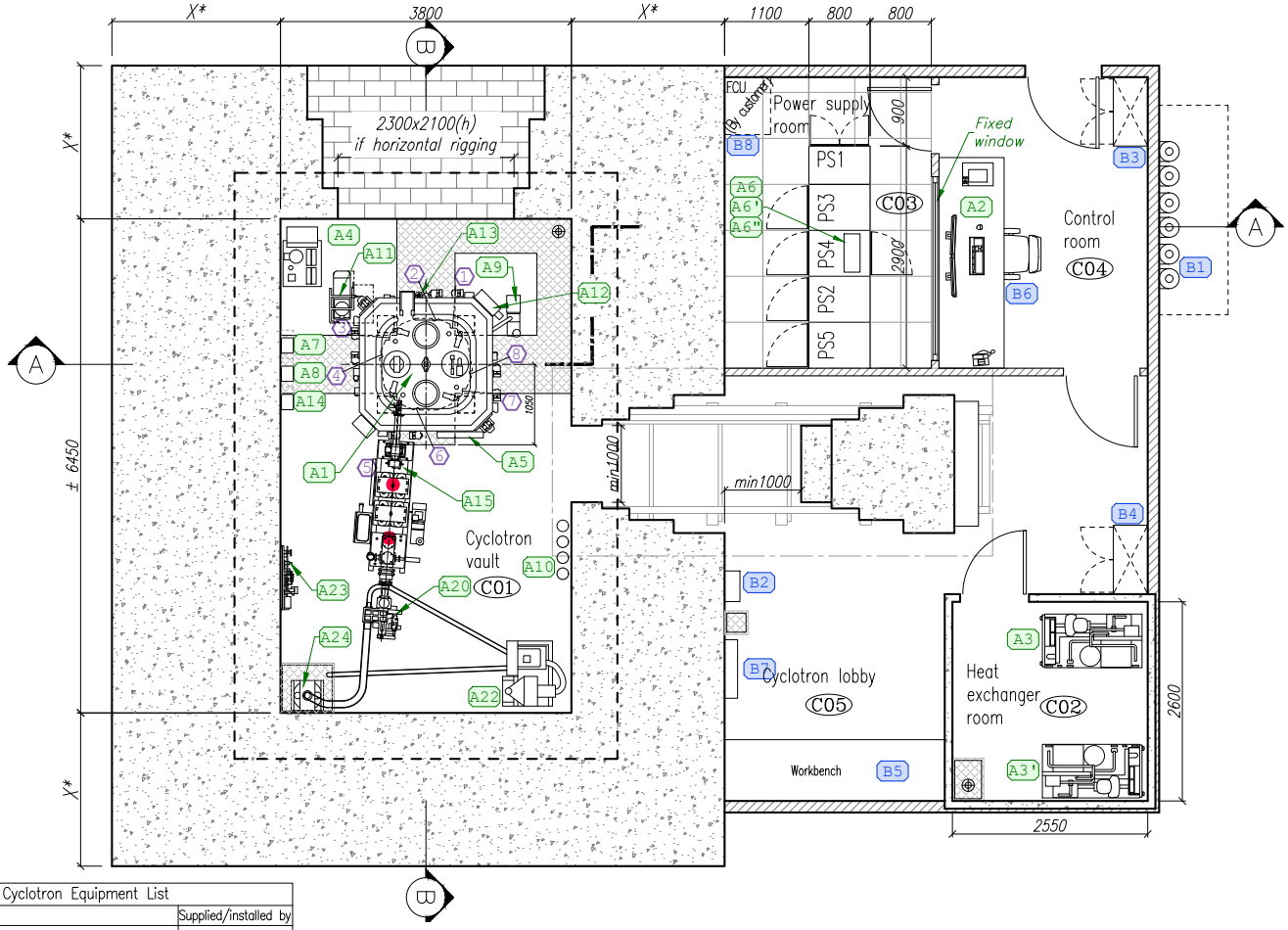
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Drawing SRG-0380 | Dimensions: mm

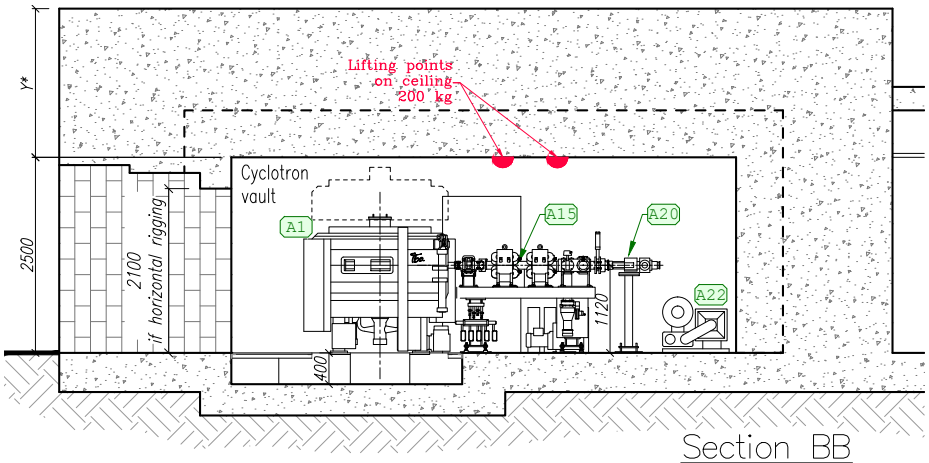
Ref.: 07.46.00.0380-G | Scale: .



Section AA



Cyclotron Equipment List		
Item	Name	Supplied/installed by
A1	Cyclone® KIUBE	IBA
A2	Control st. Thin Customer+printer+2 screens	IBA
A3	Water conditioner	IBA
A3'	Solid target Water conditioner	IBA
A4	Hydraulic Yoke lifting system	IBA
A5	Water manifold	IBA
A6	18F Syringe pumps cabinet (in PS4)	IBA
A6'	16O Rinsing Syringe pumps cabinet (in PS4)	IBA (optional)
A6''	Other Syringe pumps cabinet (13N/68Ga)	IBA (optional)
A7	Target switching input module (optional)	IBA (optional)
A8	Target switching output module (optional)	IBA (optional)
A9	Primary pump	IBA
A10	Safety pillars	IBA
A11	He compressor	IBA
A12	Plug to plug	IBA
A13	RF connection	IBA
A14	Target Gas distribution cabinet (optional)	IBA (optional)
A15	Short beam transport line	IBA
A20	Solid target Nirta® HP	IBA
A22	Blower	IBA
A23	water manifold for Nirta® HP	IBA
A24	Switching for Nirta® HP	IBA
PS1	Power Distribution Cabinet	IBA
PS2	R.F. Driver Amplifier Cabinet	IBA
PS3	Main Coils Ion Source P.S. Cabinet	IBA
PS4	Siemens P.L.C. Cabinet	IBA
PS5	BTL Power Supply Cabinet	IBA
B1	Gas cylinders supply	Customer
B2	Vault door control box	Customer
B3	Documentation storage cabinet	Customer
B4	Spare parts storage cabinet	Customer
B5	Workbench	Customer
B6	Desk + chair	Customer
B7	Cyclotron gas regulator panel	Customer
B8	Fan coil unit	Customer
1→8	Exit port number	IBA



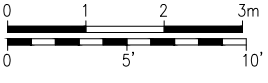
Section BB

All displayed dimensions are minimal values.

X* = 2000 to 2400mm
Y* = 1700 to 2300mm

The shielding thickness shown on this layout is indicative only. It is the responsibility of the buyer to determine the appropriate level of shielding and to execute the construction to fulfill the applicable local legislation.

The specific drawings of the scope of supply will be provided at later stage.



S.A. Ion Beam Applications

Cyclone® KIUBE

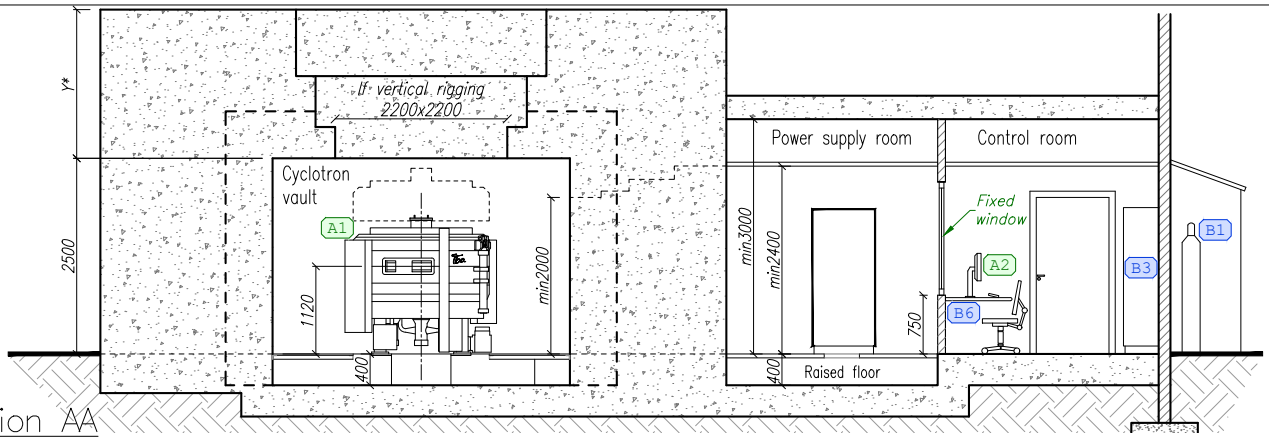
Contents:

Site configuration
KIUBE + short BTL

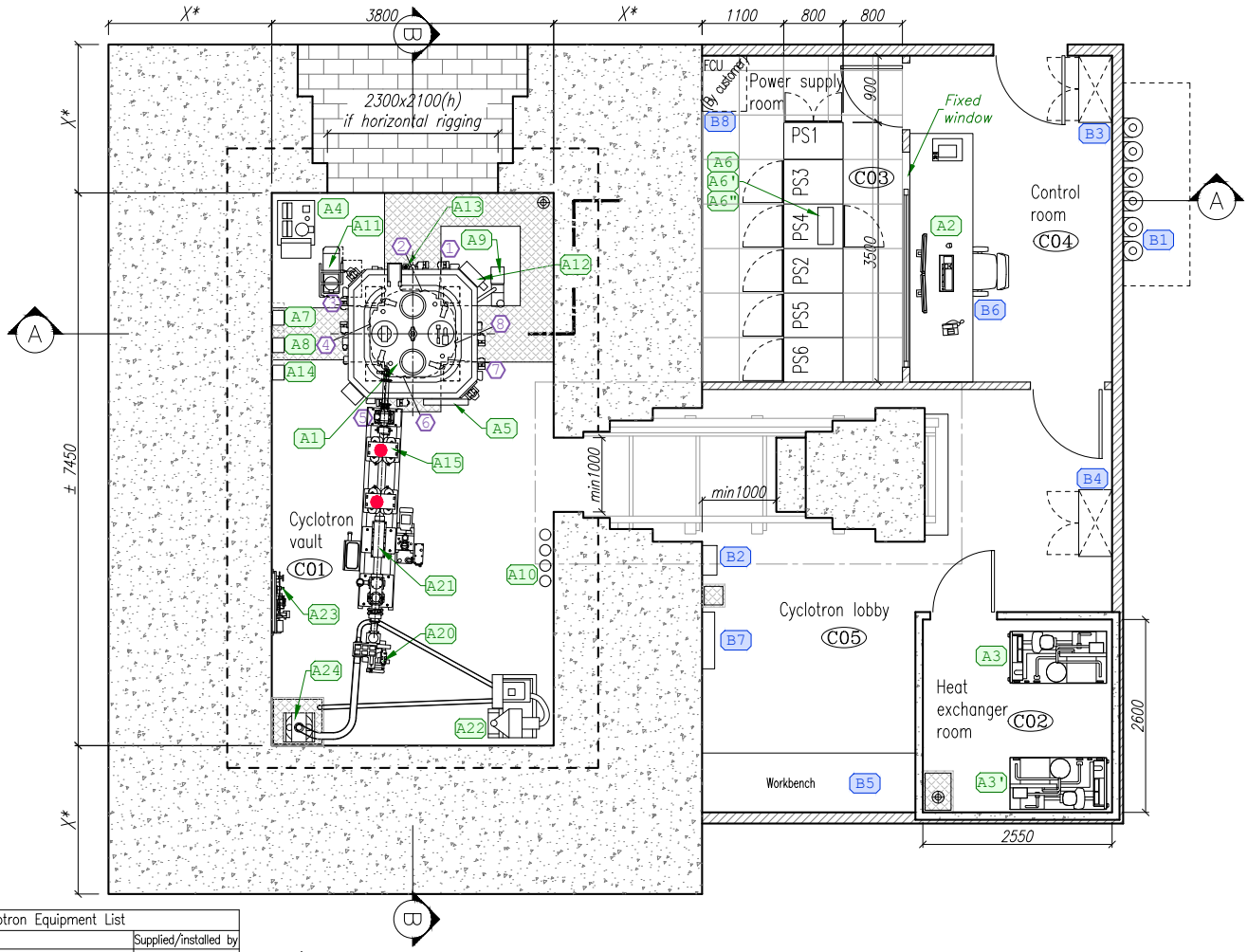
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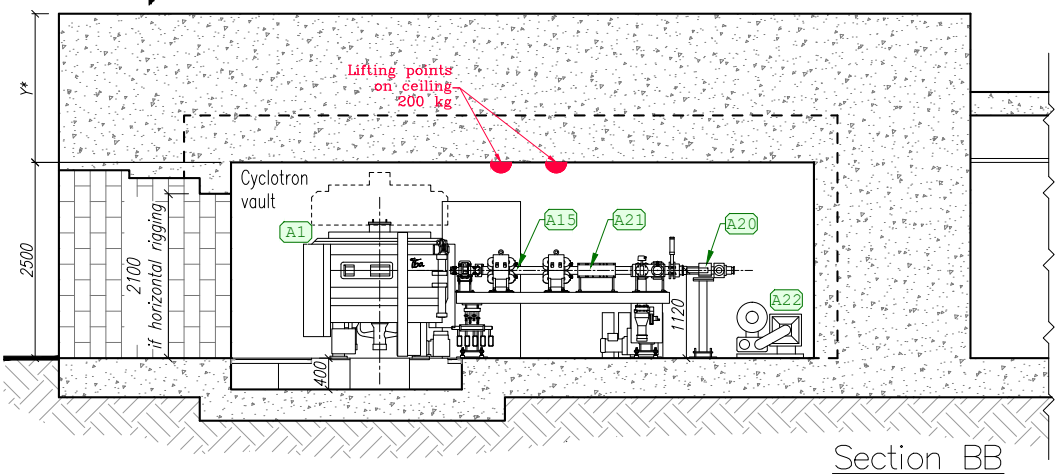
Drawing SRG-0400
Ref.: 07.46.00.0400-G

Dimensions: mm
Scale: 1/100



Section AA





Section BB

Cyclotron Equipment List		
Item	Name	Supplied/installed by
A1	Cyclone® KIUBE	IBA
A2	Control st. Thin Customer+printer+2 screens	IBA
A3	Water conditioner	IBA
A3'	Solid target Water conditioner	IBA
A4	Hydraulic Yoke lifting system	IBA
A5	Water manifold	IBA
A6	18F Syringe pumps cabinet (in PS4)	IBA
A6'	16O Rinsing Syringe pumps cabinet (in PS4)	IBA (optional)
A6''	Other Syringe pumps cabinet (13N/68Ga)	IBA (optional)
A7	Target switching input module (optional)	IBA (optional)
A8	Target switching output module (optional)	IBA (optional)
A9	Primary pump	IBA
A10	Safety pillars	IBA
A11	He compressor	IBA
A12	Plug to plug	IBA
A13	RF connection	IBA
A14	Target Gas distribution cabinet (optional)	IBA (optional)
A15	Short beam transport line	IBA
A20	Solid target Nirta® HP	IBA
A22	Blower	IBA
A23	water manifold for Nirta® HP	IBA
A24	Switching for Nirta® HP	IBA
PS1	Power Distribution Cabinet	IBA
PS2	R.F. Driver Amplifier Cabinet	IBA
PS3	Main Coils Ion Source P.S. Cabinet	IBA
PS4	Siemens P.L.C. Cabinet	IBA
PS5	BTL Power Supply Cabinet	IBA
B1	Gas cylinders supply	Customer
B2	Vault door control box	Customer
B3	Documentation storage cabinet	Customer
B4	Spare parts storage cabinet	Customer
B5	Workbench	Customer
B6	Desk + chair	Customer
B7	Cyclotron gas regulator panel	Customer
B8	Fan coil unit	Customer
1→8	Exit port number	IBA

All displayed dimensions are minimal values.

X* = 2000 to 2400mm
Y* = 1700 to 2300mm

The shielding thickness shown on this layout is indicative only. It is the responsibility of the buyer to determine the appropriate level of shielding and to execute the construction to fulfill the applicable local legislation.

The specific drawings for the scope of supply will be provided at later stage.

0 1 2 3m
0 5' 10'

S.A. Ion Beam Applications

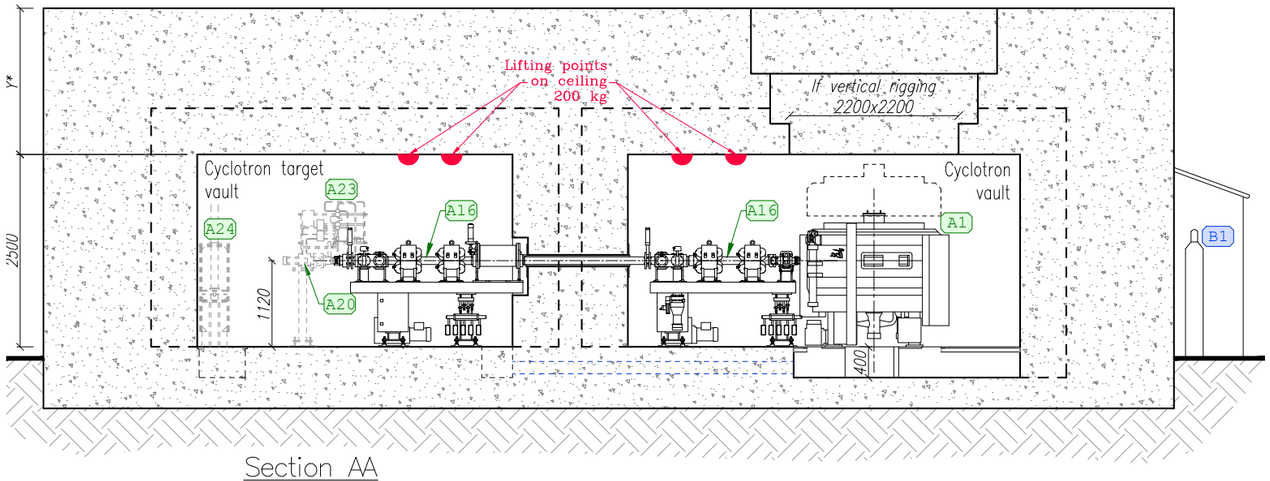
Cyclone® KIUBE

Contents:
Site configuration
KIUBE + medium BTL

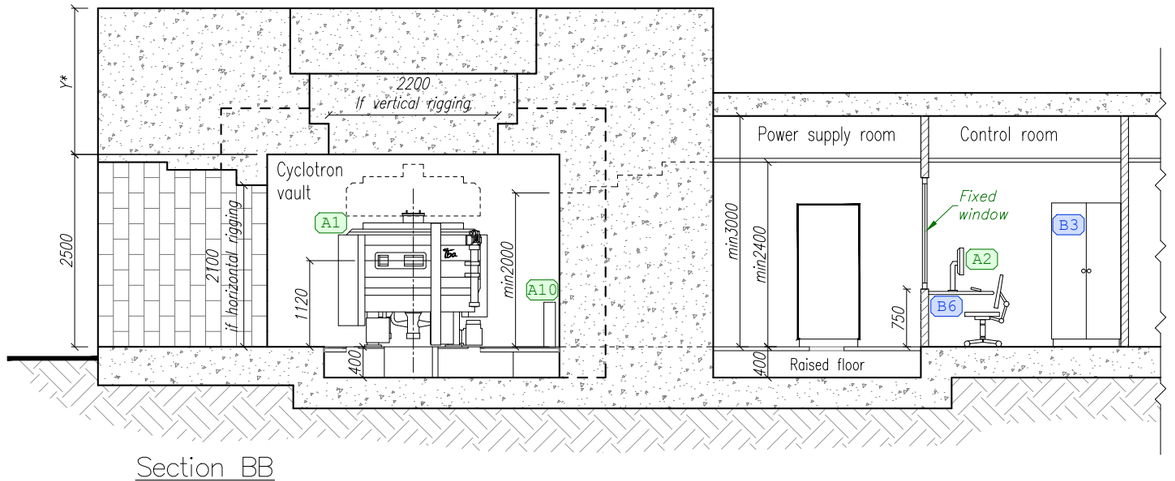
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Drawing SRG-0410
Ref.: 07.46.00.0410-G

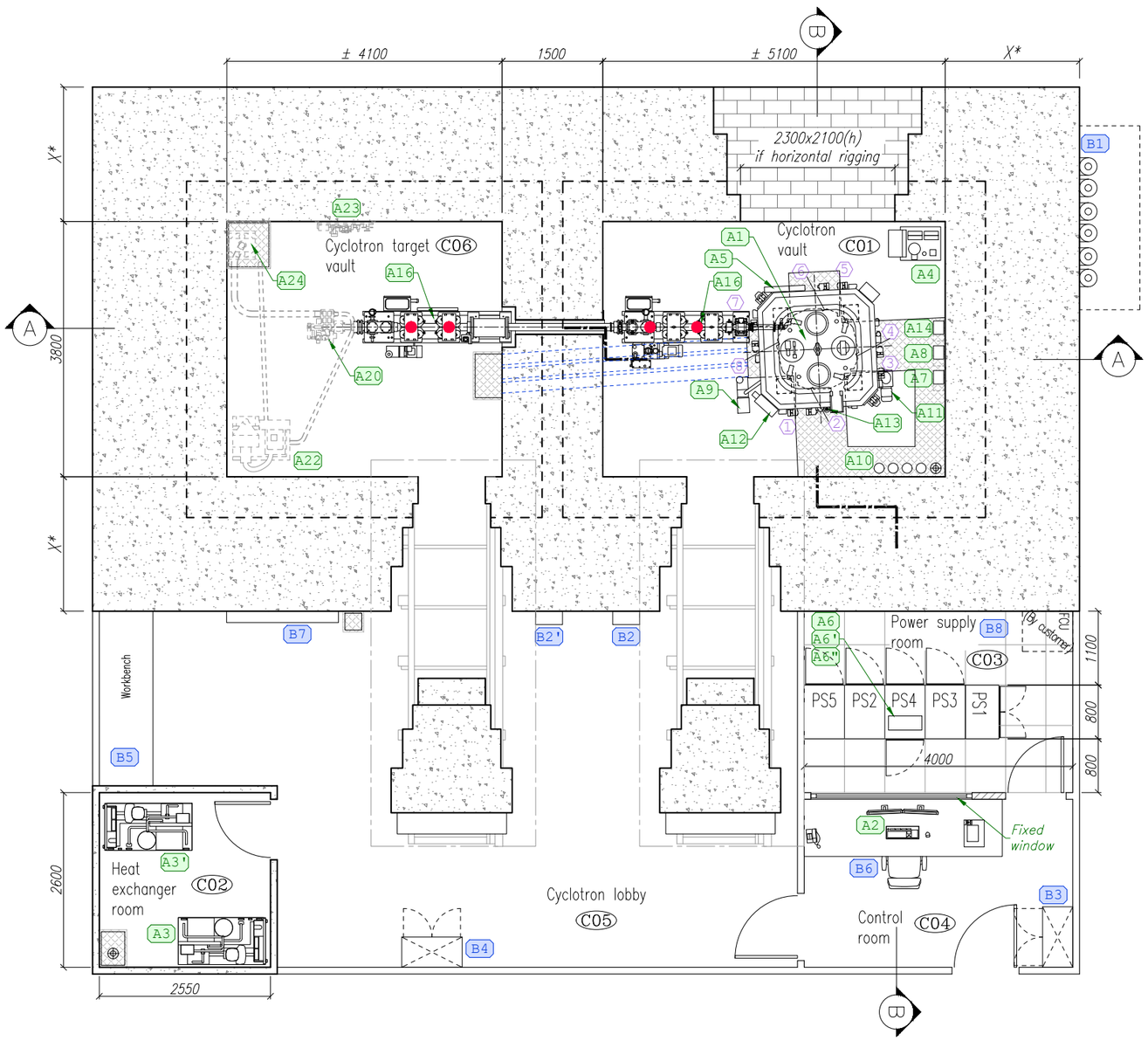
Dimensions: mm
Scale: 1/100



Section AA



Section BB



Cyclone® KIUBE	IBA
Control st. Thin Customer+printer+2 screens	IBA
Water conditioner	IBA
Solid target Water conditioner	IBA
Hydraulic Yoke lifting system	IBA
Water manifold	IBA
18F Syringe pumps cabinet (in PS4)	IBA
160 Rinsing Syringe pumps cabinet (in PS4)	IBA (optional)
Other Syringe pumps cabinet (13N/68Ga)	IBA (optional)
Target switching input module (optional)	IBA (optional)
Target switching output module (optional)	IBA (optional)
Primary pump	IBA
Safety pillars	IBA
He compressor	IBA
Plug to plug	IBA
RF connection	IBA
Target Gas distribution cabinet (optional)	IBA (optional)
Long beam transport line (example)	IBA
Solid target Nirta® HP (optional)	IBA
Blower (optional)	IBA
water manifold for Nirta® HP (optional)	IBA
Switching for Nirta® HP (optional)	IBA
Power Distribution Cabinet	IBA
R.F. Driver Amplifier Cabinet	IBA
Main Coils Ion Source P.S. Cabinet	IBA
Siemens P.L.C. Cabinet	IBA
BTL Power Supply Cabinet	IBA
Gas cylinders supply	Customer
Vault door control box	Customer
Documentation storage cabinet	Customer
Spare parts storage cabinet	Customer
Workbench	Customer
Desk + chair	Customer
Cyclotron gas regulator panel	Customer
Fan coil unit	Customer
Exit port number	IBA

displayed dimensions are minimal value

= 2000 to 2400mm
= 1700 to 2300mm

shielding thickness shown on this layout indicative only.
s the responsibility of the buyer to ermine the appropriate level of shielding to execute the construction to fulfill applicable local legislation.

The specific drawings for the scope of supply will be provided at later stage.

1 2 3meter
5' 10'

S.A. Ion Beam Applications

Cyclone® KIUBE

Contents:
Site configuration
KIUBE + long BTL

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Drawing SRG-0420
Ref.: 07.46.00.0420-G

Dimensions: mm
Scale: 1/100